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CS 411 – Software Engineering
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Software Design Specifications

For

eduassist

Version [01]

Students:

Supervised By:
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This Software Design Specification was prepared and provided as a deliverable for [Software Engineering, CS 411, term 1], and it will be used by [end user].

This document is based in part on the IEEE Recommended Practice for Software Design Descriptions.

Revision History

Name	Date	Reason For Changes	Version
All members	Oct. 21, 2024	Initial version	0.1
All members	Oct. 23, 2024	Updated section 1	0.2
All members	Nov. 25, 2024	Updated section 2 and 3	0.3
All members	Nov. 27, 2024	Updated section 4,5 and 6	0.4
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All members	Dec. **2**, 2024	Spelling and grammar mistake	0.7
All members	Dec. **3**, 2024	Complete review - Final version	1.0

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1. Introduction

The Software Design Specification (SDS) for the EDUassist project serves as a comprehensive guide for the design phase of the software development lifecycle. This document delineates the intended architecture, components, interfaces, and data structures for the EDUassist, ensuring a clear understanding of the design intent among stakeholders. By outlining the design principles and methodologies, this SDS provides a structured framework to facilitate communication and alignment among team members during the documentation phase, with no actual implementation occurring at this stage.

1.1 Purpose

The purpose of this document is to detail the design specifications for the EDUassist, which is envisioned as an academic management tool to assist students in organizing their educational responsibilities. This SDS outlines the system's architecture, identifies key components and their interactions, and specifies the interfaces that will facilitate communication within the system. By establishing a clear framework, this document aims to ensure that all stakeholders share a common understanding of the design goals and requirements, thereby guiding the development team in subsequent phases.

1.2 Scope

This document delineates the design specifications for the EDUassist, emphasizing the unique design features that align with the planned characteristics and services of the platform. It encompasses a comprehensive overview of the system’s architecture and logical design, including a conceptual illustration of the user interfaces as a prototype to demonstrate interface flow.

Additionally, the document outlines the primary functionalities through pseudocode, offering clarity on operational logic. It provides an in-depth analysis of data management, detailing the relationships between data entities and their modeling of data using :

- Entity-Relationship Diagram (ER)
- Relational Schema (ER Mapping)
- Sequence Diagrams
- Flow chart
- DFD

This document serves as a critical resource for software developers, project managers, and quality assurance teams involved in the theoretical design of the EDUassist, ensuring a comprehensive understanding of its foundational aspects

1.3 Definitions, Acronyms, and Abbreviations

Terminology	Definition
EDUassist	A platform designed to assist students in managing studies and collaborating using AI

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	tools.
Study Plan	A personalized schedule tailored to students' syllabus and preferences for tracking progress.
AI Recommendations	Suggestions generated by the system to help users access relevant learning resources.
Whiteboard Collaboration	A shared virtual workspace enabling students to interact and annotate in real time.
Admin	A system user responsible for managing user accounts, monitoring progress, and sending alerts.
Student	The primary user of EDUassist, utilizing features like study plans and collaborative tools.
Notifications	Alerts sent by the system to inform users of deadlines, updates, or recommendations.
External Tools	Third-party apps like Google Calendar or Zoom integrated into the platform.

Table 2 Terminology

Acronym	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
OTP	One-Time Password
LMS	Learning Management System
UML	Unified Modeling Language
DFD	Data Flow Diagram
SRS	Software Requirements Specification
SDS	Software Design Specification
UI	User Interface
ERD	Entity-Relationship Diagram
QA	Quality Assurance

Table 3 Acronym

Abbreviation	Meaning
Dashboard	The main interface for users to interact with the system.
Task Tracker	A feature for monitoring tasks, deadlines, and progress.
Collab	Short form for collaboration in the whiteboard feature.
RecSys	Abbreviation for Recommendation System.
Processes	Actions within the DFD, such as generating study plans or managing user profiles.
Entities	External actors in the DFD, such as students, admins, or third-party tools.

Table 4 Abbreviation

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2. System Overview

EDUassist is an advanced academic management platform designed to support students and administrators by providing tools for effective study planning, resource management, and collaborative

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learning. The primary goal of the system is to simplify academic tasks through personalized recommendations, AI-assisted learning tools, and seamless integration with external services such as Google Calendar.
The platform's core strengths lie in its intuitive interface, robust security measures, and adaptability to individual learning preferences. EDUassist ensures that users can access all features securely while maintaining a high level of user satisfaction.

Common Functionalities
There are two common functionalities available for both students and administrators:
Login/Sign-Up: Secure access to the platform with support for password recovery.
Profile Management: Update personal information and preferences.

Admin Functionalities
- User Management: Create, modify, and delete user accounts for students.
- System Monitoring: Track and generate reports on system performance and user activities.
- Content Management: Upload, edit, and organize educational resources.
- Integration Management: Configure integrations with external tools like Google Calendar.
- Generate Reports: Access analytics and insights into user progress and engagement.
- Admin Profile: View and update personal information.

Student Functionalities
- Personalized Study Plans: Create, view, and modify study plans tailored to academic goals and deadlines.
- AI-Assisted Whiteboard: Collaborate with peers in real-time, annotate, and visualize learning materials.
- Video Generation: Generate short explanatory videos from whiteboard sessions for later review.
- Task Tracking: Track academic tasks and receive reminders for upcoming deadlines.
- Access Resources: Browse a library of study materials and receive AI-driven recommendations based on progress and preferences.
- Collaborative Tools: Share and collaborate on projects with peers through interactive features.
- External Integration: Sync tasks and events with Google Calendar for better organization.
- Profile Management: Update personal data and adjust study preferences.

System Features
1. Authentication Services: Secure login and password recovery, including two-factor authentication.
2. Data Management: Store and manage user data, study plans, and progress tracking securely.
3. AI-Powered Assistance: Offer personalized recommendations, study suggestions, and intelligent insights.
4. Integration Services: Seamlessly connect with third-party tools like Learning Management Systems and cloud platforms.
5. Performanc**e Tracking: Monitor user interactions and provide usage statistics for system optimization.**

3. Design Considerations

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This section outlines the design considerations for the EDUassist system, focusing on assumptions, dependencies, and constraints that may impact the design process.

3.1Assumptions and Dependencies

- **Related Software or Hardware:**
  - **Software Dependencies:** The system depends on compatibility with widely used third-party tools like Google Calendar, Zoom, and LMS platforms for seamless integration. REST APIs and OAuth 2.0 are used to manage secure connections and data synchronization.
  - **Hardware Assumptions:** EDUassist does not require specialized hardware. It assumes users will access the platform through standard devices such as laptops, tablets, and smartphones running modern operating systems.

- **End-User Characteristics**
  The primary users of EDUassist are students and educators with varying levels of technical expertise. The interface and features are designed to cater to users who may not possess advanced technical skills, ensuring usability and inclusivity.

- **Possible and/or Probable Changes in Functionality**:

  - **Scalability Requirements**: Future scalability will involve accommodating a growing user base without compromising system performance. The modular design allows for iterative upgrades.
  - **Feature Extensions**: Anticipated updates may include support for additional external tools, expanded language options, or enhanced collaboration features based on user feedback.

3.2 General Constraints

- **Hardware or Software Environment**: EDUassist is designed to work on modern devices like computers, tablets, and smartphones, using updated operating systems and web browsers such as Chrome, Safari, and Edge. However, older devices or outdated software might cause slower performance or limit some features.
- **End-User Environment**: The system needs to work well in different conditions, including areas with slow internet speeds or users with devices that have lower capabilities. This means the platform should be lightweight, quick, and able to deliver the main features smoothly, no matter the environment.
- **Data Repository and Distribution Requirements**: The platform relies on a secure and scalable database to manage things like user profiles, study plans, and AI recommendations. It must ensure that data is handled safely and efficiently, even when lots of users are online at the same time, to avoid delays or losing information.
- **Security Requirements**: Strong security features like data encryption and two-factor authentication are necessary to keep user information safe. These measures are essential to prevent breaches and build trust with users.

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- **Performance Requirements**: EDUassist has to manage many users at once, especially for features like collaborative whiteboards and task tracking. This means the system must be designed to use server resources wisely and keep everything running smoothly during busy times.
- **Interoperability Requirements**: The platform integrates with tools like Google Calendar and Learning Management Systems (LMS). To make this work, it needs to use standard APIs and protocols so that data can sync and move between systems without any issues.

By addressing these limitations, the EDUassist will provide a secure, user-friendly, and scalable platform that meets user needs while maintaining compliance with technical and regulatory standards. While still ensuring that EDUassist remains reliable and adaptable to the diverse needs of its users and the demands of modern technology.

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4. User Interface Design

4.1 Overview of User Interface

From the user's perspective, the EDUassist system provides an intuitive and streamlined interface to manage various academic tasks and responsibilities. Upon logging in, users are greeted with a personalized dashboard that displays key information such as upcoming tasks, deadlines, study plans, and notifications. The system allows users to easily navigate through different sections, such as task tracking, study plan creation, and resource access.

Users can create, update, and track their study plans, with the system providing personalized recommendations based on their academic goals and progress. The task management functionality enables users to add, edit, and mark tasks as complete, while receiving reminders for upcoming deadlines. Additionally, the system includes collaborative tools like the whiteboard, where users can annotate, draw, and share notes in real time with others.

Throughout these processes, the system provides clear feedback to the user. For instance, when a task is completed, a success message appears, confirming the update. In the case of errors, such as incorrect input or missing information, the system displays error messages to guide users toward the necessary corrections. The interface also offers visual cues for tasks and events, ensuring users are always informed about their academic progress and upcoming obligations. This feedback ensures a smooth and efficient user experience, empowering users to effectively manage their academic life.

4.2 Interface Design Rules

The EDUassist application will adhere to the following interface design rules to produce a well-designed, intuitive, and user-friendly operating interface:

1. **Consistency**
- Maintain a consistent layout, interaction patterns, and visual elements throughout the interface.
- Ensure that users can anticipate the behavior of various elements and build mental models of the system.
- Standardize the use of typography, color schemes, and iconography across the application.
2. **Simplicity**
- Reduce complexity and maintain a simple interface.
- Prioritize clarity and avoid extraneous cognitive load or visual clutter.
- Design the interface to be easy to learn and navigate.
3. **Visibility**
- Clearly display and make easily accessible any pertinent information about the system's status.
- Utilize visual cues like color, contrast, and typography to highlight important details.
- Provide users with feedback to help them understand the system's reactions and the results of their actions.
4. **Feedback**
- Give users prompt and insightful feedback as they engage with the interface.
- Provide feedback in the form of textual messages, animations, or visual cues.
- Ensure that the feedback explains the system's reaction and informs users of the result of their actions.
5. **User Control**

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- Give users authority over the features and interface.
- Provide easy-to-use navigation, undo/redo functionality, and customization options.
- Allow users to explore and interact with the system according to their preferences.

6. **Error Prevention and Handling**

- Employ constraints, validation, and unambiguous instructions to assist users and reduce errors.
- When mistakes do occur, provide helpful error messages along with recommendations for fixing them.
- Design the interface to avoid errors as much as possible.

7. **Accessibility**

- Ensure that people with disabilities can utilize the interface.
- Provide alternate forms of interaction, such as keyboard navigation and screen reader compatibility.
- Adhere to accessibility guidelines and standards to ensure an inclusive user experience.

8. **Responsiveness**

- Maintain functionality and usability across different platforms and resolutions.

9. **Aesthetic Integrity**

- Design the interface with a visually appealing aesthetic that aligns with the application's purpose and target audience.
- Use appropriate imagery, color palettes, and typography to create an engaging user experience.
- Ensure that the aesthetic elements do not detract from functionality or usability.

By adhering to these interface design rules, the EDUassist application will provide a well-designed, intuitive, and user-friendly experience, enabling students and administrators to effectively interact with the system and achieve their academic goals.

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4.3 Screen Images

4.3.1 Log in:

Login/Sign-In Interface of the EDUassist platform. This interface serves as the gateway for users, providing secure access to the system. Users are required to enter their credentials, such as a username and password, into clearly labeled fields. The interface includes validation to ensure the accuracy of the input, displaying error messages for incorrect credentials or missing fields. Once the user successfully logs in, they are redirected to a personalized dashboard, tailored to their role (student or administrator). This interface prioritizes simplicity and security, with additional options for password recovery and assistance for new users who need to create an account.

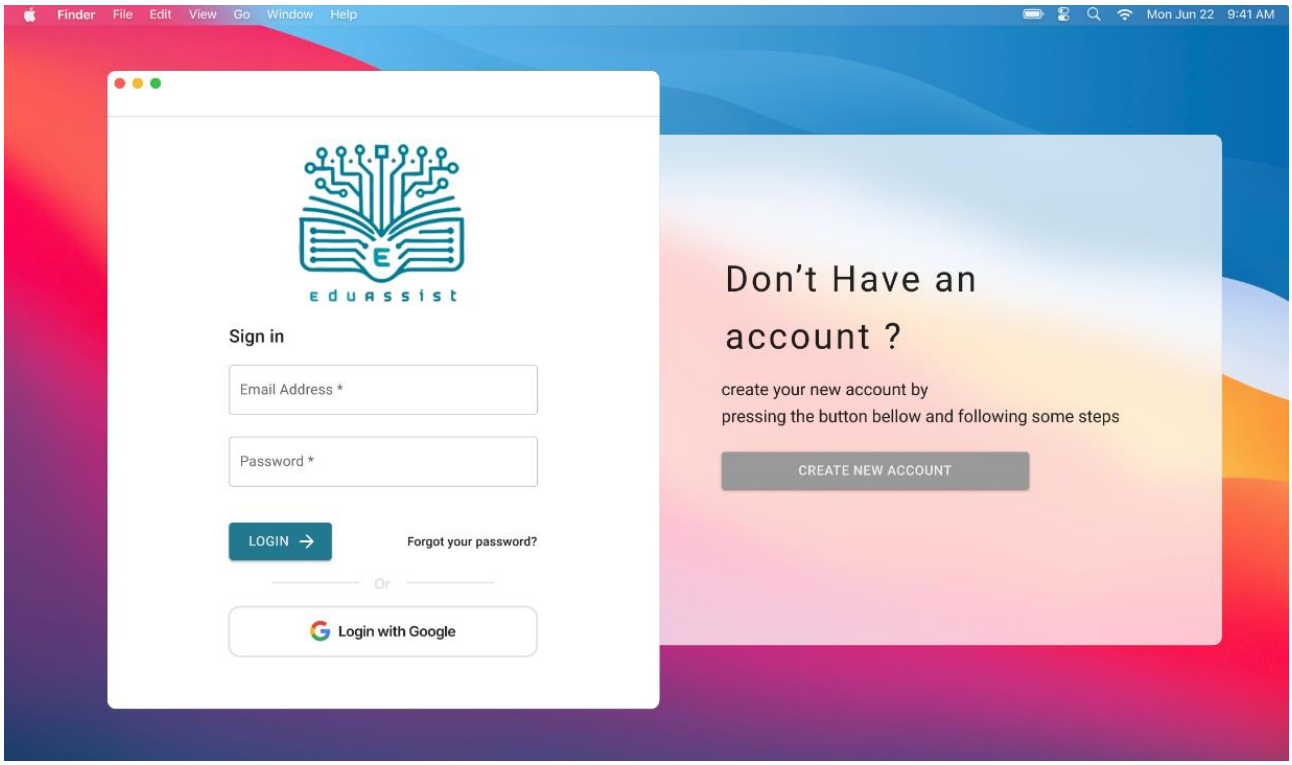


Figure 1 sign in interface

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4.3.2 log in with google:

Log in with Google Interface. This interface provides users with the option to log in using their Google account credentials, streamlining the authentication process. A prominent "Login with Google" button is displayed, allowing users to initiate a secure OAuth-based connection with their Google account. Once clicked, the system redirects the user to Google's authentication page, where they can confirm their account details. After successful validation, users are automatically redirected to their EDUassist dashboard. This interface enhances convenience by eliminating the need to remember separate passwords while ensuring secure and efficient access through Google's trusted authentication services.

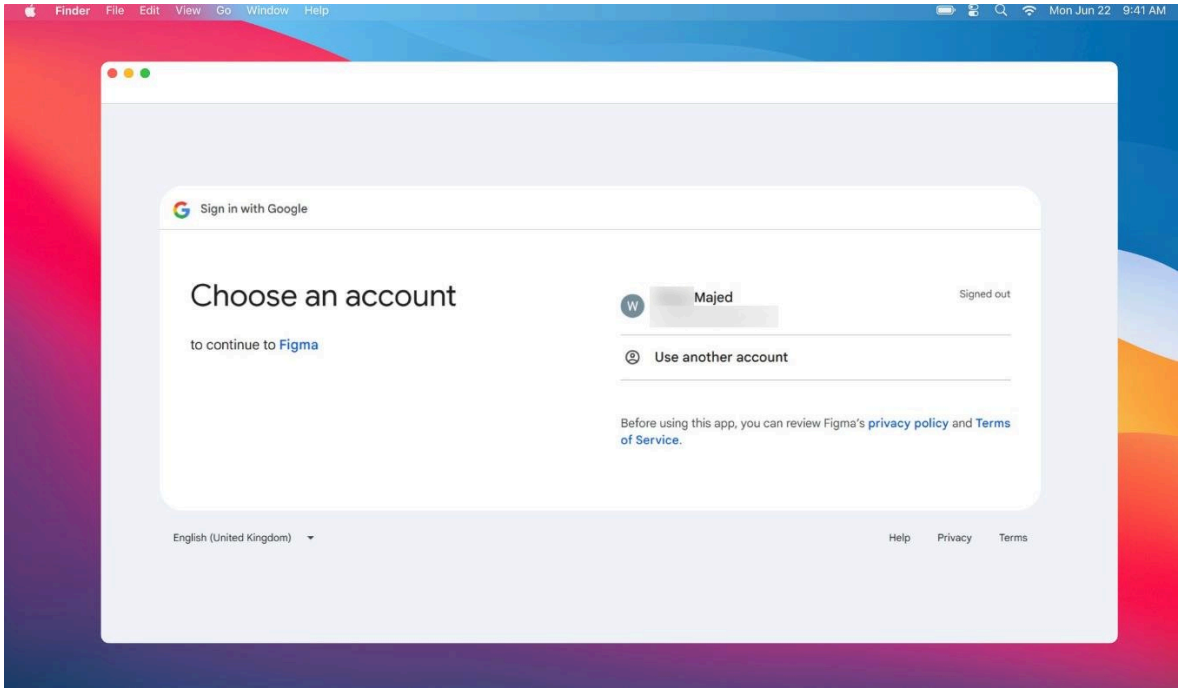


Figure 2 sign in with google p1

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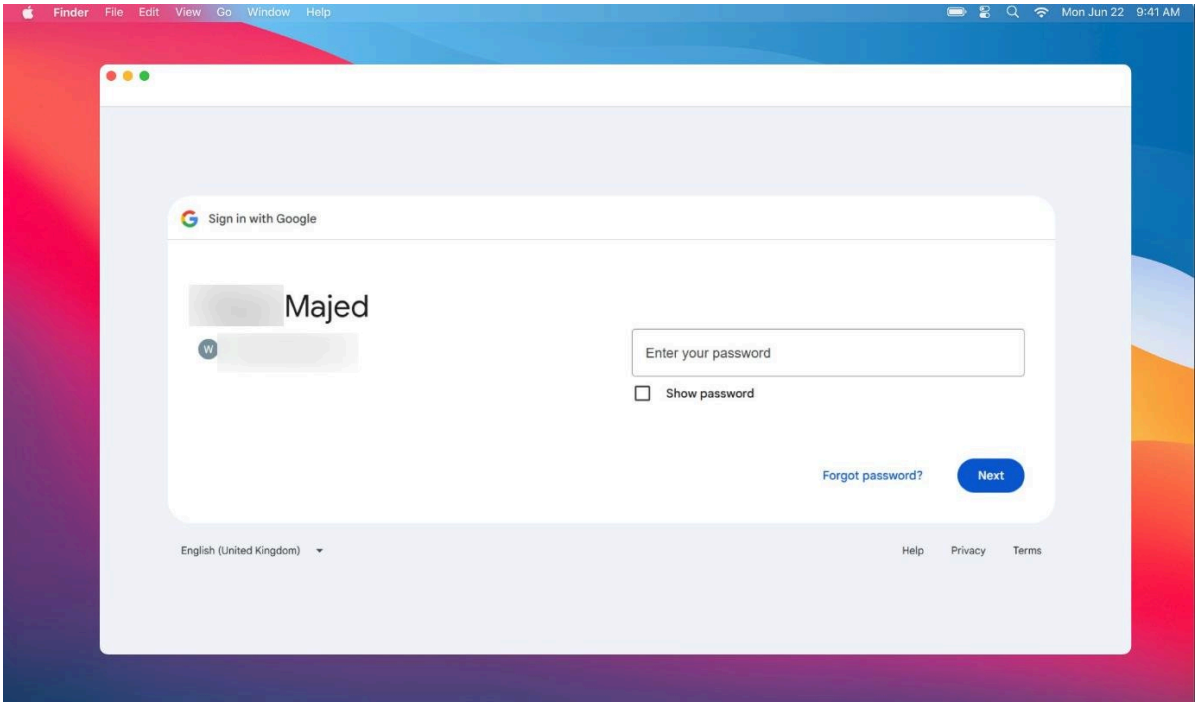


Figure 3 sign in with google p2

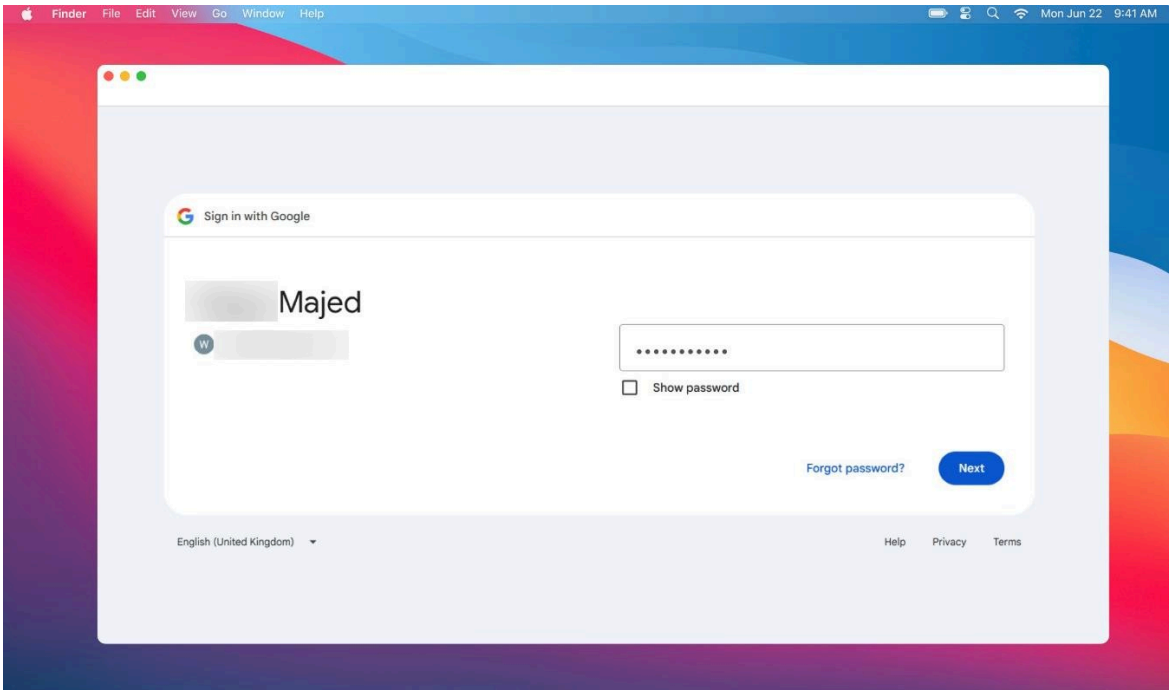


Figure 4 sign in with google p3

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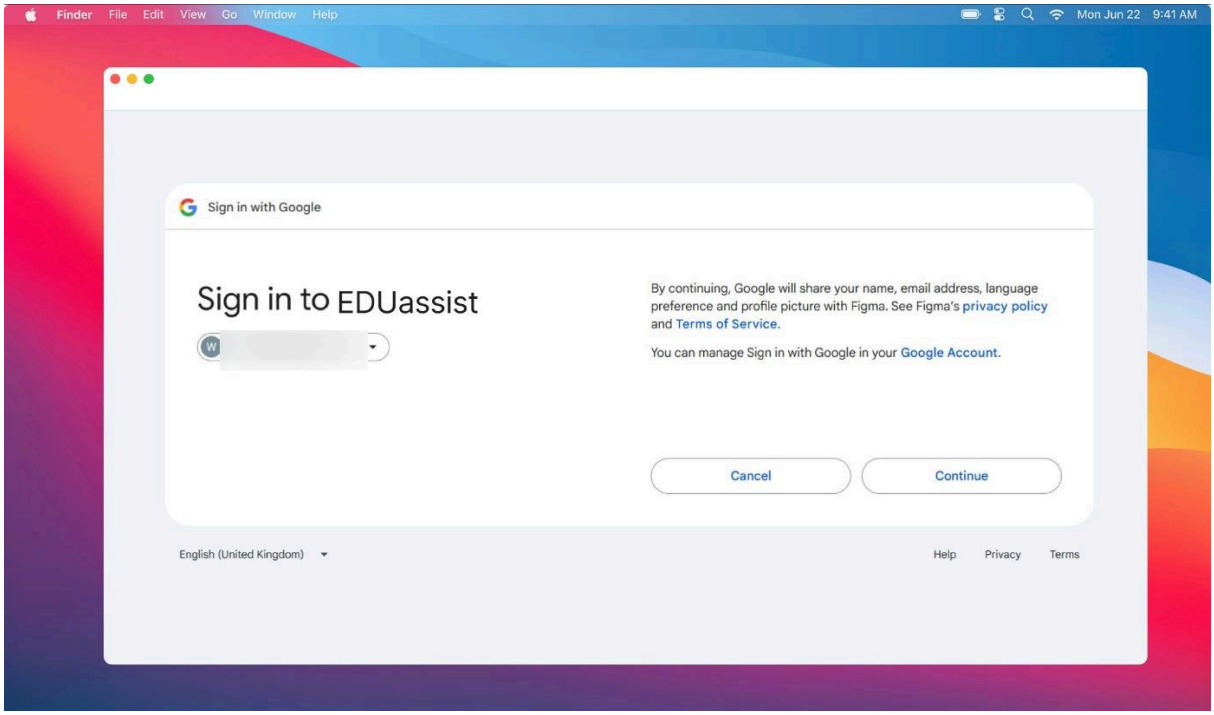


Figure 5 sign in with google p4

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4.3.3 sign up:

Sign-Up Page interface is tailored for new users, specifically students, to register on the EDUassist platform. This interface presents a clear and intuitive form requiring inputs such as the user's full name, email address, and a secure password. Validation mechanisms ensure that fields like email and password meet the system's requirements, such as a valid email format and password strength. The interface includes a prominent "Sign Up" button to submit the form and initiate the account creation process. If the provided details are incorrect or incomplete, error messages are displayed to guide the user in resolving the issues. Additionally, the page offers a convenient link to the login interface for users who already have an account. The design focuses on simplicity and efficiency, enabling students to quickly and securely join the platform.

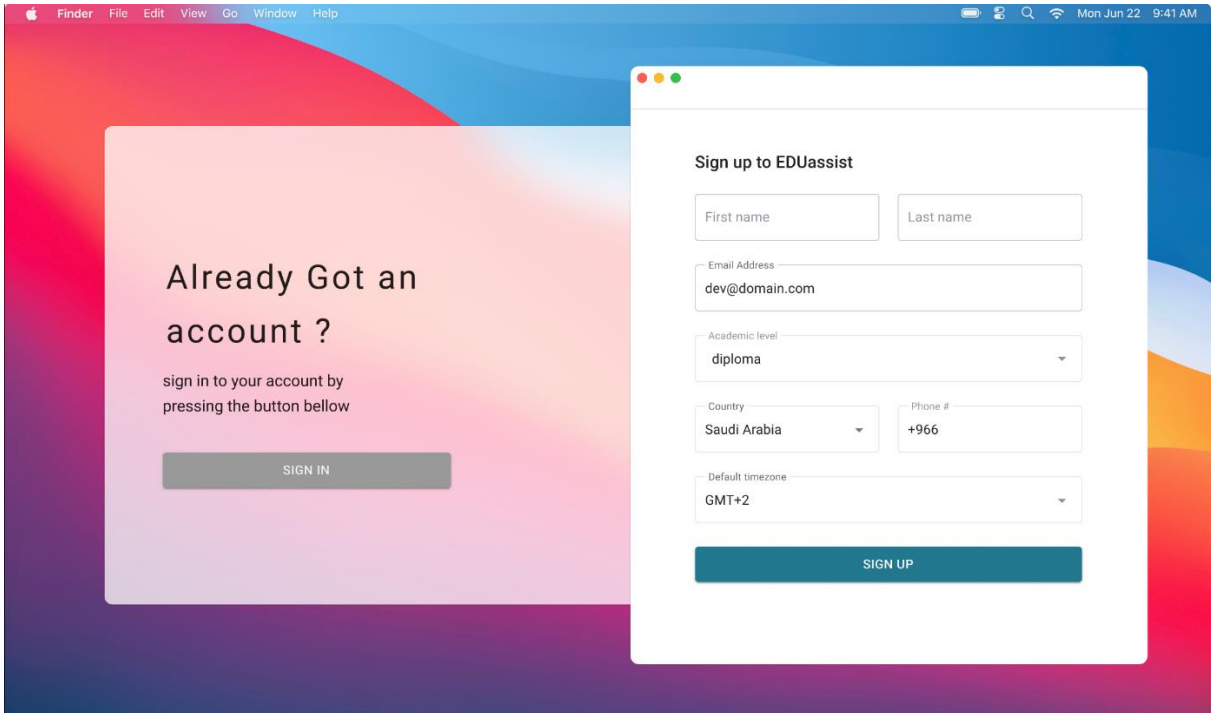


Figure 6 sign up interface



Figure 7 Varification code massage

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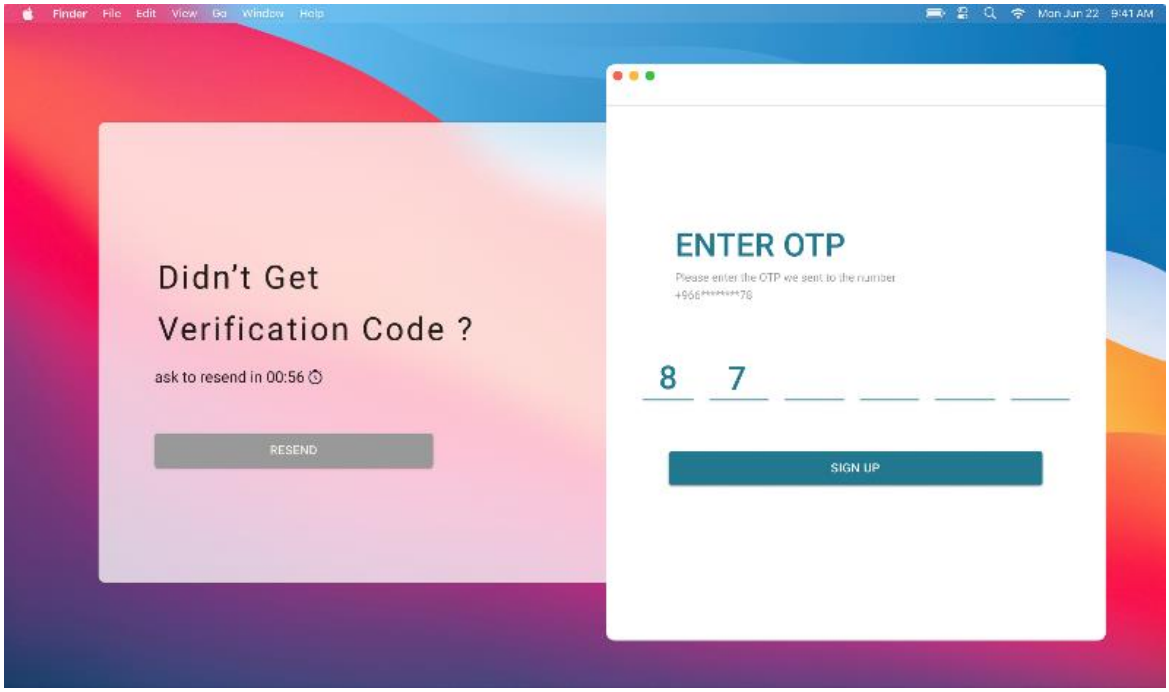


Figure 8 OTP interface

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4.3.4 home page:

The Home Page interface serves as the central hub for users after logging into the EDUassist platform. It is designed to provide a comprehensive overview of the system's features and functionality, tailored to the student's needs. The interface includes a clean layout with quick access to essential tools such as task tracking, study plans, and resource recommendations. Key elements like notifications, upcoming deadlines, and personalized progress metrics are prominently displayed to keep the user informed and organized.
Navigation is intuitive, with a menu or icons allowing users to seamlessly switch between features such as the whiteboard, resource library, and calendar integration. The interface may also highlight AI-driven insights or tips to enhance productivity. Overall, the Home Page is designed to balance aesthetics and functionality, ensuring a user-friendly experience that supports academic success

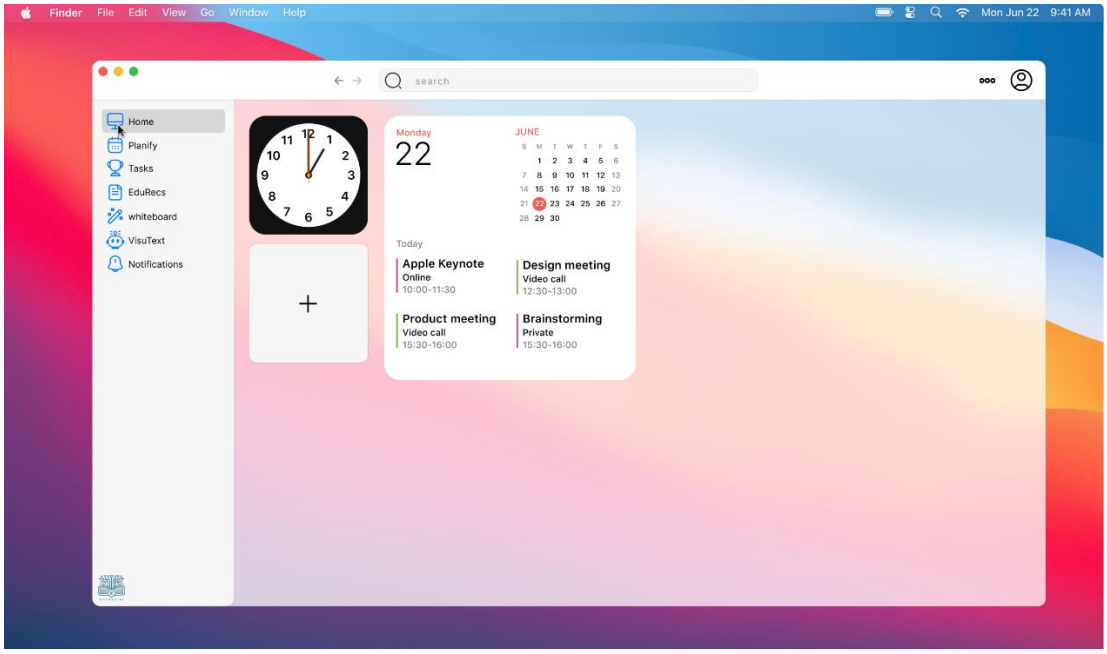


Figure 9 home page interface

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4.3.5 Recommendation:

The EduRecs Interface likely refers to the educational resource recommendation feature of the EDUassist platform. This interface is designed to provide students with personalized learning resources based on their study plans, progress, and preferences. It displays a curated list of resources such as articles, videos, or other study materials that are relevant to the student's academic goals.
The interface typically includes a search bar for users to filter through resources or browse categories, allowing for easy access to materials. Each resource is accompanied by a brief description and a link for further exploration or download. Additionally, AI-driven recommendations may be integrated to suggest content based on the student's past activities or learning patterns, helping them discover new, valuable resources to enhance their study sessions. The EduRecs interface is designed to support academic growth by delivering relevant and timely learning materials

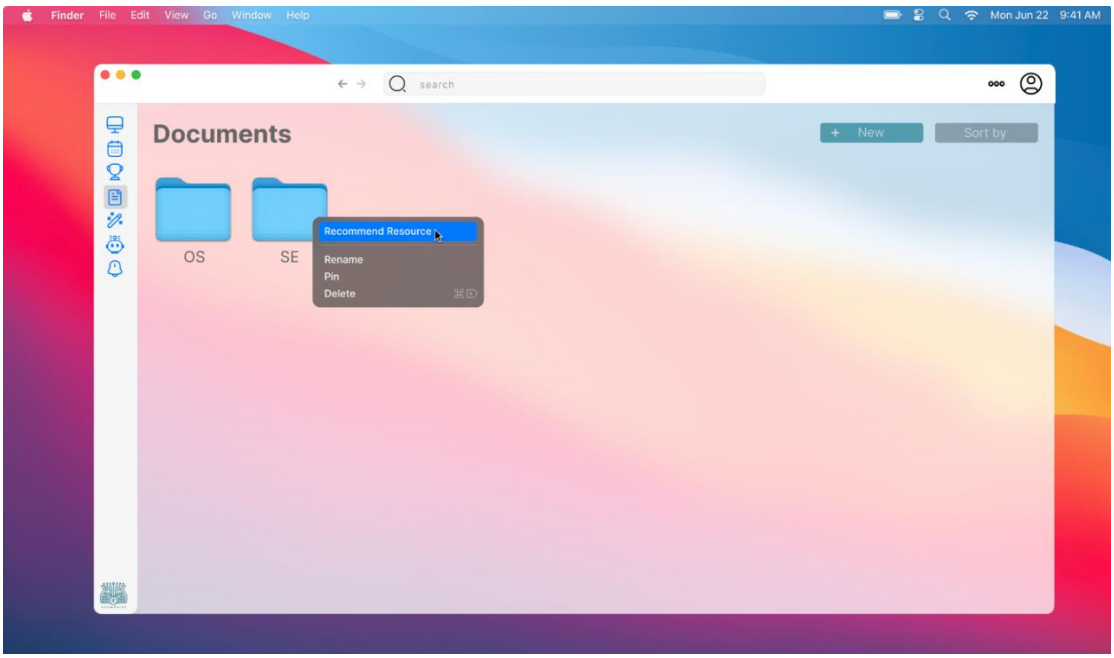


Figure 10 Documents Resource recommendation p1

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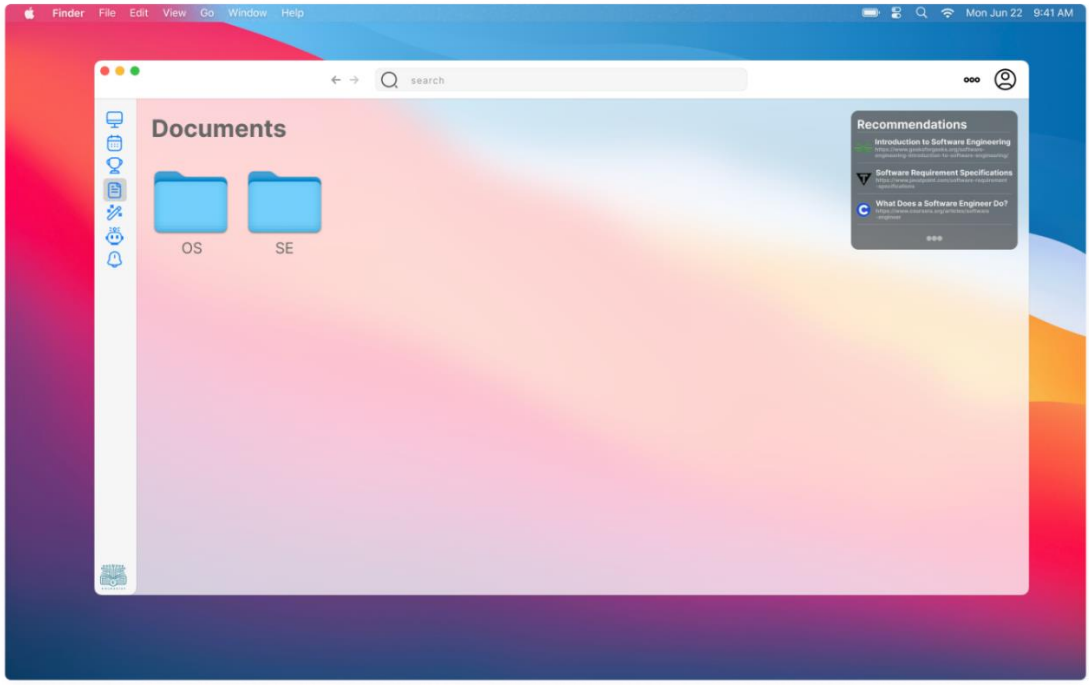


Figure 11 Documents Resource recommendation p2

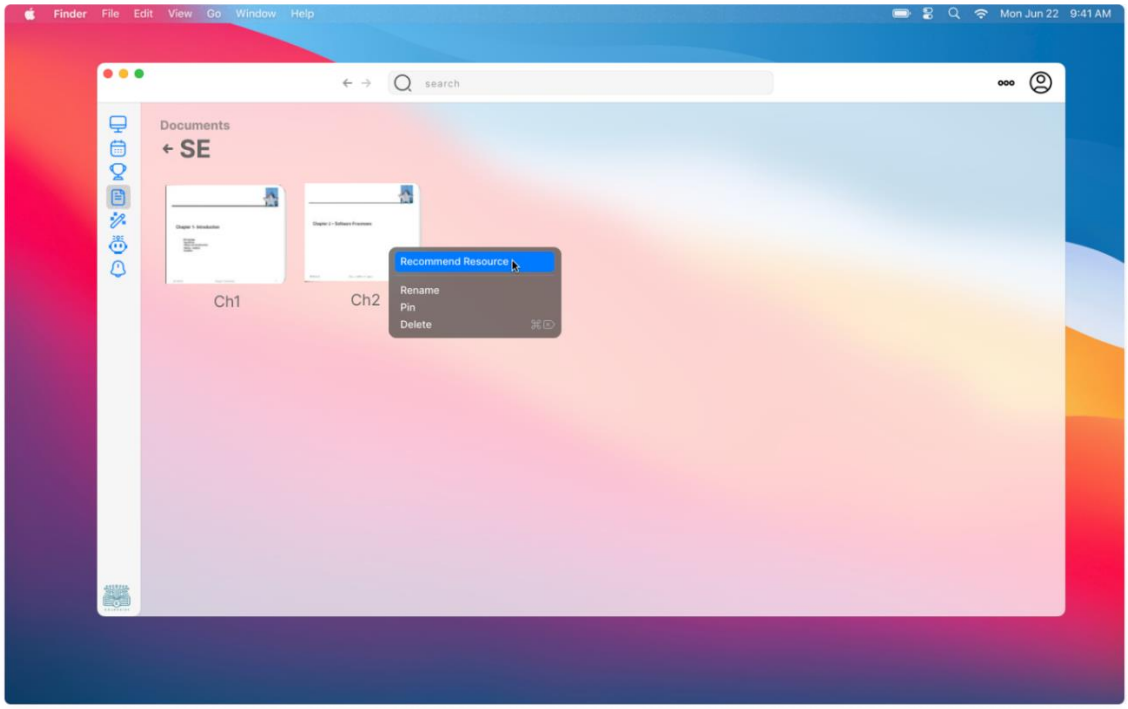


Figure 12 Files Resource recommendation p1

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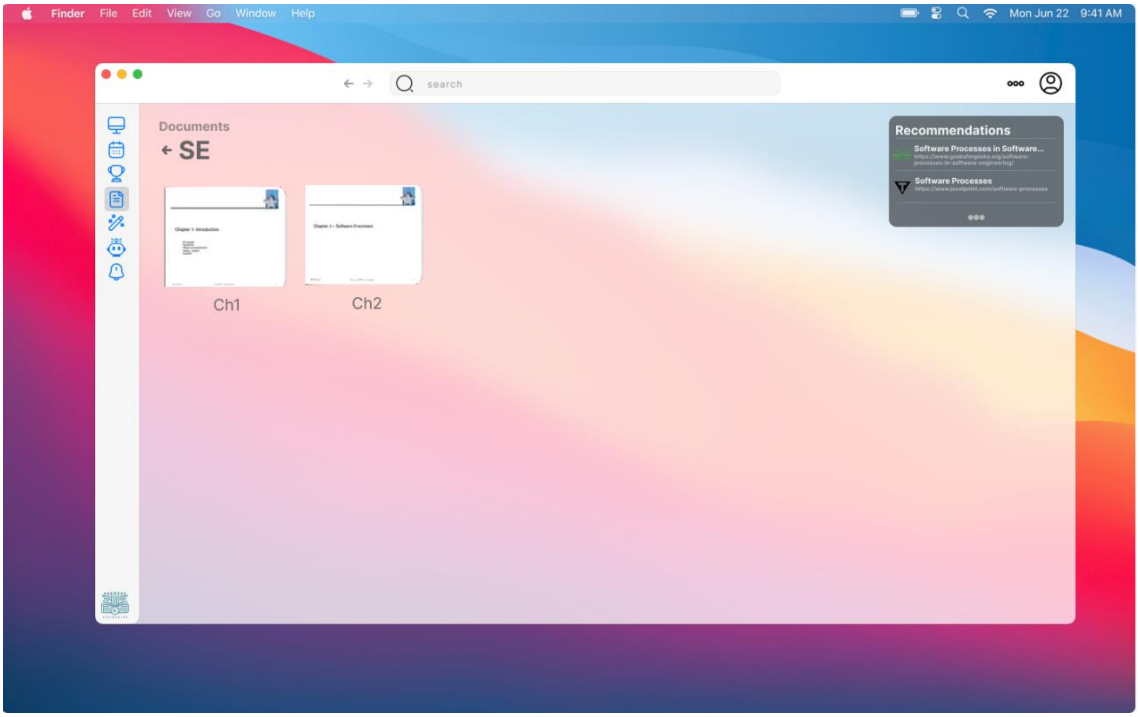


Figure 13 Files Resource recommendation p2

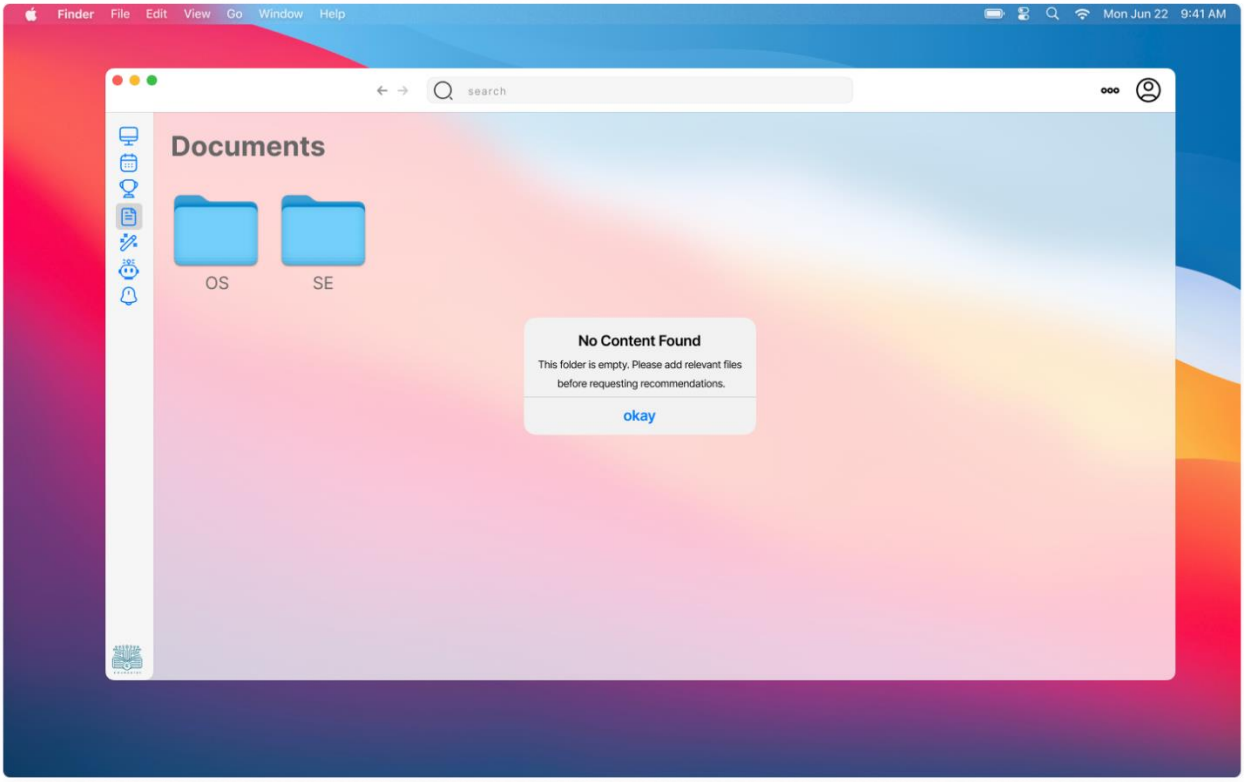


Figure 14 Empty document Resource recommendation

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4.3.6 Visualization:

The VisuText Interface is designed to help students convert textual content into visual formats, such as infographics, diagrams, or charts. This interface allows users to input text or upload study materials, and the system processes the content to generate a visual representation that simplifies complex information and enhances understanding. The interface includes options to customize the visual output, such as selecting specific diagram types, color schemes, or layouts. This tool is especially useful for students who learn better through visual aids, helping them grasp difficult concepts more easily. The VisuText interface prioritizes simplicity and clarity, providing an intuitive experience to transform study content into easily digestible visual formats.

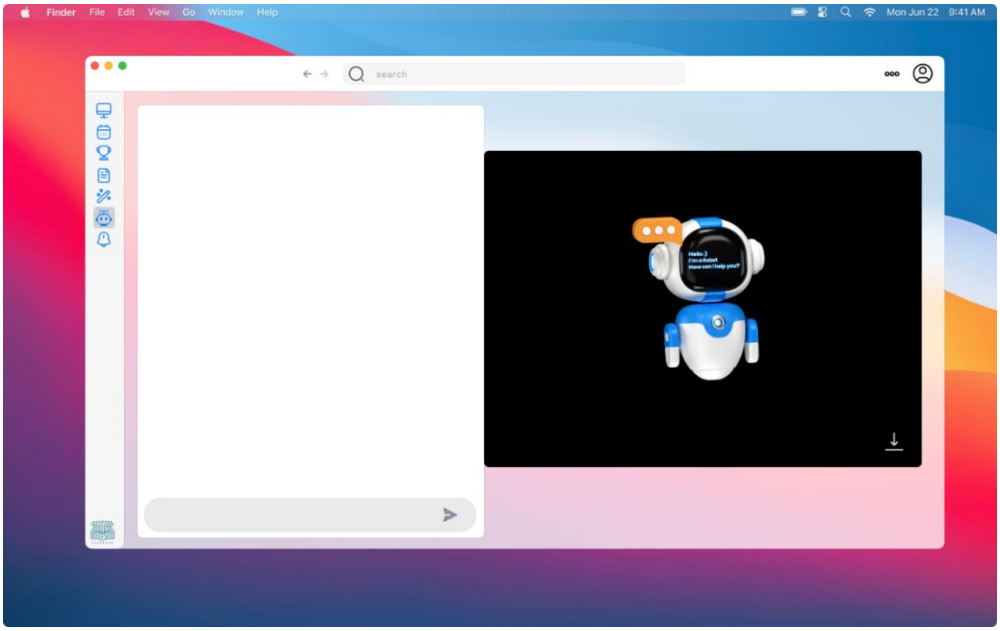


Figure 15 Visual Representation of Text p1

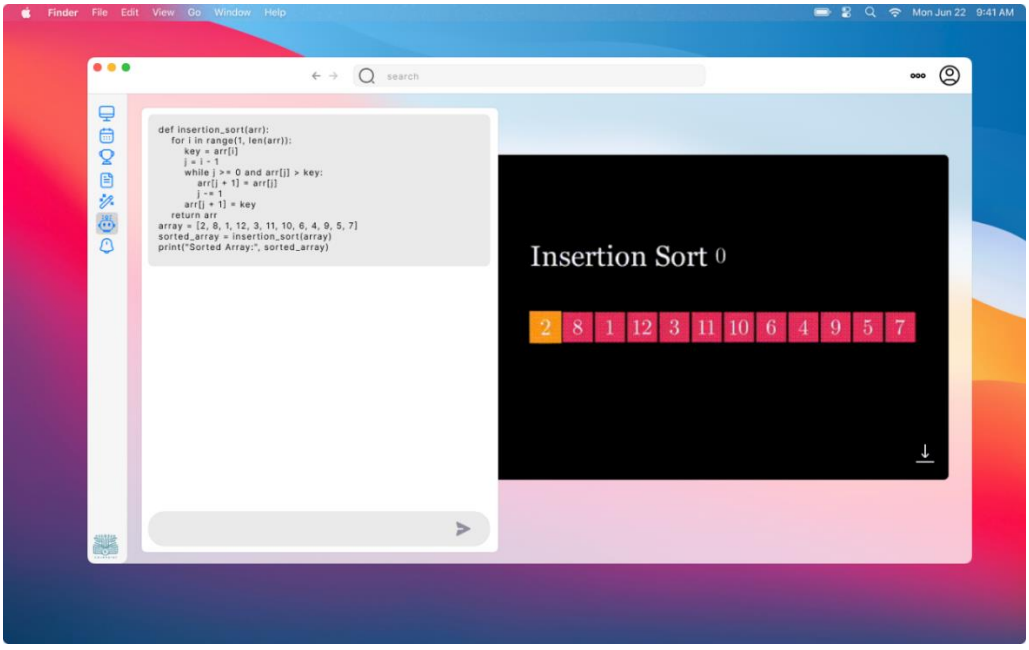


Figure 16 Visual Representation of Text p2

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4.3.7 Whiteboard:

The Whiteboard Interface is designed to facilitate real-time collaboration and note-taking for students. This interactive tool allows users to draw, write, and annotate content on a shared digital whiteboard, making it ideal for brainstorming, explaining concepts, or collaborating on group projects. The interface provides various drawing tools, such as pens, highlighters, and text boxes, which users can use to create and edit content. Users can also upload images, diagrams, or other study materials to enhance their notes and explanations. Changes made to the whiteboard are instantly synchronized across all participants in the session, allowing for seamless collaboration. The whiteboard can be cleared, saved, or exported as needed. This interface supports both individual and group work, making it a versatile tool for enhancing learning and collaboration. The design is focused on simplicity and ease of use, ensuring that students can quickly engage with the tool and contribute to the whiteboard.

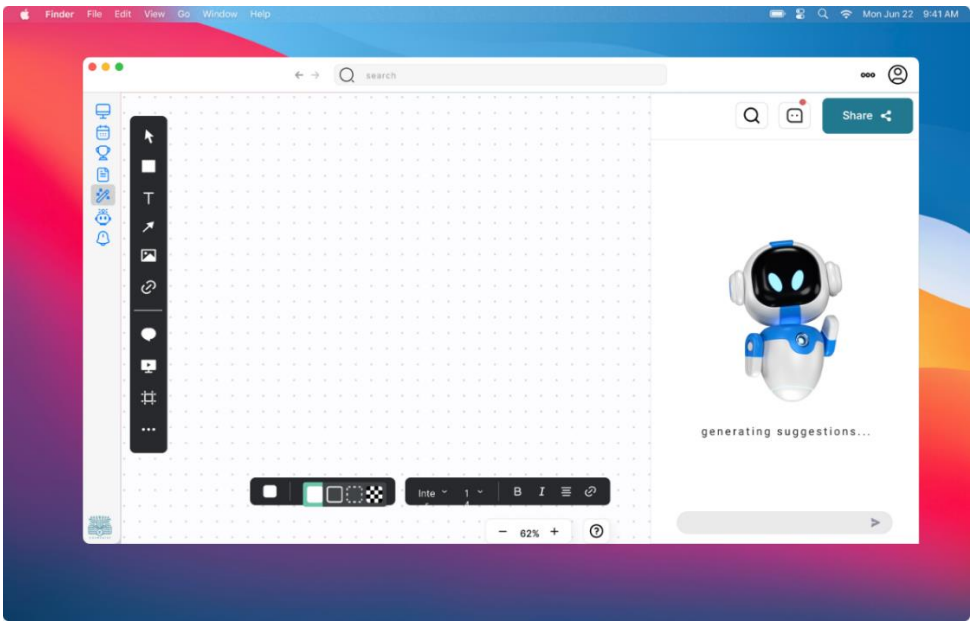


Figure 17 AI-assist whiteboard p1

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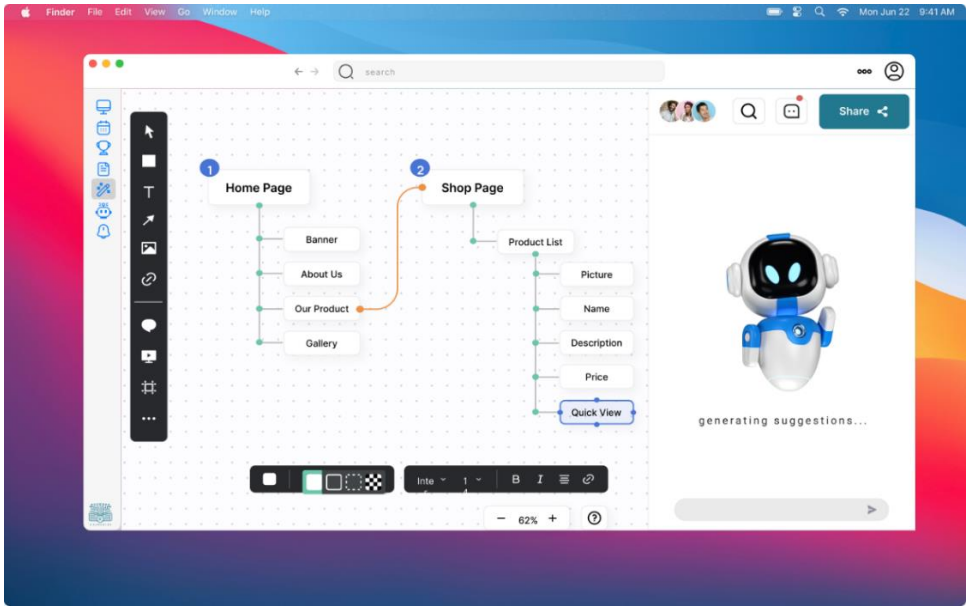


Figure 18 AI-assist whiteboard p2

4.3.8 Planify:

Integrating a calendar interface with Google Calendar allows users to sync and manage their events seamlessly

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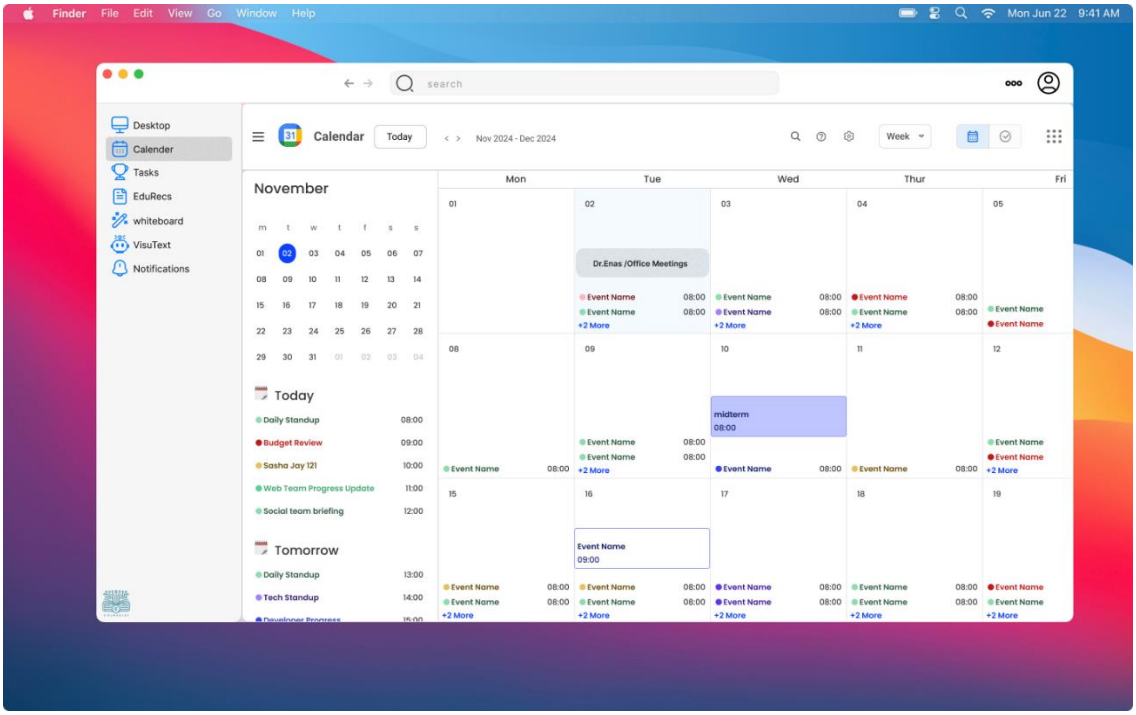


Figure 19 Google Calendar Integration

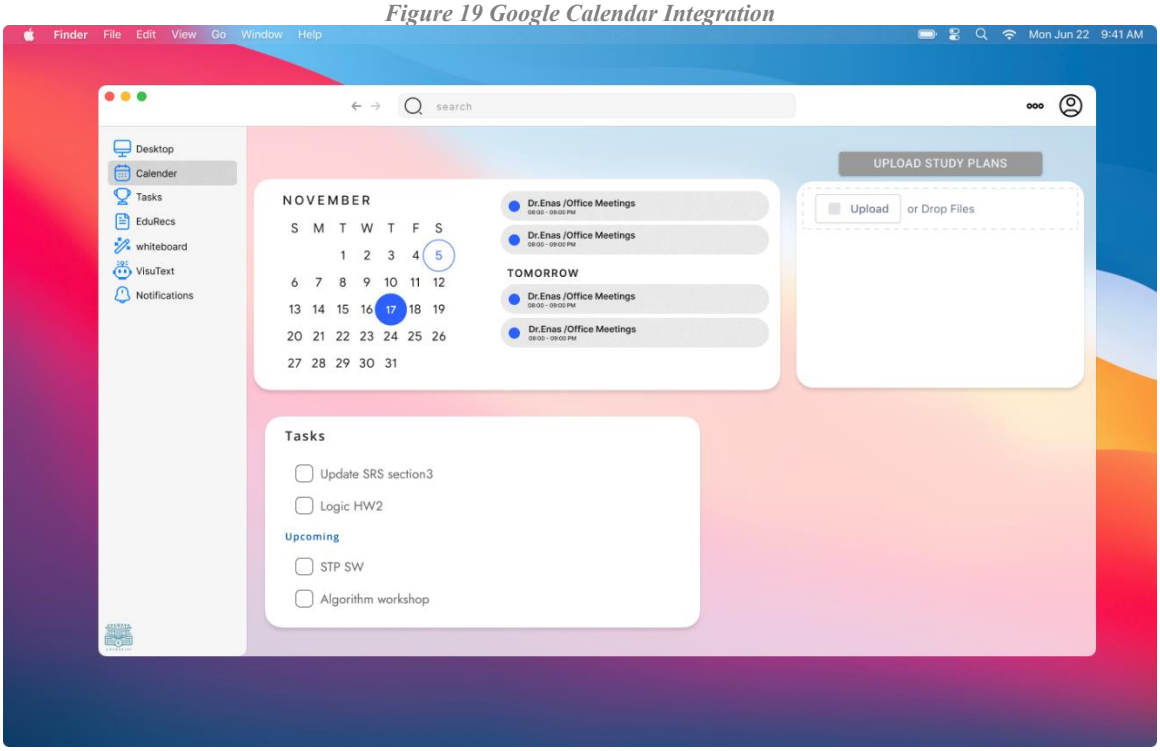


Figure 20 *Calendar*

The interface serves as a productivity tool, allowing users to manage tasks and deadlines effectively. It features sections for daily activities, displaying urgent reminders and upcoming assignments. Users can easily track what needs immediate attention, helping to prioritize their workload and stay organized. The layout enhances visibility with color-coded alerts, ensuring important tasks are not overlooked.

[planify p1]

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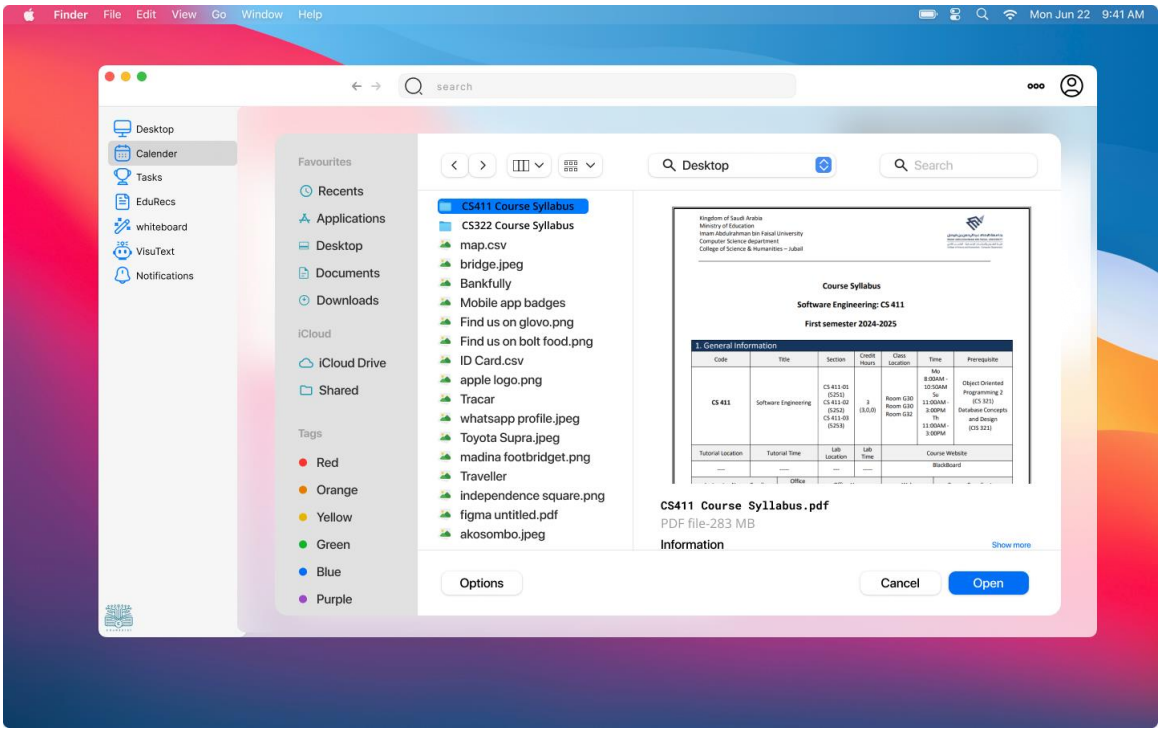


Figure 21 Upload Planify p2

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4.3.9 task board:

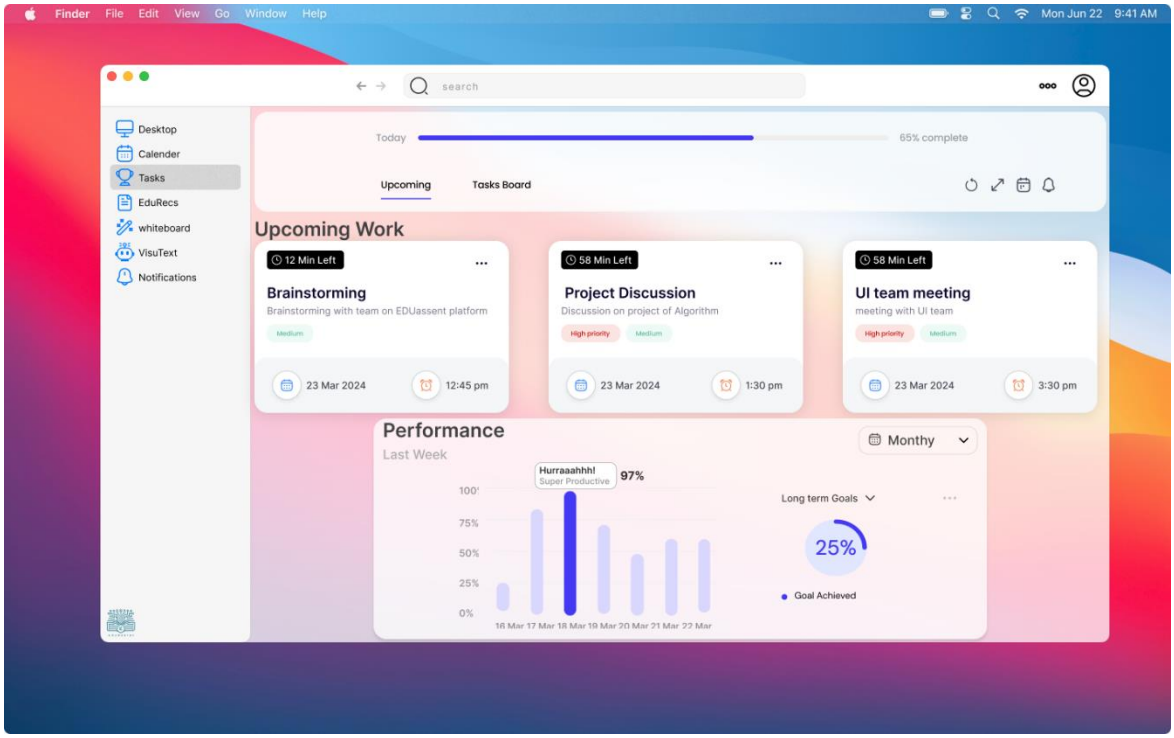


Figure 22 Tasks interface 1

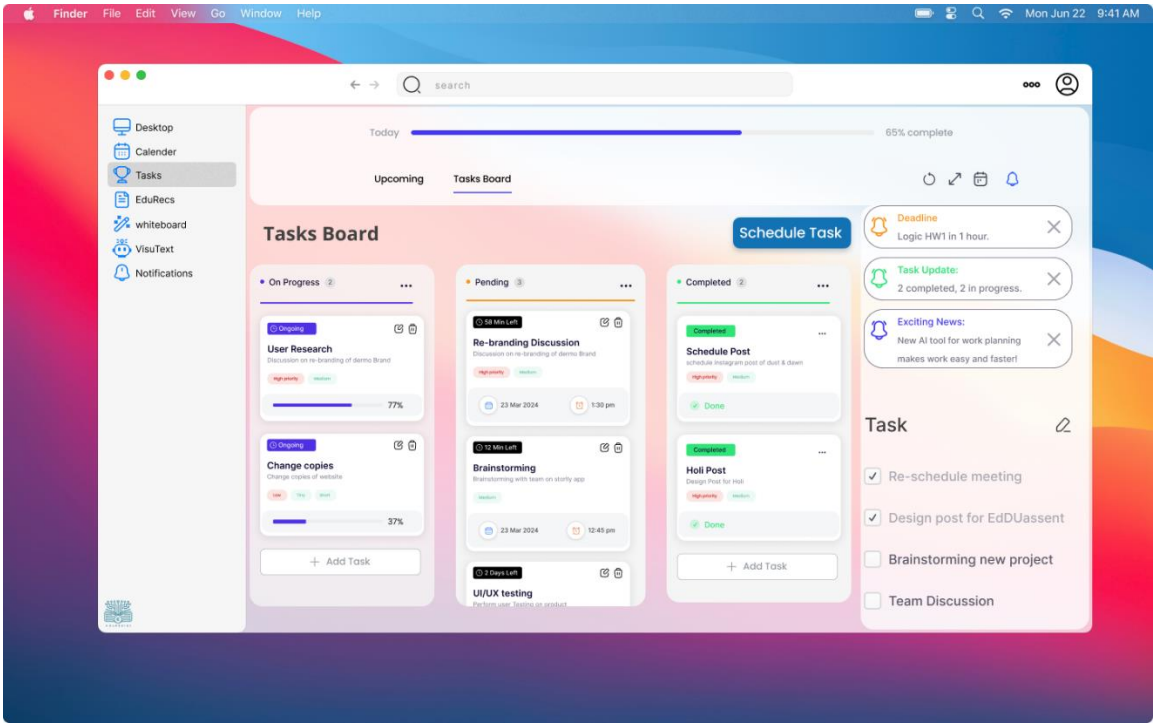


Figure 23 Tasks interface 2

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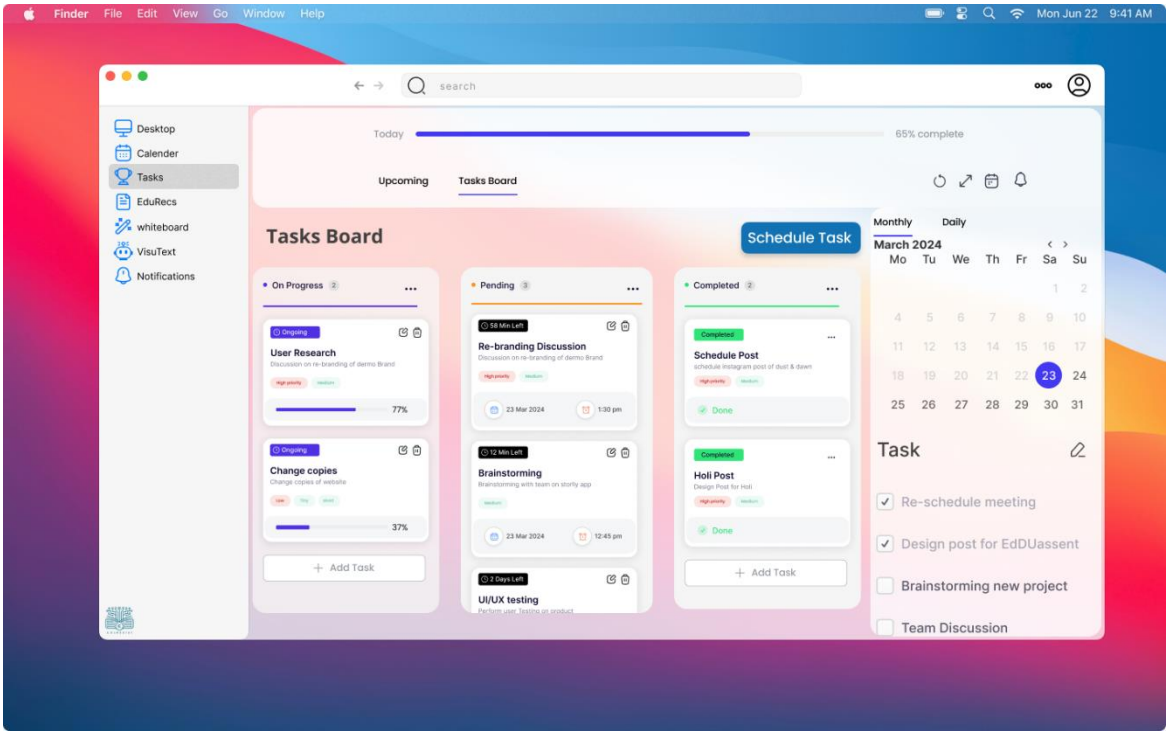


Figure 24 Tasks interface 3

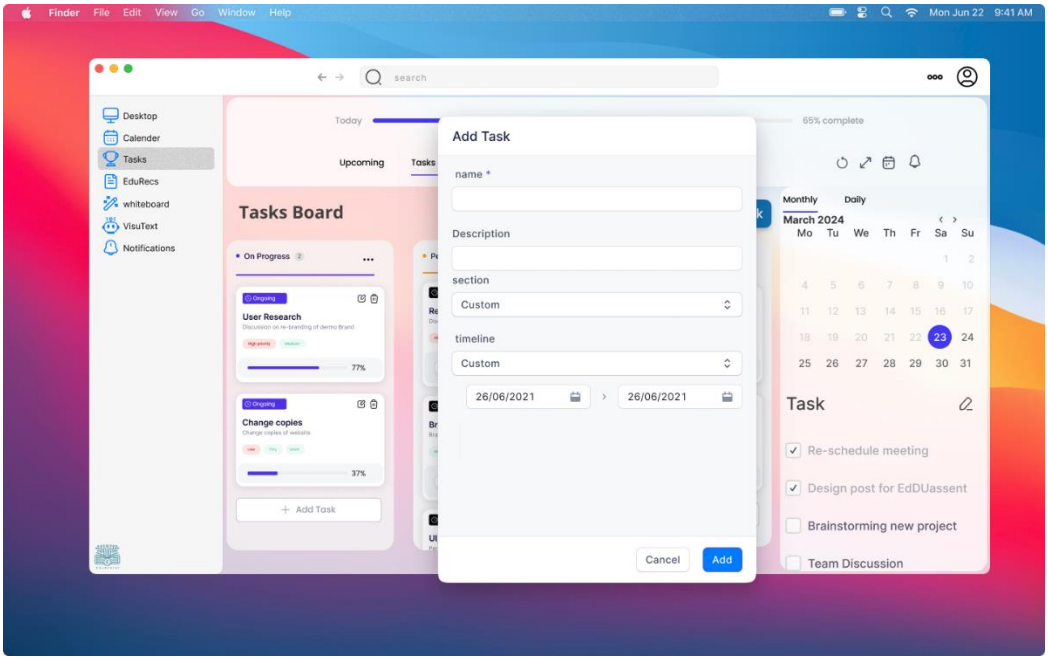


Figure 25 Add task

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4.3.10 Notifications:

The interface functions as a productivity tool, enabling users to manage tasks and deadlines efficiently. It highlights daily activities, urgent reminders, and upcoming assignments, helping users prioritize their workload. The layout features color-coded alerts for improved visibility, ensuring important tasks are easily noticed.

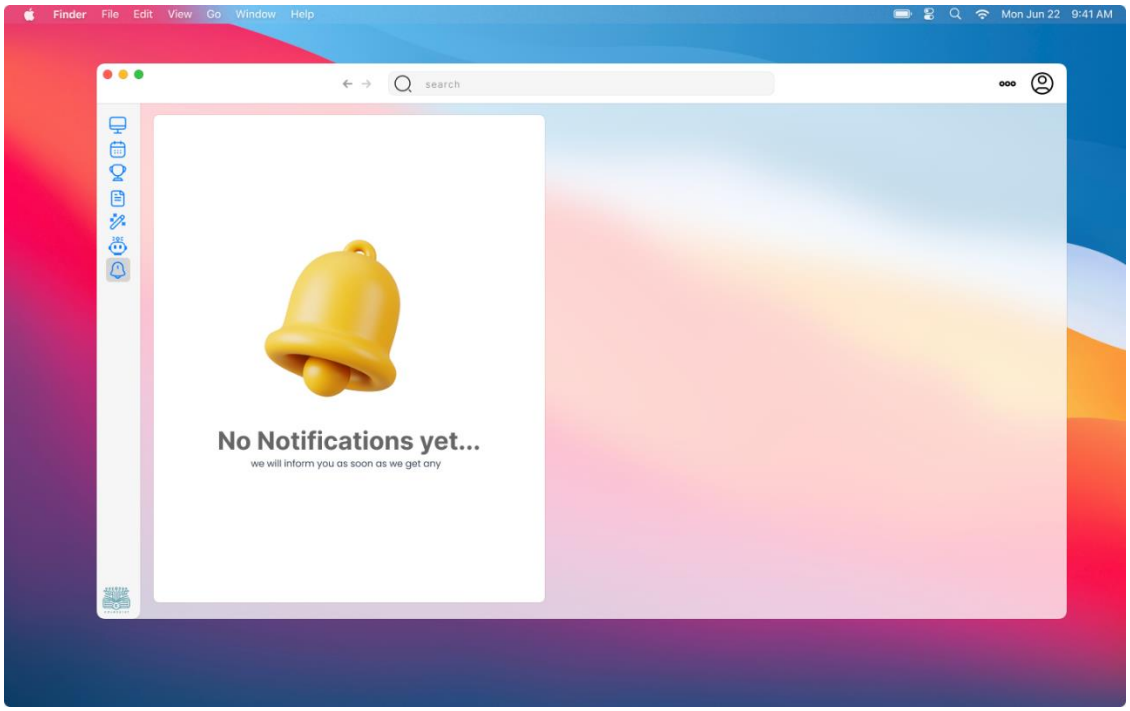


Figure 26 Notification Notifications p1

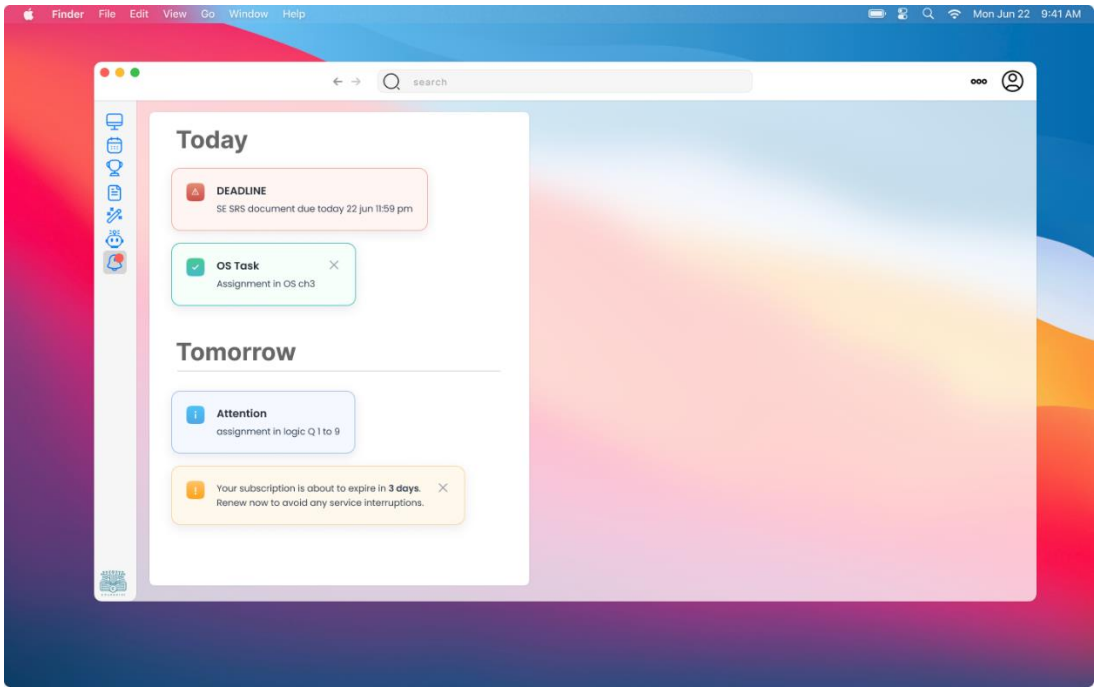


Figure 27 Notification p2

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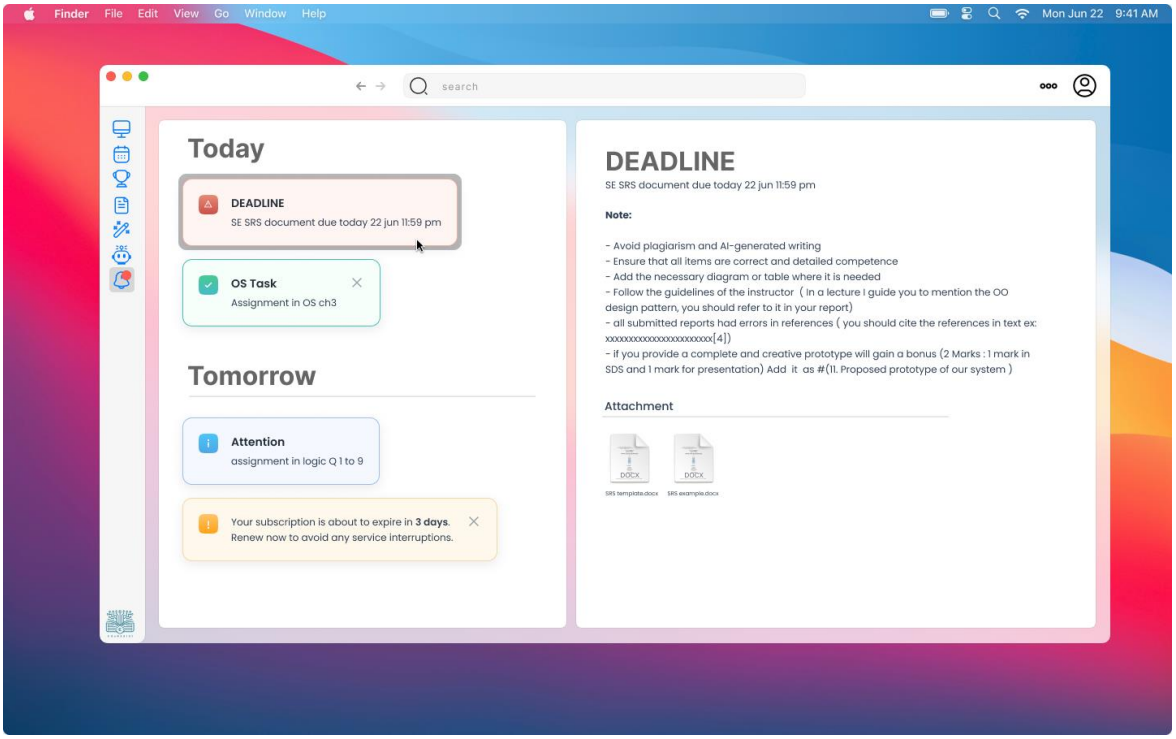


Figure 28 Notification p3

4.3.11 User profile:

The user profile interface allows individuals to input and manage personal information, including first and last names, email address, gender, academic level, language, and time zone. It provides a straightforward layout for users to easily edit their details, ensuring that their profiles are up-to-date and accurately reflect their preferences. The design emphasizes simplicity and accessibility, enhancing user experience by facilitating quick modifications and clear visibility of essential information.

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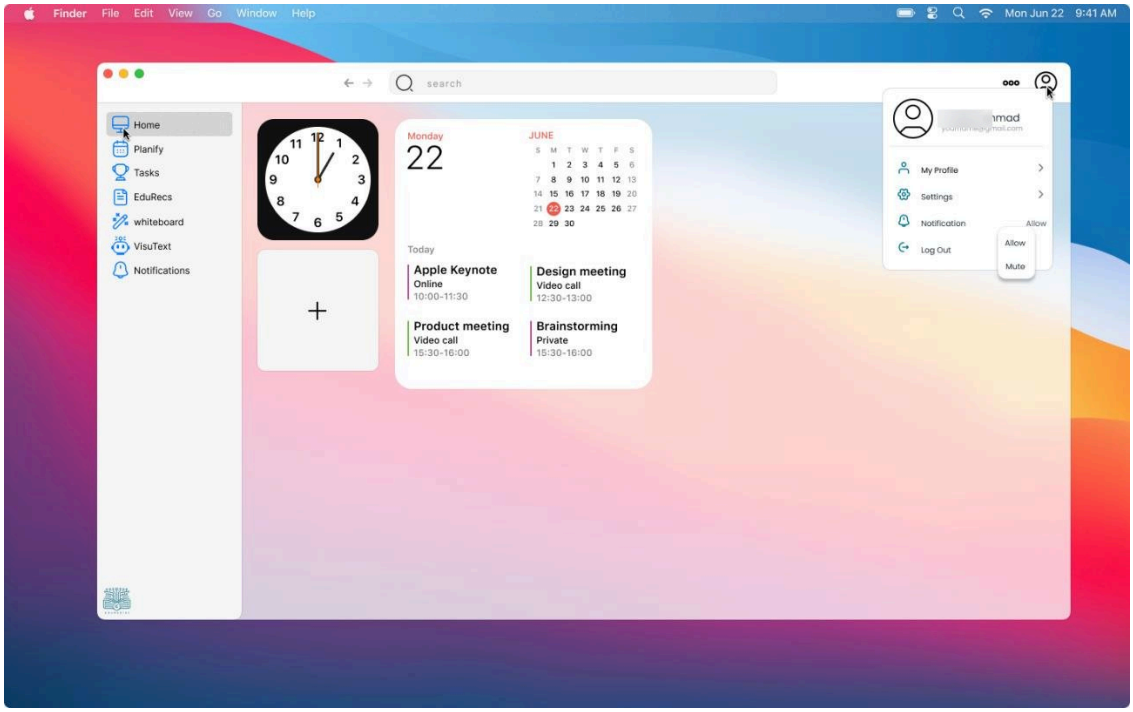


Figure 29 pop profile

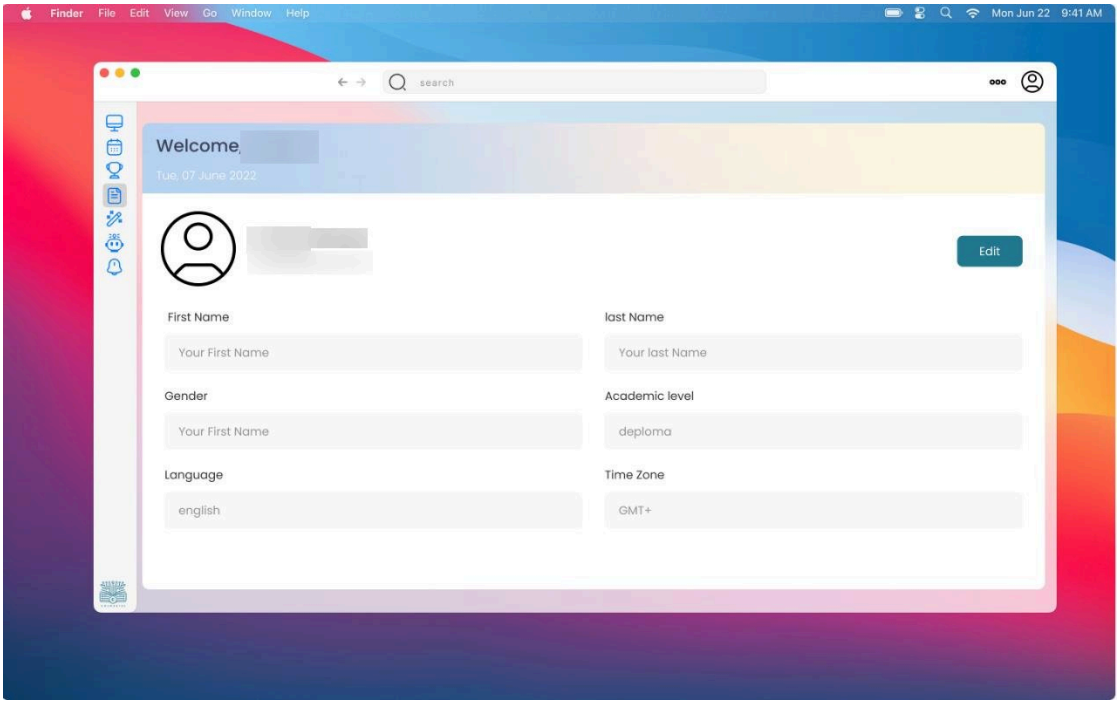


Figure 30 Profile Settings

4.3.12 Admin dashboard :

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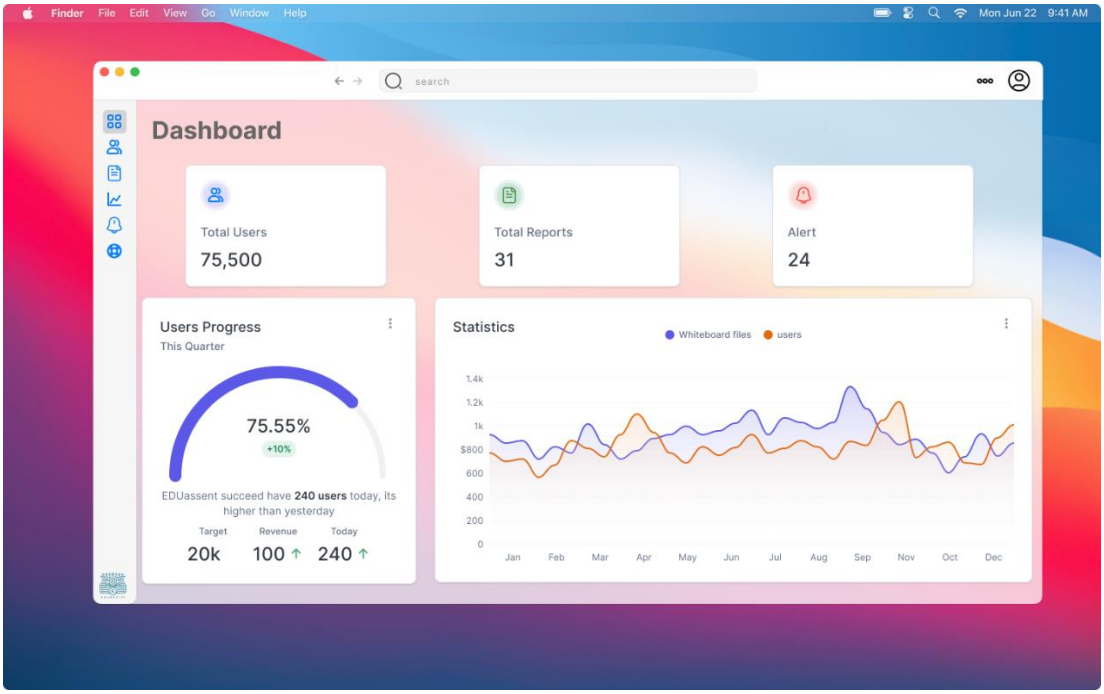


Figure 31 Admin DashBoard

The dashboard interface displays essential performance metrics in a user-friendly layout. It includes key statistics such as total users, reports, and alerts, providing a quick overview of platform activity. A progress gauge illustrates user engagement, while graphical representations visualize trends and comparisons over time. This design facilitates efficient access to critical information, supporting informed decision-making for users and administrators alike.

4.3.13 Admin user management :

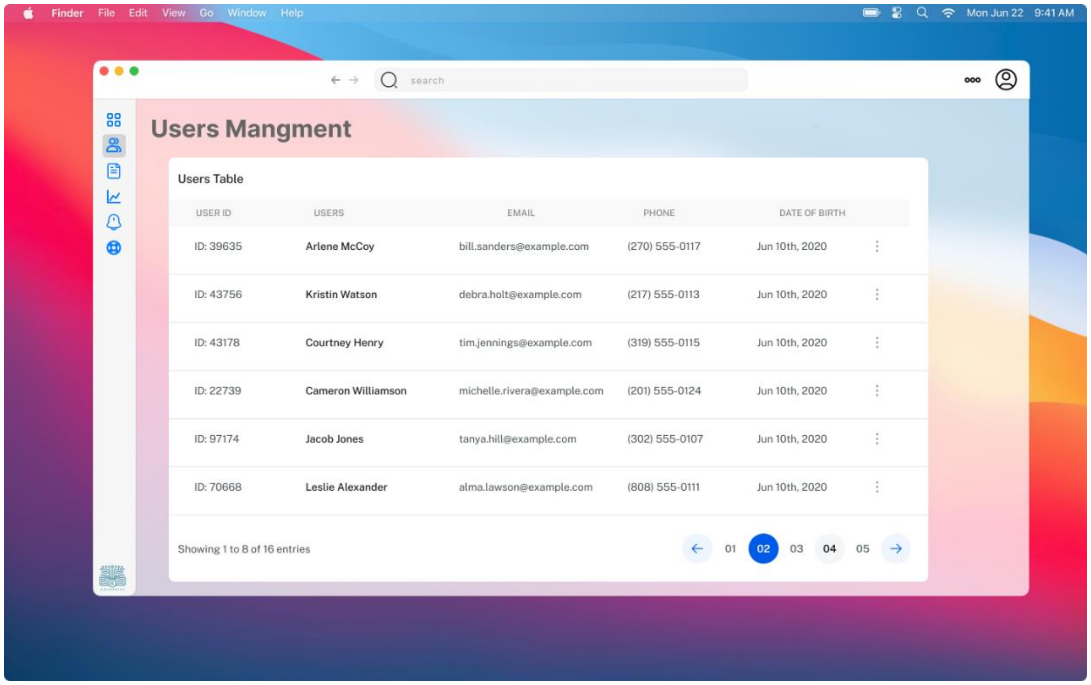


Figure 32 Admin Mange users

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The Users Management interface provides a structured view of user information within the platform. It features a table that lists key details for each user, including names, email addresses, and phone numbers, alongside their respective registration dates. The table is designed for easy navigation, allowing administrators to manage user data efficiently. Pagination controls at the bottom enable users to navigate through multiple entries, ensuring streamlined access to the complete list of users. Overall, this interface enhances user management capabilities by presenting vital information in a clear and organized manner.

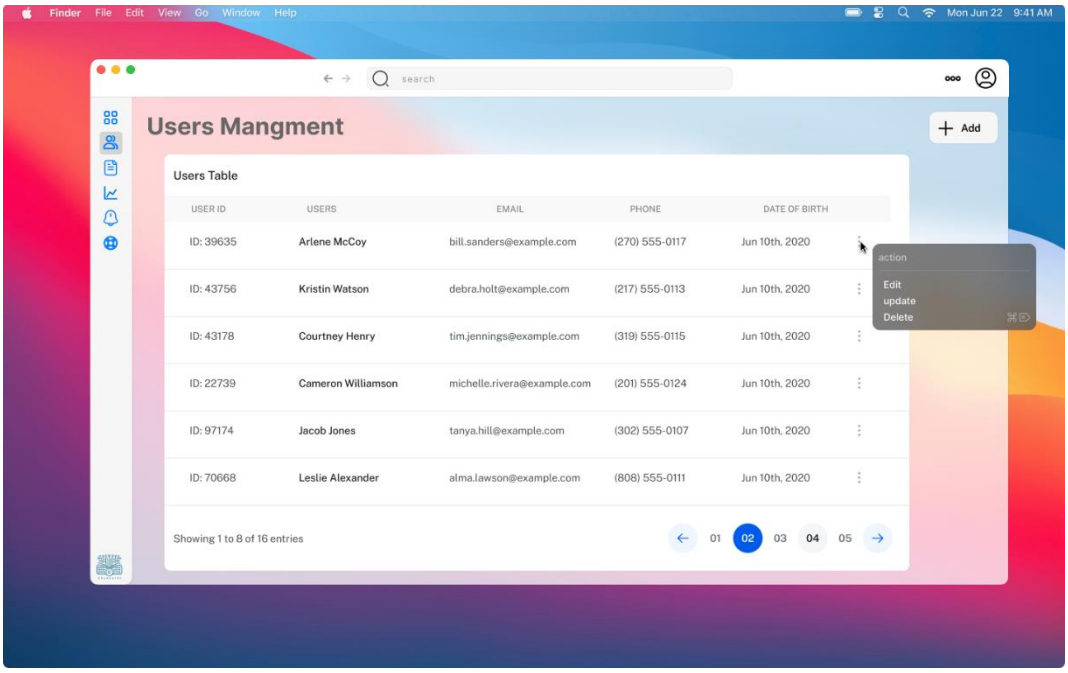


Figure 33 Users Mangment 1

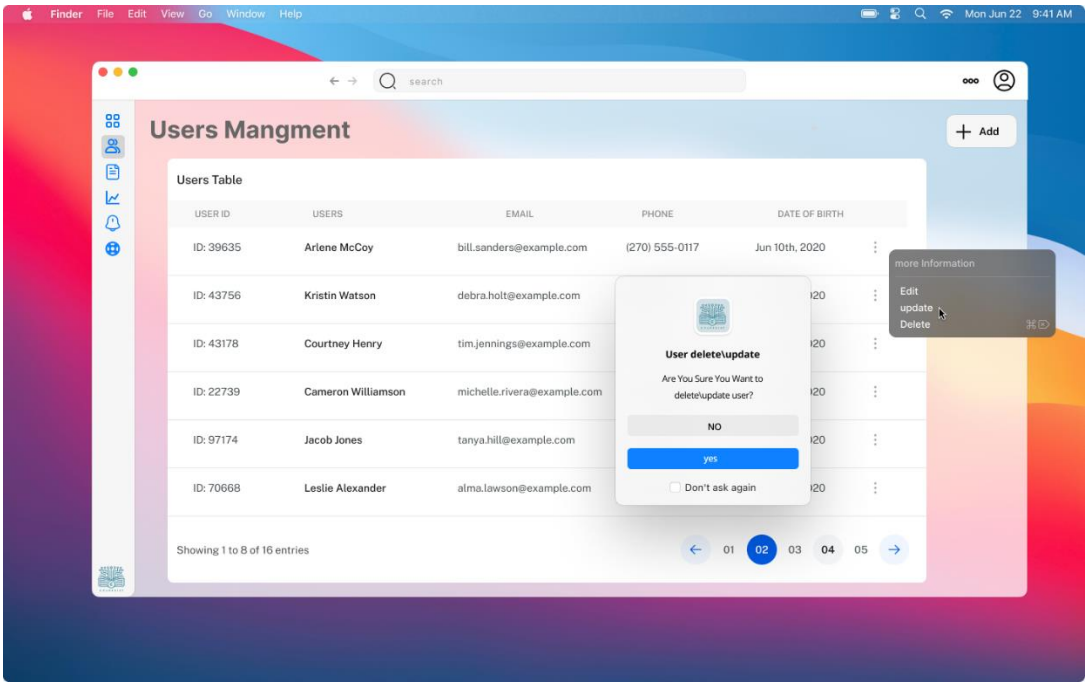


Figure 34 Users Mangment 2

4.3.14 Report Generation:

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The Admin Report Generation interface allows administrators to create detailed reports based on user activity and platform metrics.The interface is designed for ease of use, often featuring a straightforward layout with clear buttons to initiate the report generation process. Once generated, reports can usually be exported in various formats, enhancing data accessibility and facilitating informed decision-making.

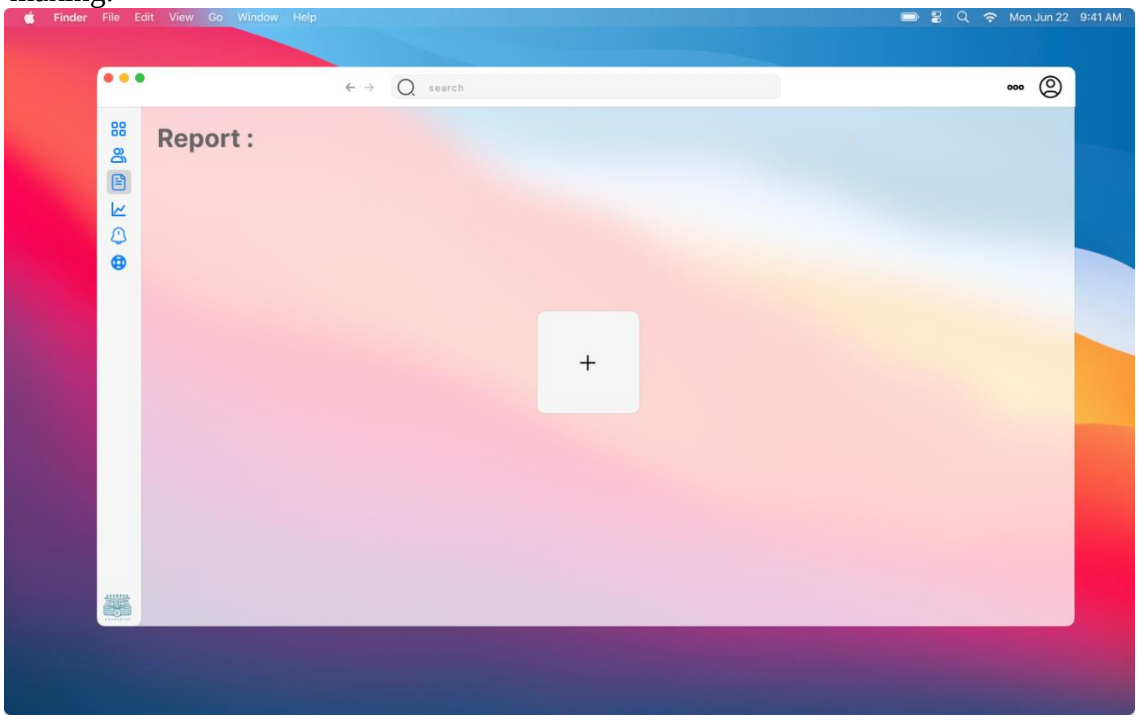


Figure 35 Figure Report interface 1

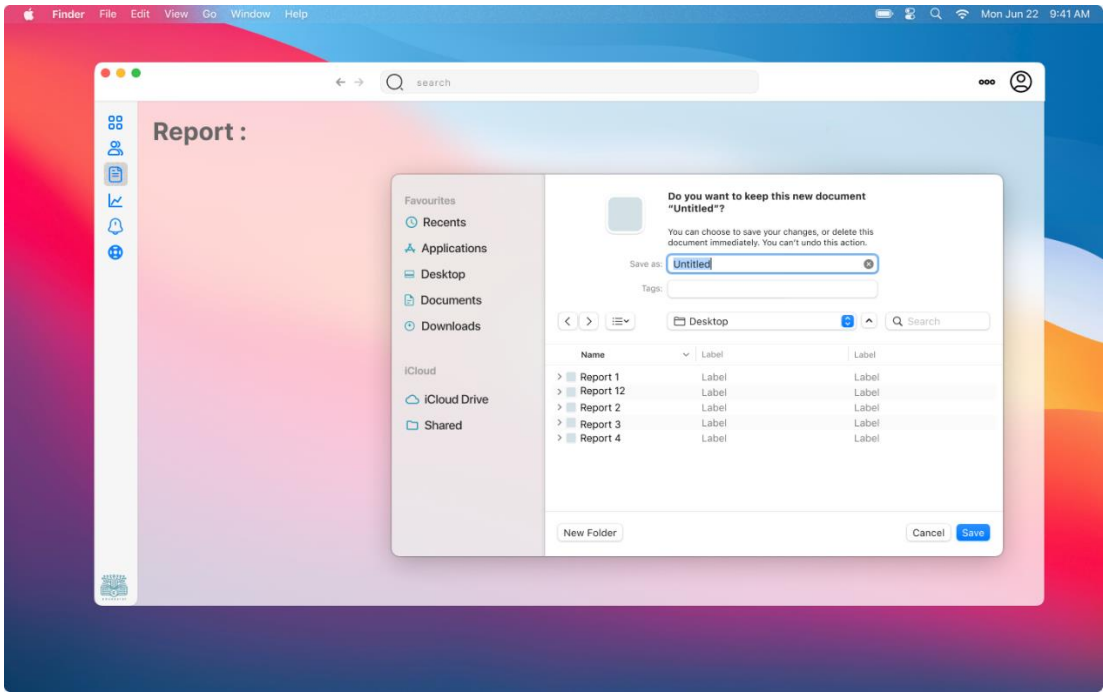


Figure 36 Report interface 2

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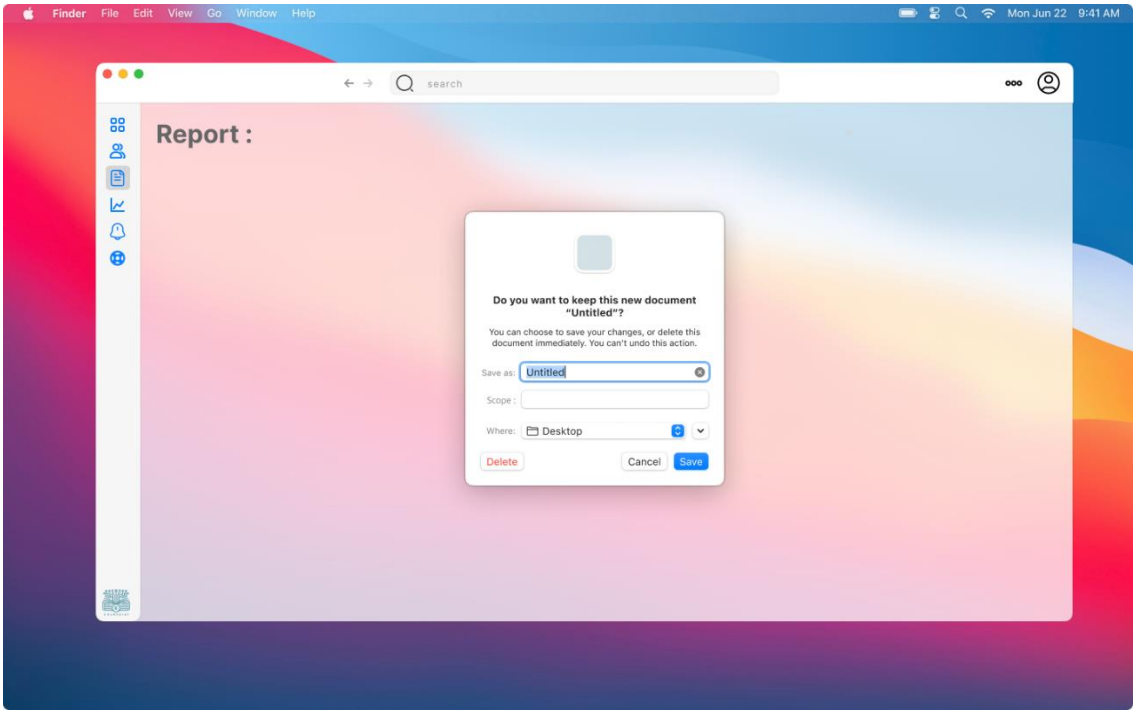


Figure 37 Report interface 3

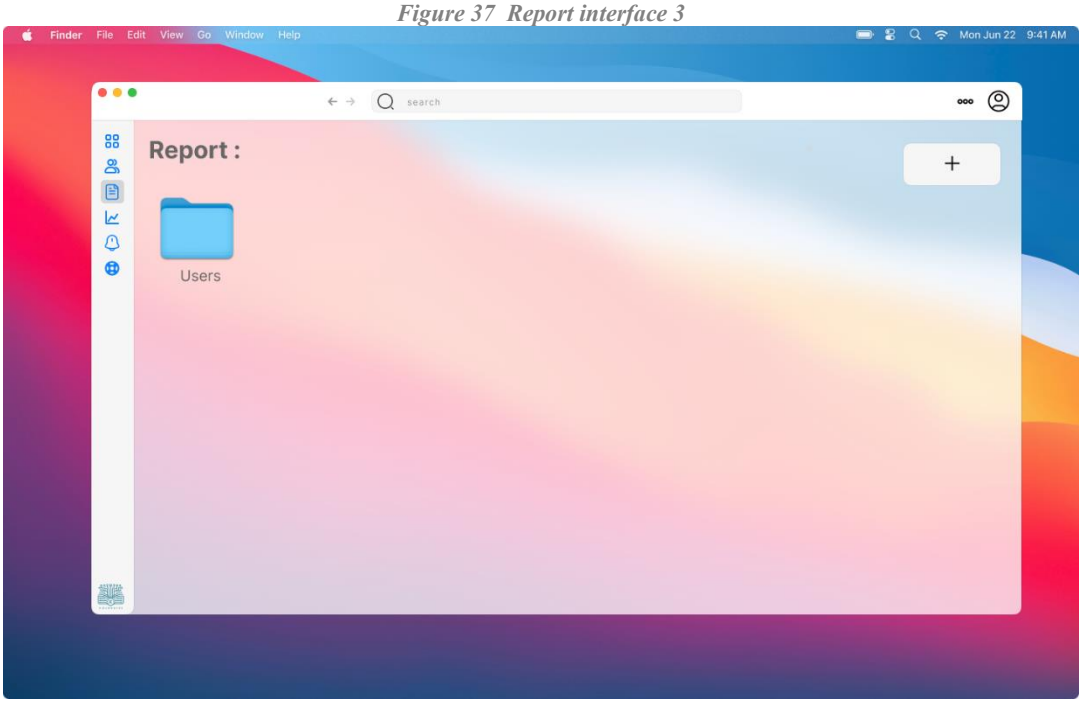


Figure 38 Report interface 4

4.3.15 Statics

The EDUassent Statistics interface provides a visual overview of platform performance metrics over various timeframes, including all-time, past 12 months, 30 days, and 24 hours. It features two main graphs: one that compares the number of whiteboard files created to user counts, and another that highlights overall user engagement trends. The interface allows users to select specific time periods for analysis, facilitating a deeper understanding of usage patterns and trends. This design enables administrators to easily track performance and make data-driven decisions based on user activity.

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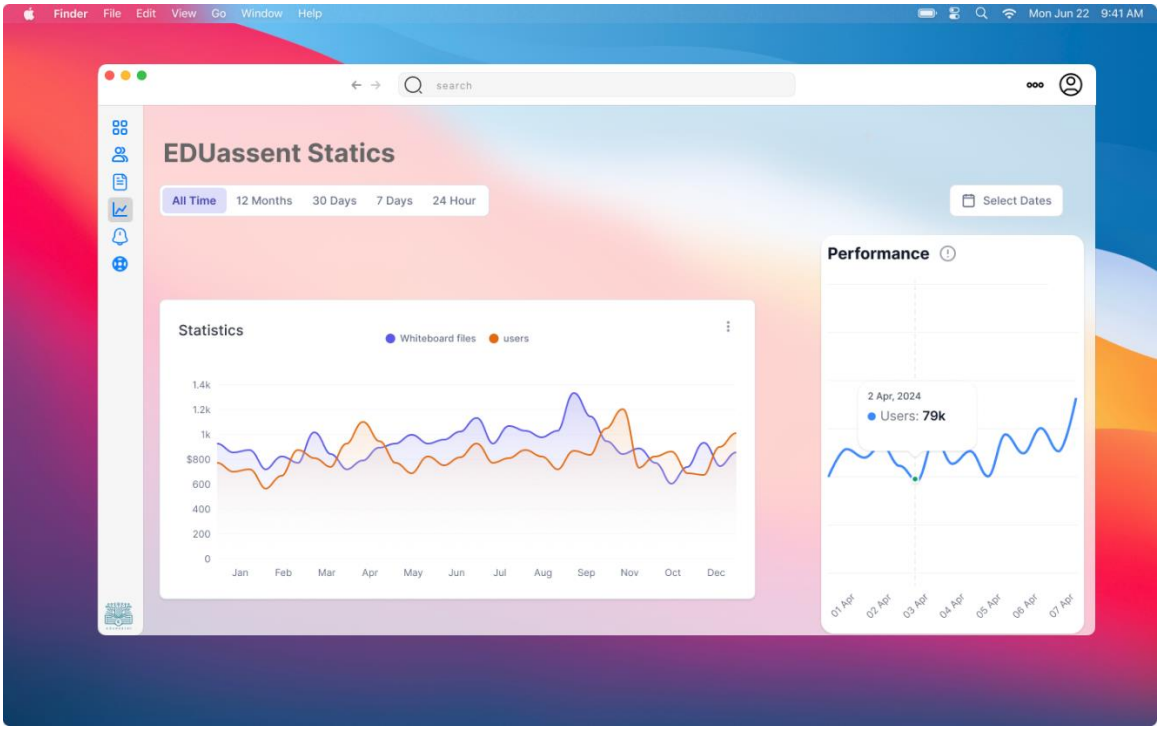


Figure 39 Statistics Interface

4.3.16 Update Platform:

The EDUassent Update Scheduling interface facilitates the management of software updates. It features a calendar view for selecting dates, highlighting the next scheduled update .

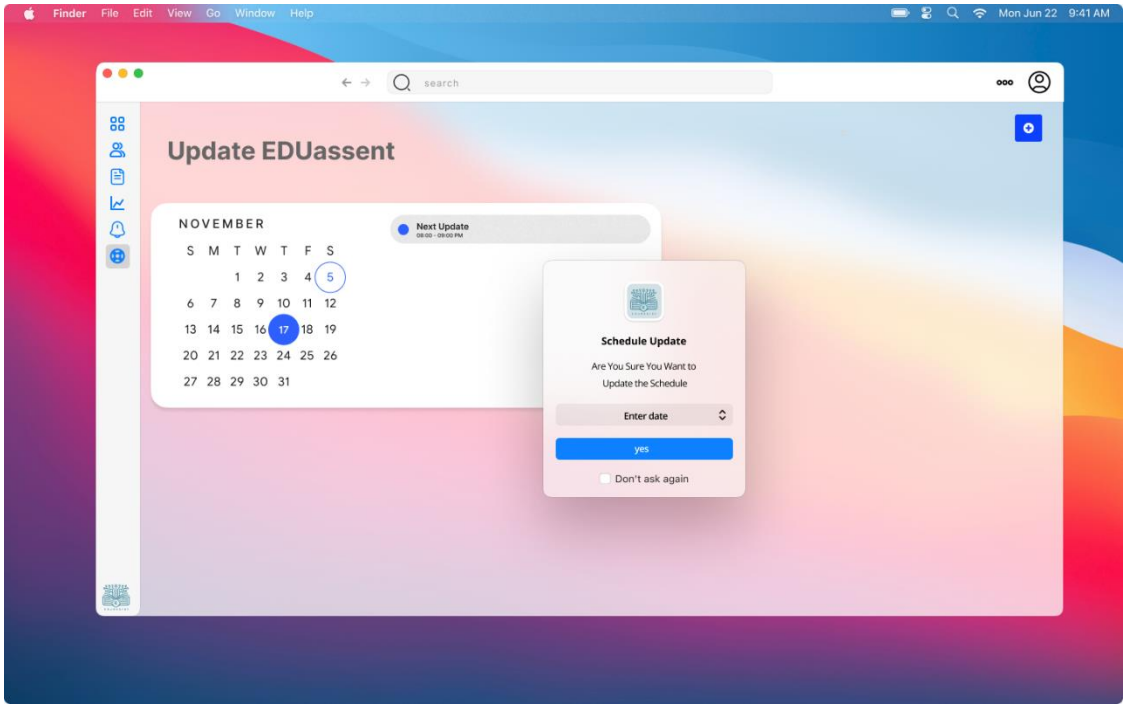


Figure 40 EDUassent Update Scheduling

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4.3.17 Alert -Feedback :

The EDUassent Recent Feedback interface displays a list of Users feedback entries, including details such as Feedback ID, customer name, email, description, status, and available actions. Each entry is organized in a table format, making it easy to navigate. The status column indicates whether feedback is processing, delivered, or cancelled, allowing for quick assessment of each case

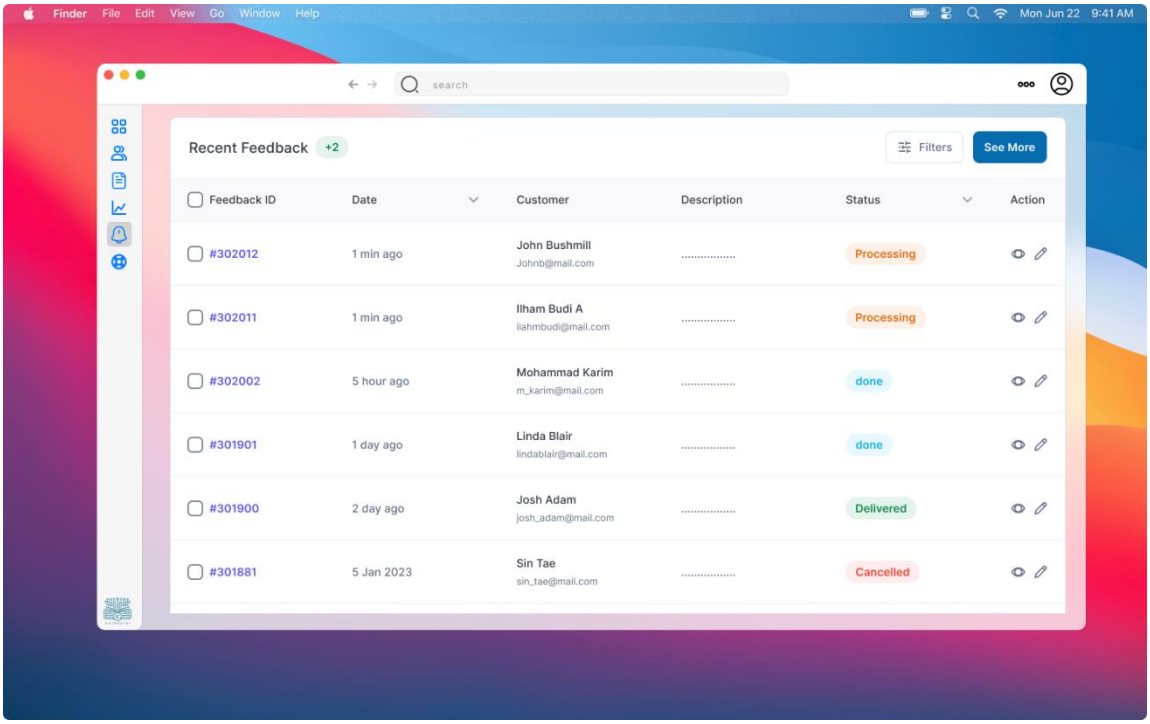


Figure 41 Alert -Feedbacks interface

4.3.18 Admin profile:

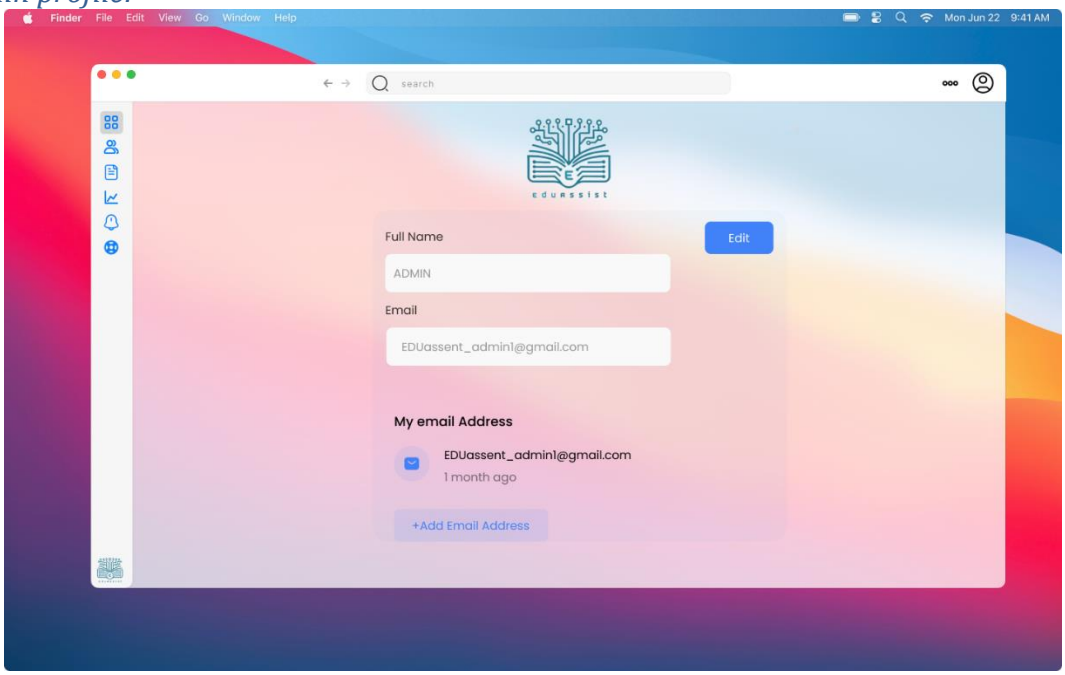


Figure 42 Profile

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4.4 Screen Objects and Actions

Object	Type	Actions
Common Function		
Get Started Button	Button	Opens the task creation modal to add a new task.
		If the user selects "Create Account," navigate to the sign-up screen.
Login Button	Button	For students: Navigate to the user's homepage.
		For admins: Navigate to the admin dashboard.
Verification Code Dialog	Dialog	Accepts a 4-digit verification code received via SMS for account verification.
Notification Icon	Icon	Displays new notifications or reminders.
		If clicked, navigates to the notifications page to display details.
Home Clock	Widget	Displays the current time.
		Automatically syncs with the user's local system clock.
User Function		
Planify Calendar	Interactive Calendar	Display Google Calendar events.
		Allow users to upload or integrate study plans.
		Add or edit events directly within the application.
Whiteboard Tools	Toolbar (Interactive)	Enable users to draw, add shapes, text, and images.
		Save or share whiteboard designs.
		Access suggestions for improvements from AI recommendations.
Task Board	Interactive Panel	Shows ongoing, pending, and completed tasks.
		Allows adding, editing, or deleting tasks.
Profile Edit Fields	Text Field	Users can edit their personal information such as name, gender, and academic level.
Admin Function		

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Admin Dashboard Widgets	Information Widget	Display statistics, user progress, total reports, and alerts for admins.
Admin User Table	Table	Lists all users with their details (ID, name, email, phone, and date of birth).
		Admin can edit, update, or delete user information.
Report Section	File Management Panel	Admin can create, save, or delete reports.
		Admin can organize reports in folders.
Statistics Page	Chart View (Graph)	Displays data trends for user activities (e.g., whiteboard usage, login counts).
		Allows admin to filter by time periods.
Update Scheduling	Scheduler Dialog	Allows admins to set the next update schedule for the system.
		Displays a confirmation dialog for the selected date.
Alert Feedback Panel	Table	Lists recent feedback from users.
		Admin can filter feedback by status (e.g., processing, done, delivered, or canceled).
Admin Profile Page	Profile Management Form	Admin can edit their name and email.
		Admin can add additional email addresses for system notifications.

Table 5 Screen Objects and Actions

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4.5 Other Interfaces

4.5.1 Forgat password:

Forgot Password Interface is designed to help users recover access to their EDUassist account if they forget their login credentials. This interface provides a straightforward process for password recovery. Users are prompted to enter the email address or phone number associated with their account. Once submitted, the system validates the information, and if it matches a registered account, a recovery code or reset link is sent to the provided contact method.

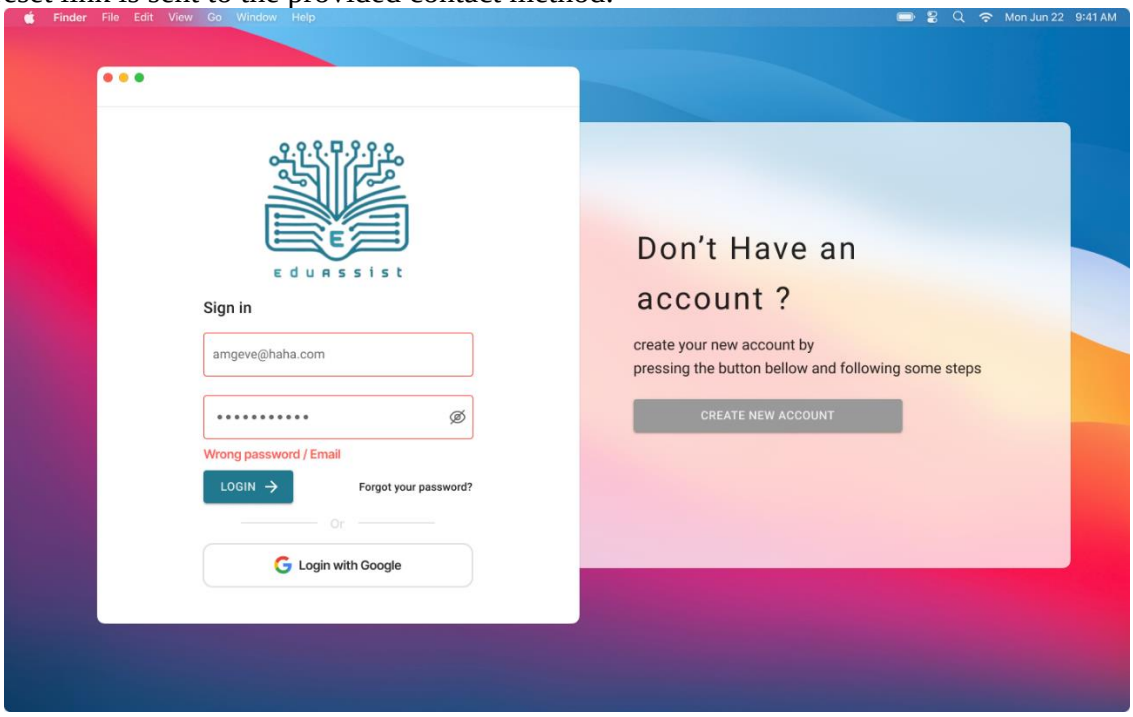


Figure 43

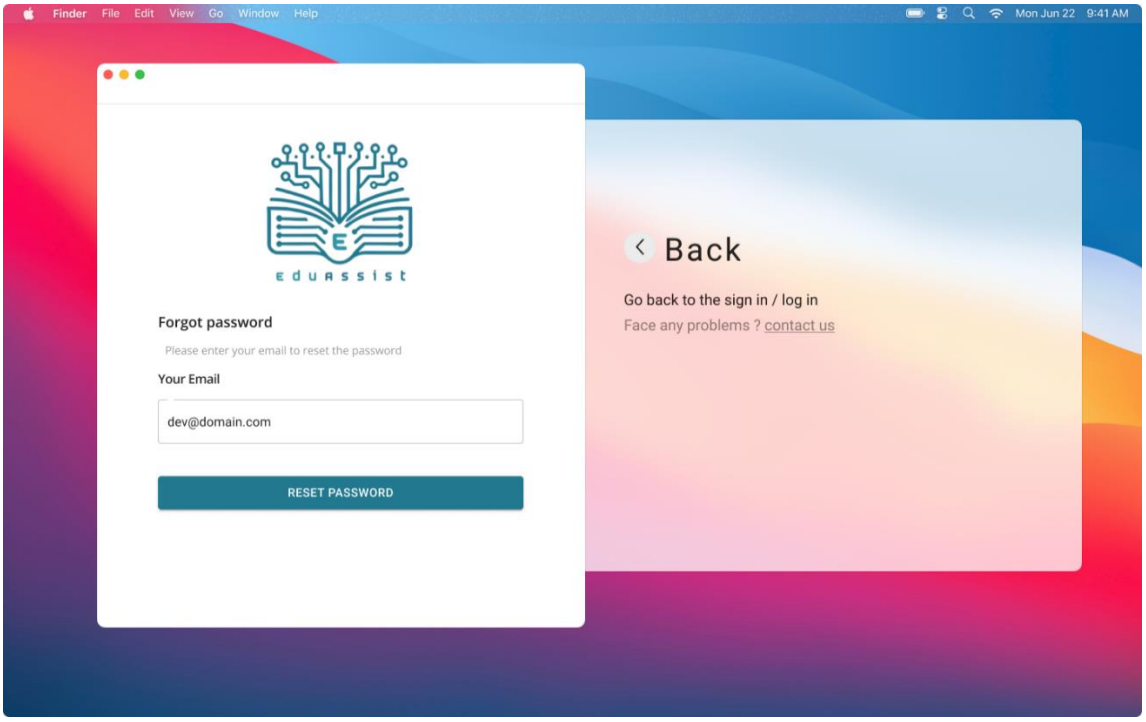


Figure 44

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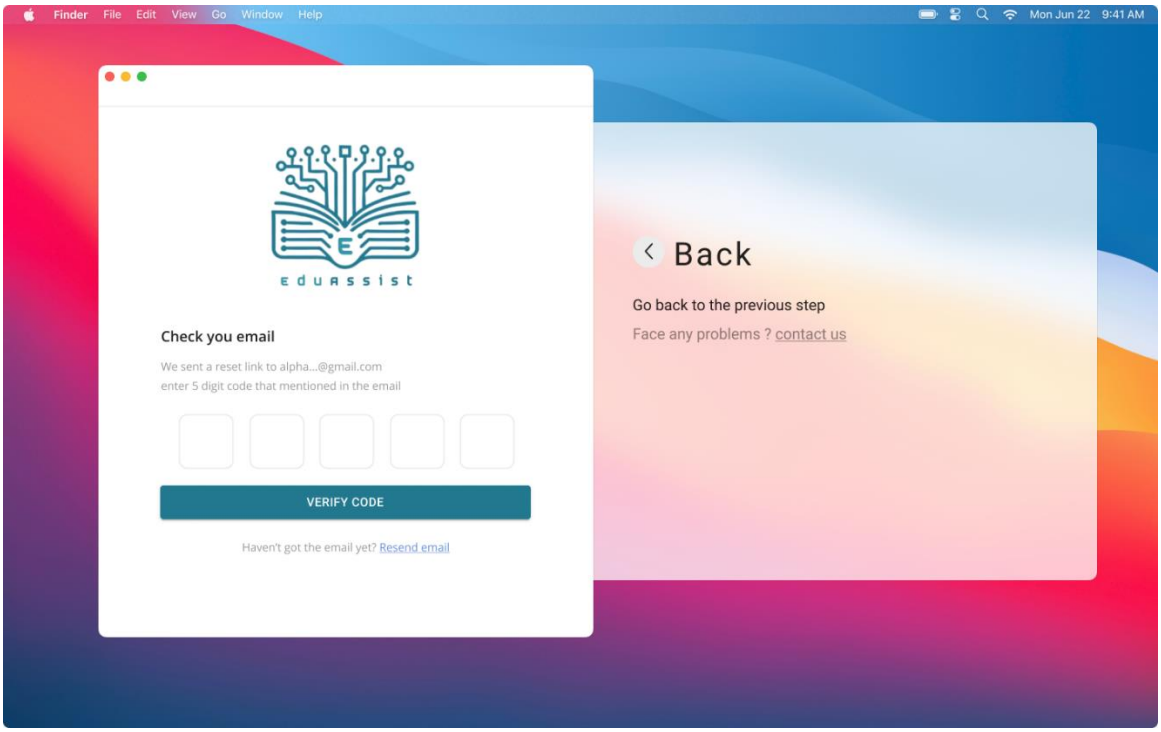


Figure 45

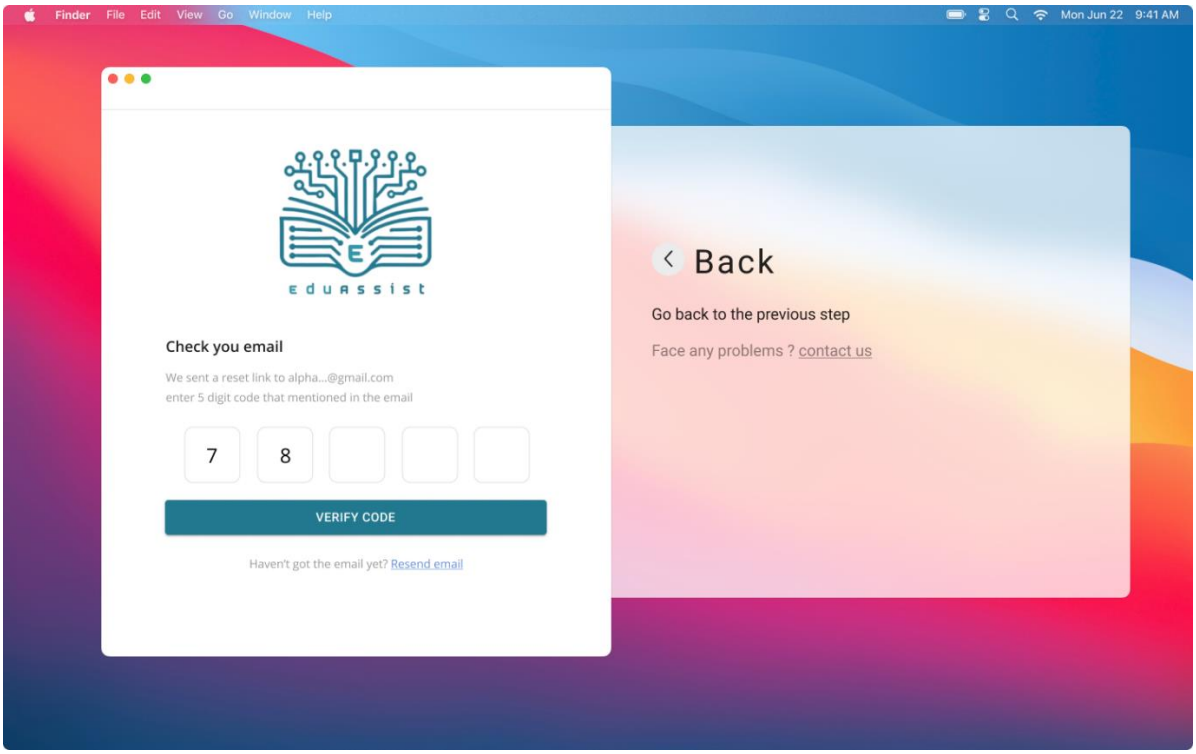


Figure 46

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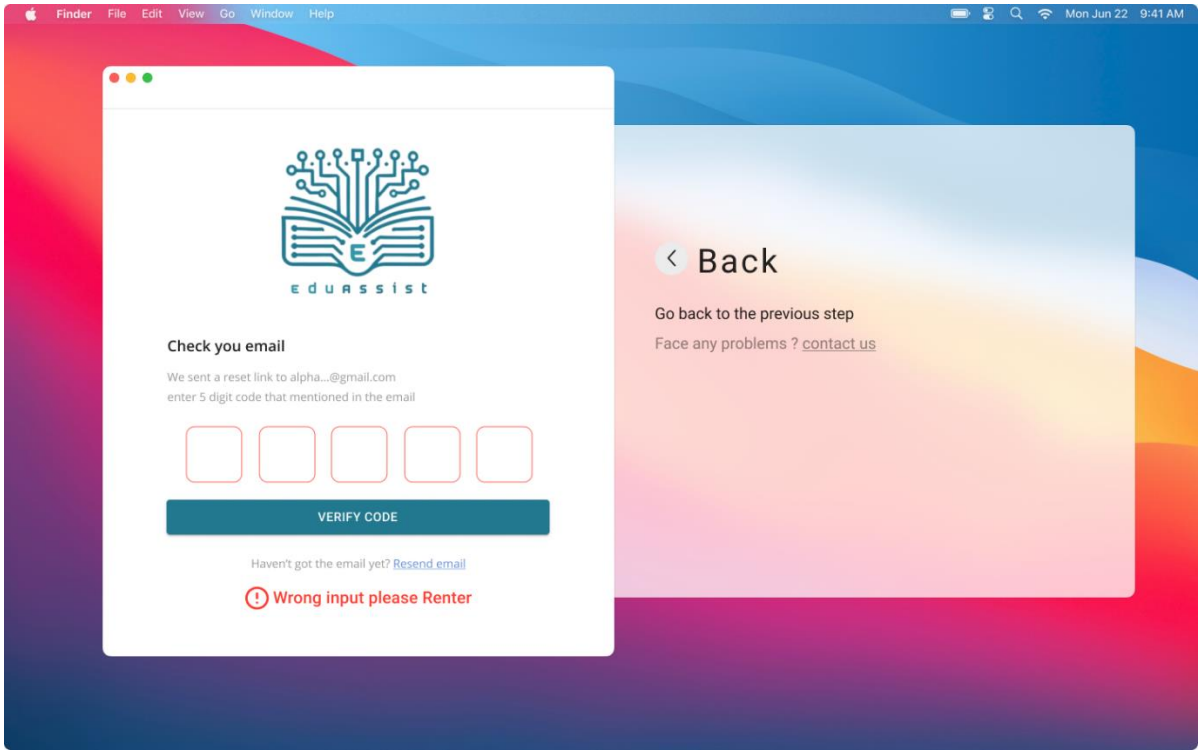


Figure 47

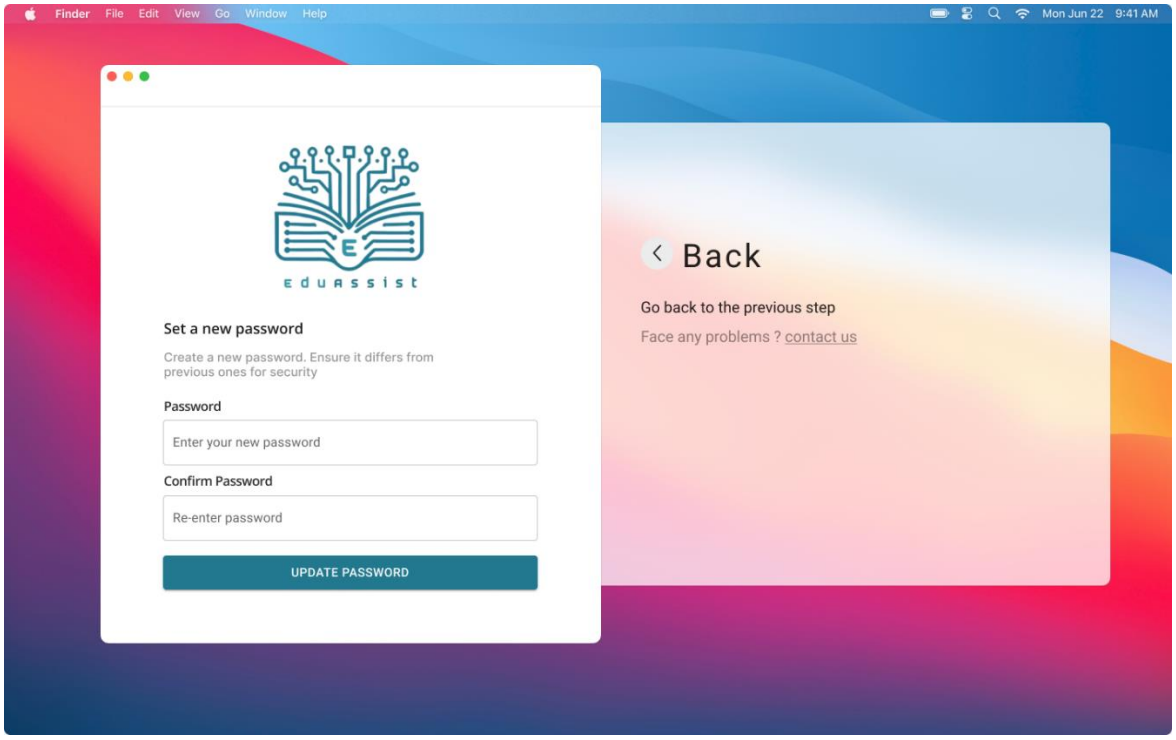


Figure 48

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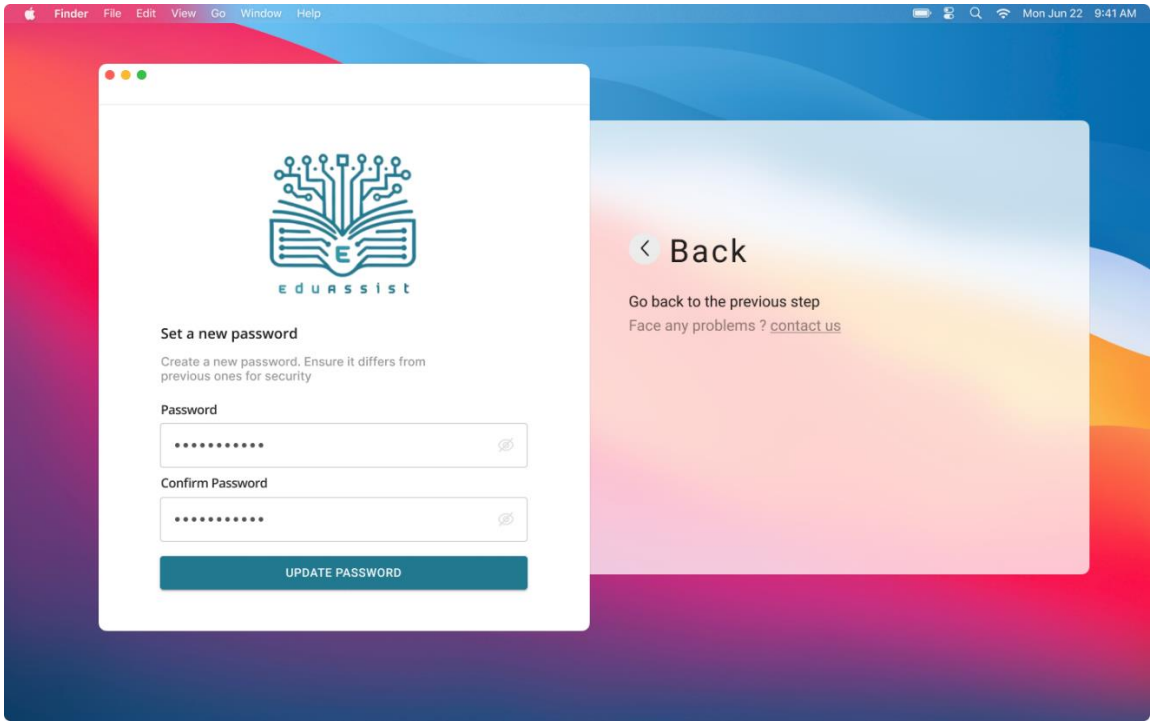


Figure 49

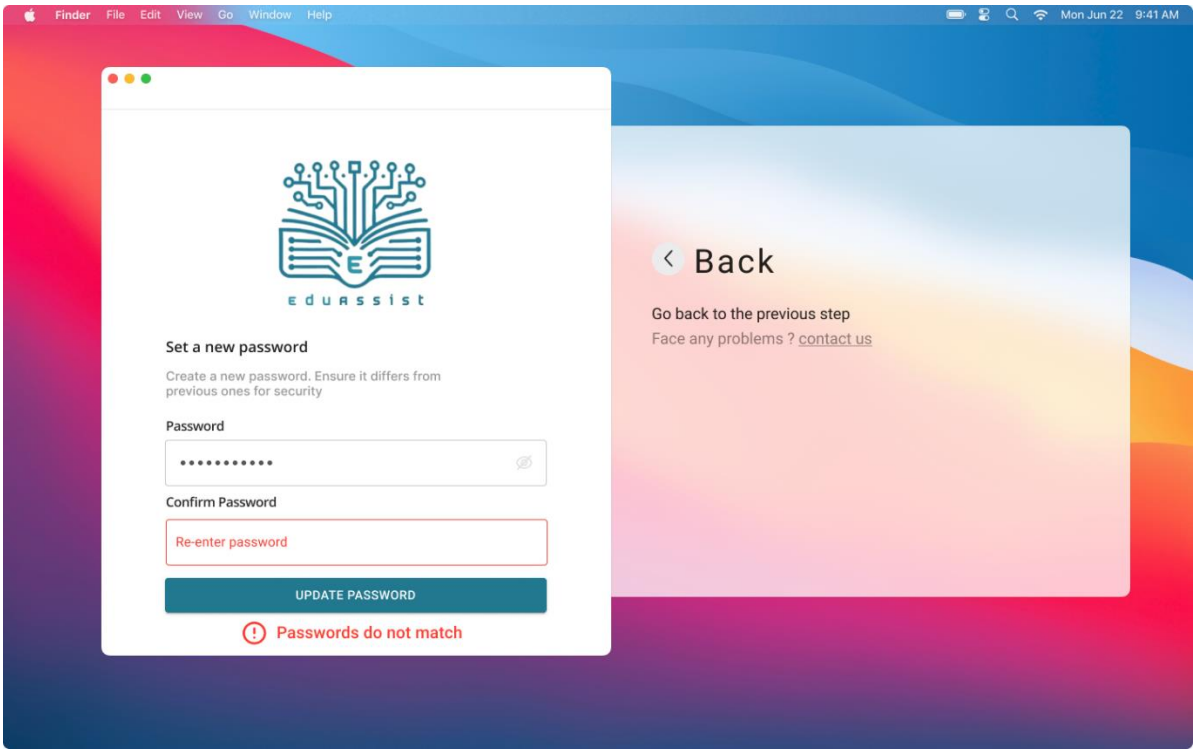


Figure 50

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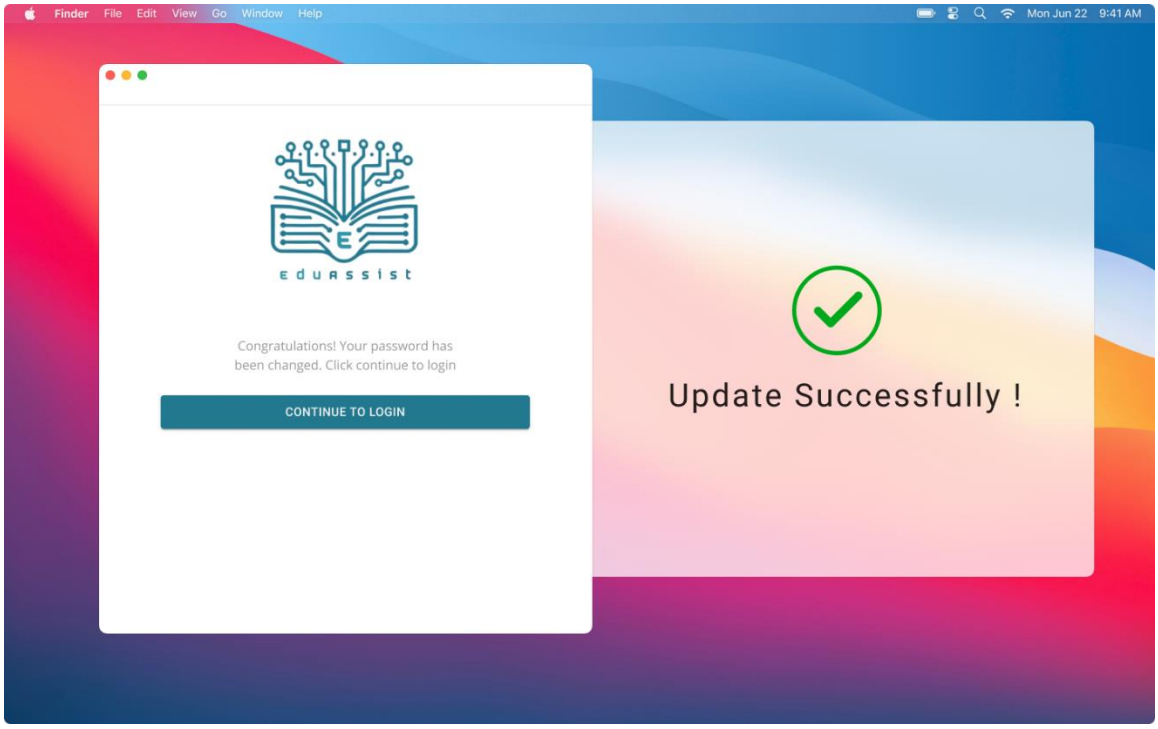


Figure 51

4.5.2 Wrong password:

The Wrong Password Interface for Google Login appears when a user enters incorrect credentials while logging into EDUassist via Google. The interface displays an error message prompting the user to check their password, with a link to Google's "Forgot Password" page for account recovery. This allows users to reset their password and regain access to their Google account, ensuring secure and seamless login to the EDUassist platform.

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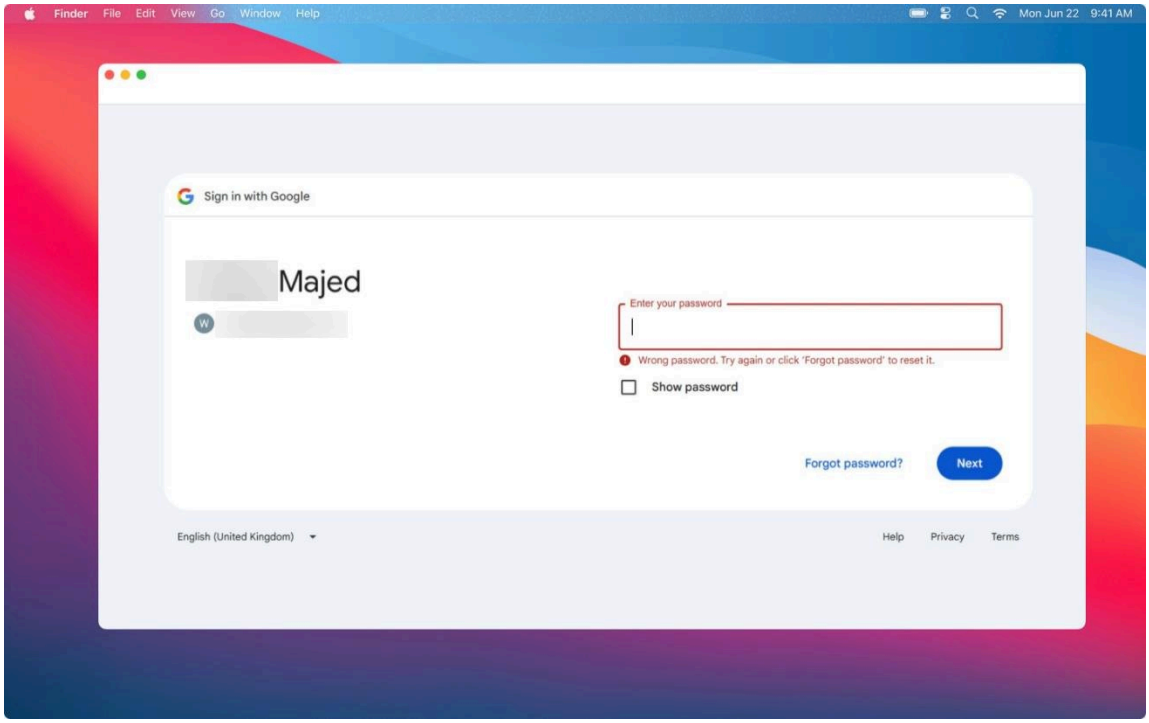


Figure 52 Error

4.5.3 Wrong input:

The Wrong Input Interface at Sign-Up appears when a user enters invalid or incomplete information during the registration process. This interface displays error messages for fields such as email format errors, password strength violations, or missing required details. The user is prompted to correct the errors, ensuring all fields are properly filled out before proceeding with account creation. This helps maintain data accuracy and security while guiding users through the sign-up process.

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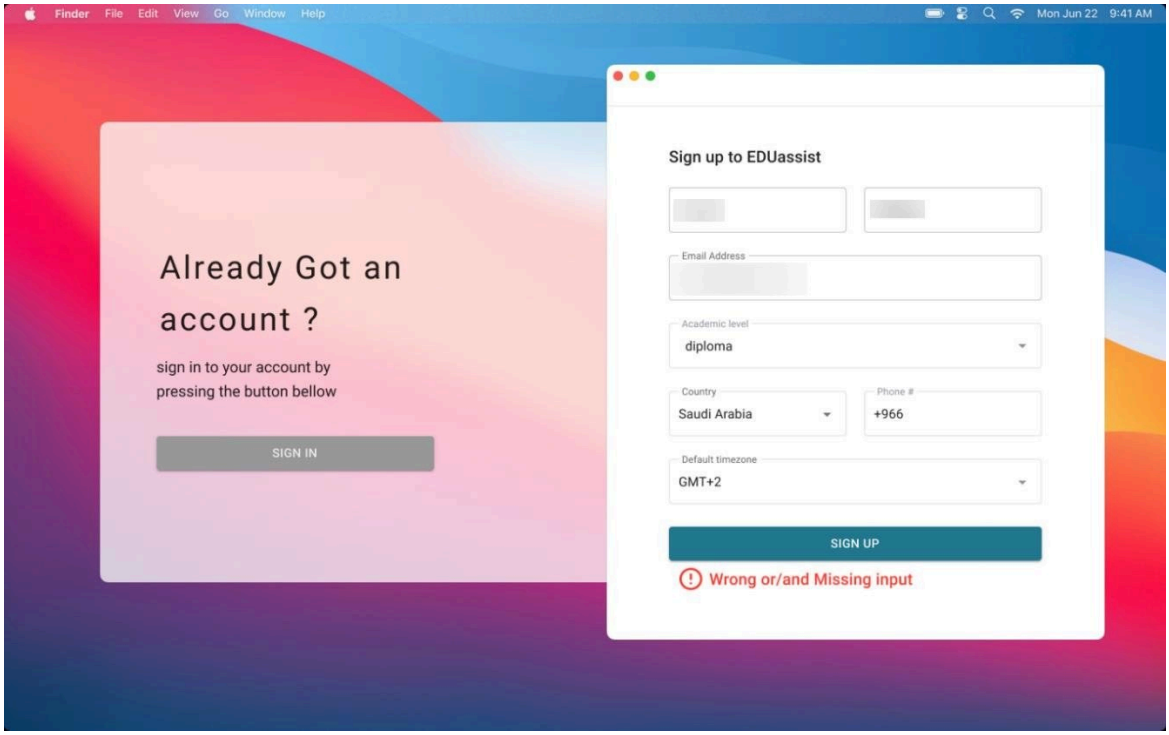


Figure 53 error Signin

4.5.4 Wrong OTP:

The Wrong OTP Interface appears when a user enters an incorrect one-time password (OTP) during the verification process. An error message is displayed, prompting the user to check the OTP and try again, ensuring only valid OTPs allow account access.

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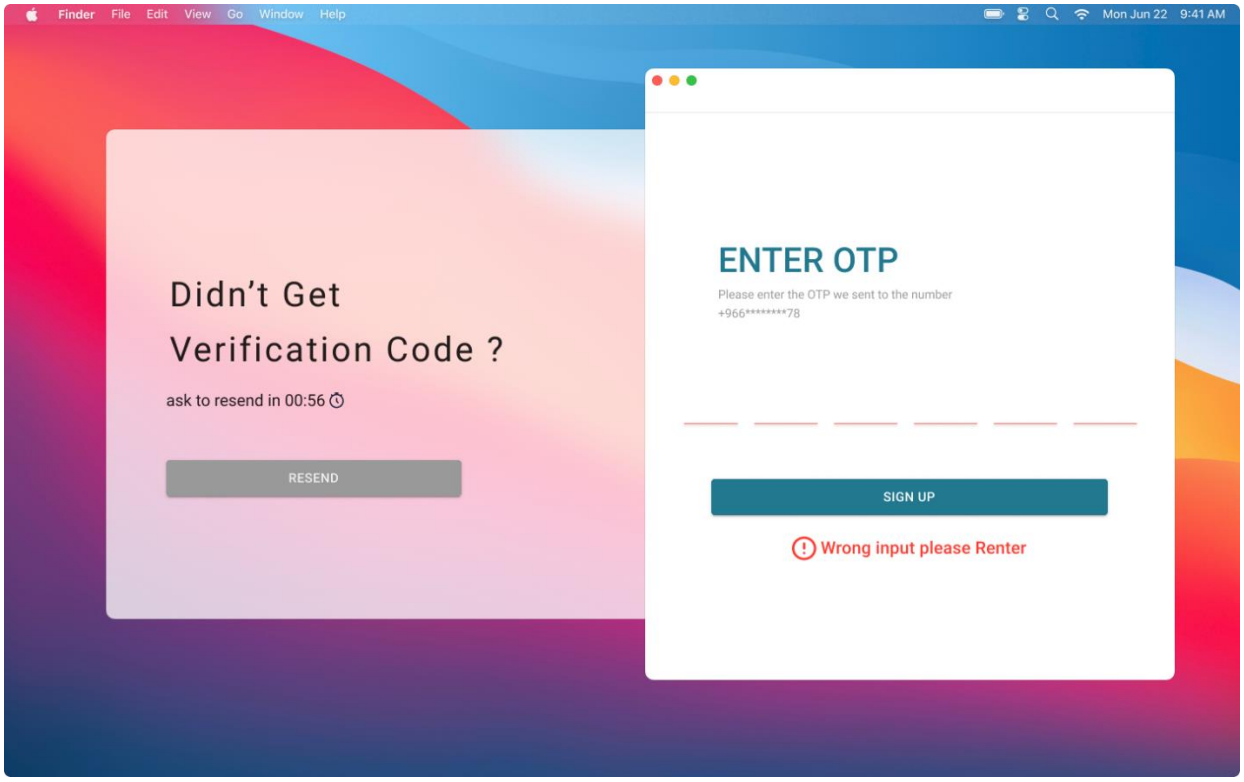


Figure 54 OTP Error

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4.5.5 Error add Tasks:

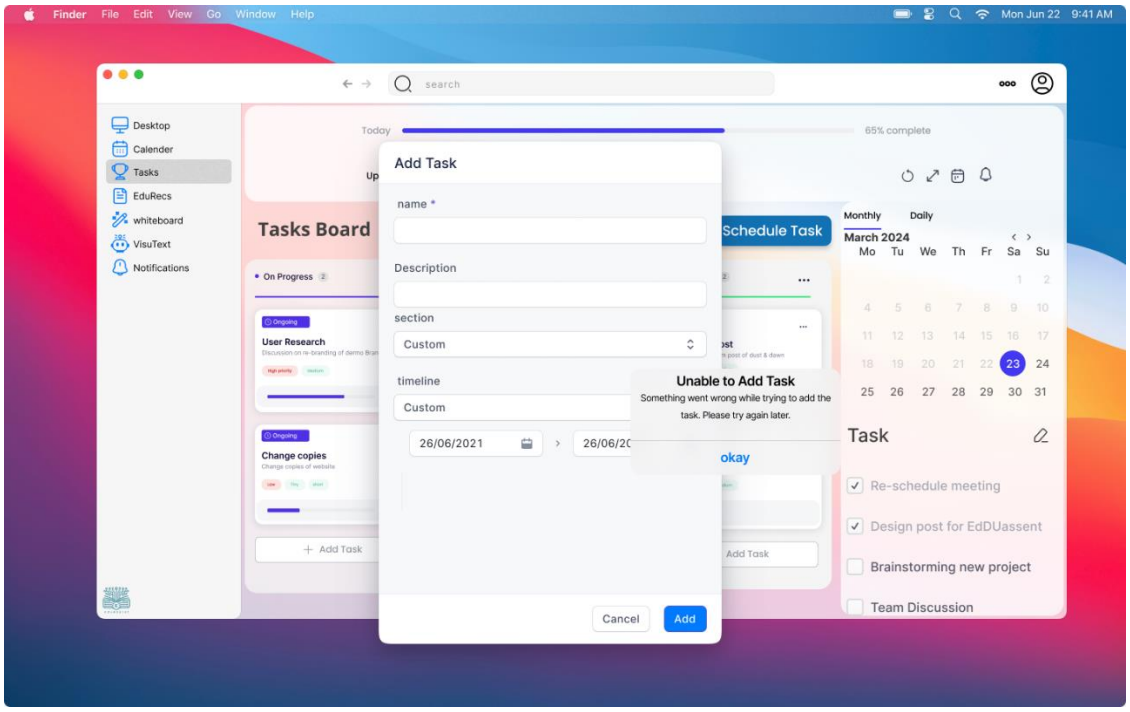


Figure 55 error add tasks

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4.5.6 Error report generations :

The EDUassent Report Creation interface shows a prompt for saving a new document titled "Untitled." Users are presented with options to input a name and tags for the report. An error message indicates that the report could not be created, suggesting that users check their input. The interface includes buttons for deleting the document and saving it, ensuring that users can manage their documents effectively. The layout is clean and straightforward, allowing for easy navigation and clear communication of actions needed.

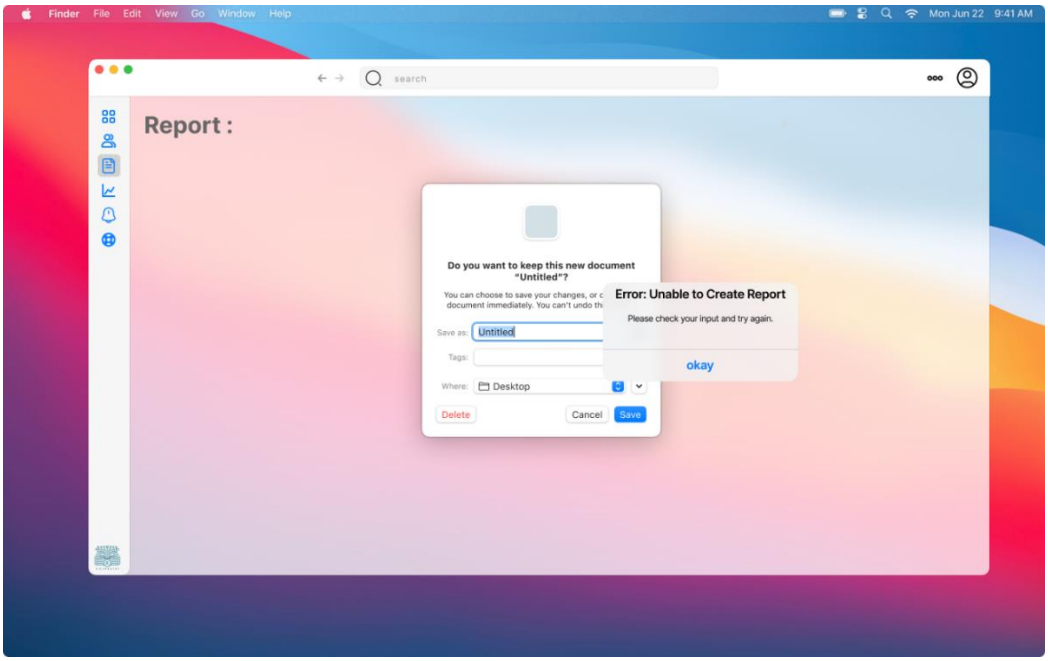


Figure **56** *error message*

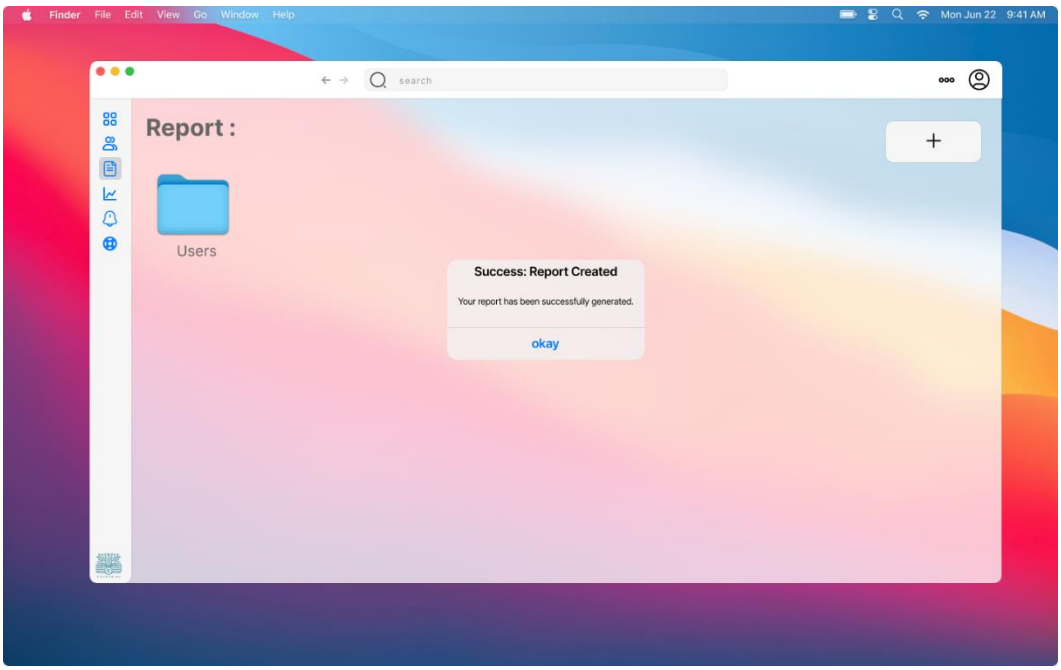


Figure 57 Success

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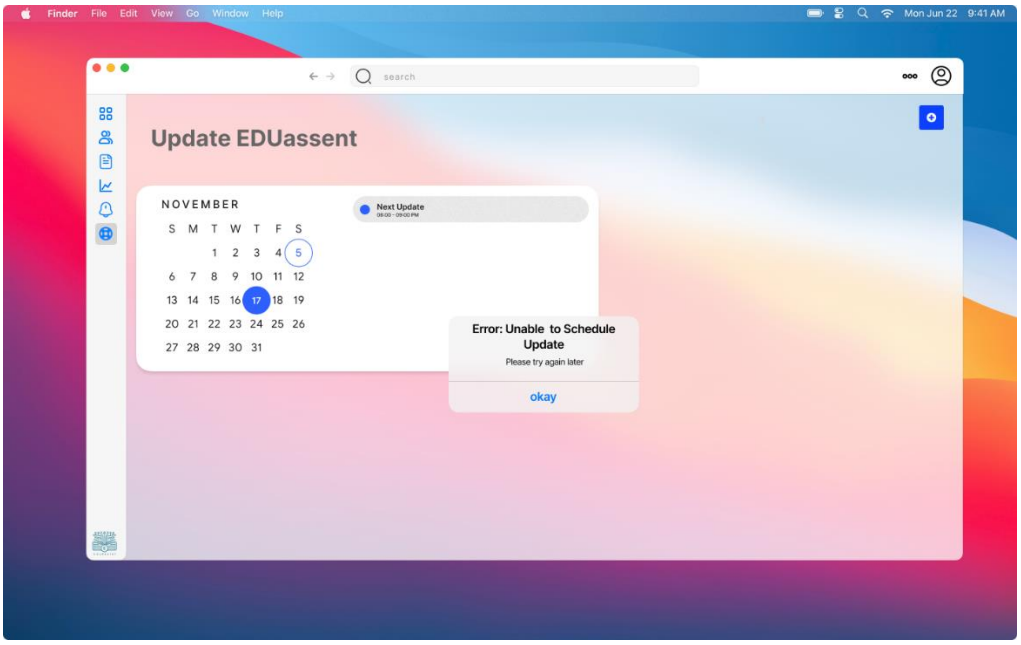


Figure 58 Error in Update Interface

An error message appears, indicating that it was unable to schedule the update and prompting the Admin to try again later.

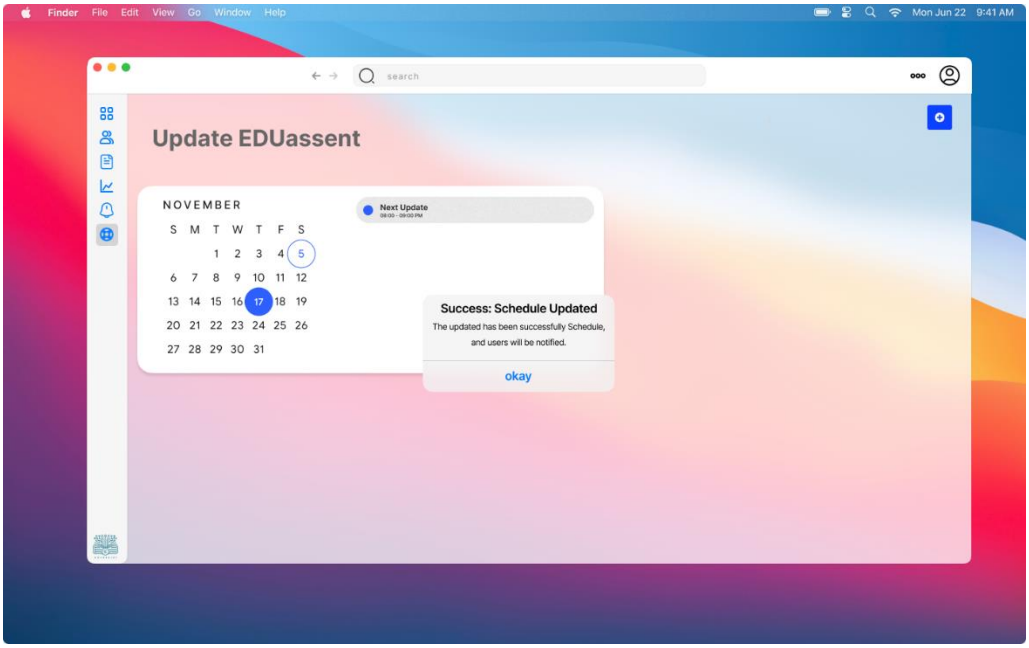


Figure 59 message

Copyright © 2024 by CS Year 4.

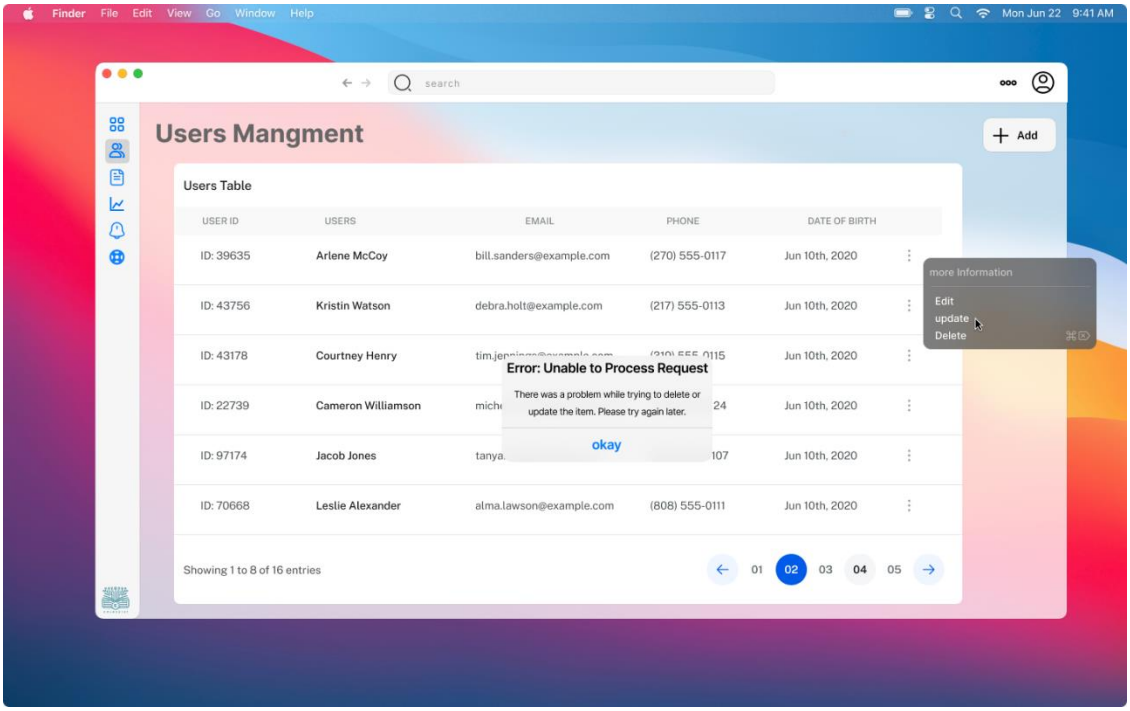


Figure 60 error Users management

An error message appears in the center, stating "Error: Unable to Process Request," and prompts the user to click "okay." The interface includes options for editing, updating, or deleting user entries, providing a straightforward way to manage user data.

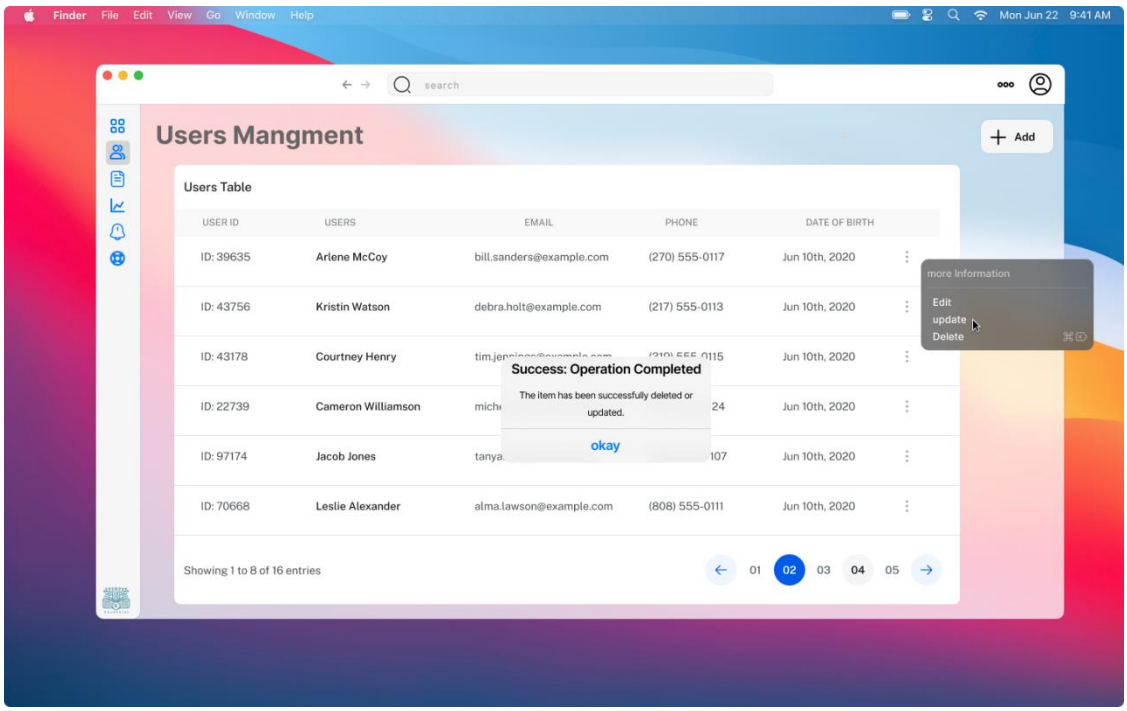


Figure 61 success messege

Copyright © 2024 by CS Year 4.

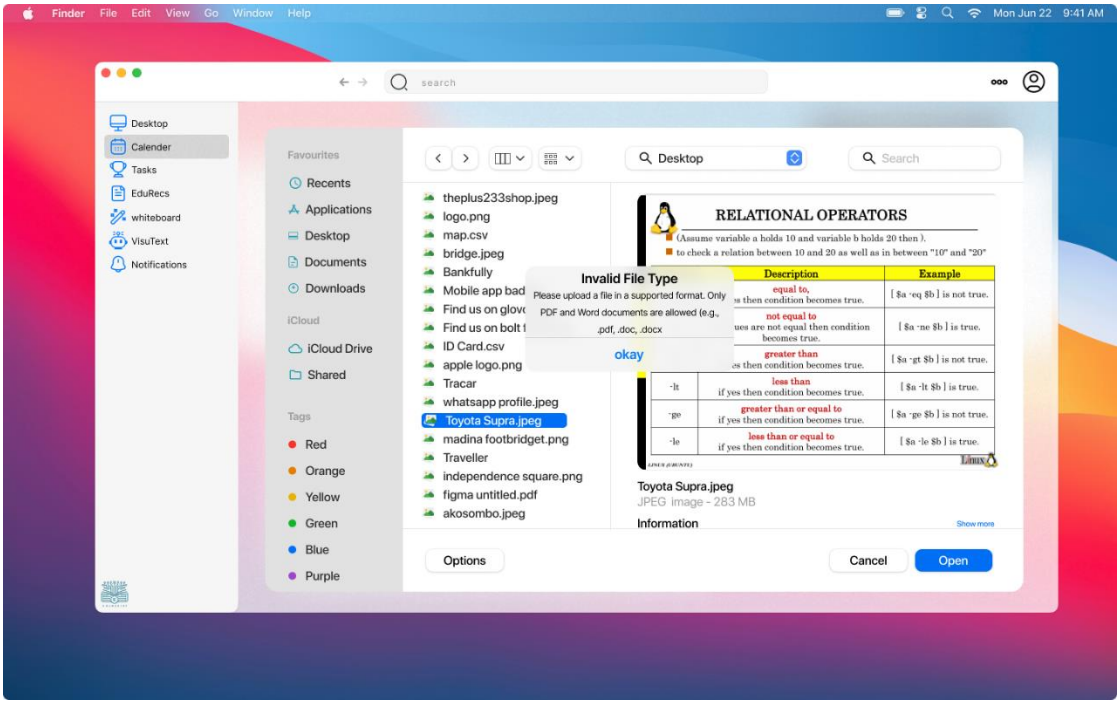


Figure 62 error invalid Filed Type

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5. System Architecture

This section provides a high-level overview of the EDUassist system architecture, outlining how the functionality and responsibilities were divided into subsystems or components and how these components collaborate to fulfill the system's goals. The architecture focuses on scalability, modularity, and maintainability to ensure adaptability to evolving user needs. The subsections below break down the architectural design approach, the system's overall architecture, and the subsystem-level decomposition.

5.1 Architectural Design Approach

The architectural design of EDUassist employs a layered modular approach to achieve scalability, maintainability, and flexibility. This design approach ensures a clear separation of concerns, with responsibilities divided into three primary layers: the Presentation Layer, the Application Layer, and the Data Layer. These layers interact seamlessly to deliver the system’s core functionalities while maintaining independence for ease of development and testing.

The Presentation Layer is responsible for managing user interactions. It provides interfaces for students and administrators, including the login/sign-up interface, user dashboard, and task management interface. By isolating user interactions in this layer, the system ensures that changes to the user interface do not affect core business logic.

The Application Layer forms the system’s core, handling all business logic and coordinating communication between the Presentation Layer and the Data Layer. This layer includes critical components such as the Authentication Service, Study Plan Generator, Collaborative Whiteboard, AI-Powered Visual Tools, and Task Management Module. These components work together to process user inputs, generate outputs, and manage workflows.

The Data Layer is tasked with managing data storage, retrieval, and integration. It consists of a Database for structured data, a File Repository for unstructured content like whiteboard notes, and Integration APIs for external services such as Google Calendar. This centralized data management ensures security, consistency, and efficient data access.

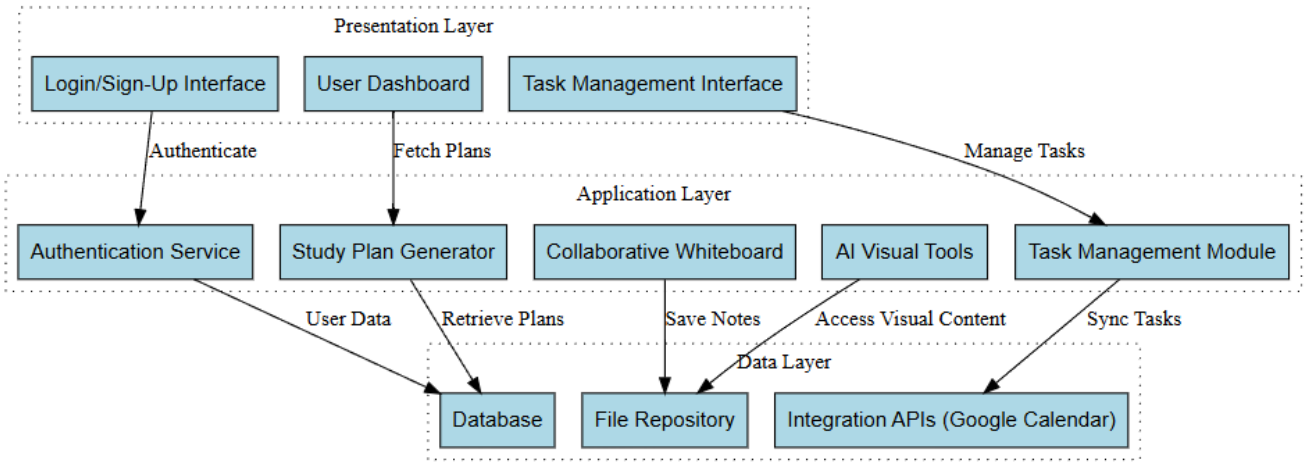


Figure 63 System Architecture Diagram

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To enhance the architecture's efficiency and maintainability, design patterns such as Model-View-Controller (MVC) in the Presentation Layer and the Repository Pattern in the Data Layer are employed. These patterns help organize the system's structure, making it easier to scale and adapt to future needs. The system also adopts a client-server model, ensuring robust communication between the user-facing components and backend services.

This layered and modular architecture was chosen to address EDUassist's need for a scalable and adaptable platform. Alternative designs, such as monolithic architectures, were considered but rejected due to their limitations in handling complexity and scaling efficiently. By following this approach, EDUassist provides a flexible, reliable, and maintainable framework that meets current requirements while allowing for future growth.

5.2 Architectural Design

The EDUassist system adopts a layered architecture comprising four primary layers: Presentation Layer, Application (Business) Layer, Persistence Layer, and Data Layer, each ensuring modularity, scalability, and maintainability with a clear separation of concerns between user interaction, business logic, and data management. The **Presentation Layer** acts as the user interface, managing all user interactions by processing inputs, displaying outputs, and communicating with the Application Layer to facilitate functionalities such as secure login, personalized dashboards, and task management, implemented using modern frameworks like JavaFX or ReactJS. The **Application Layer** serves as the core intermediary, processing user requests, applying business rules, and coordinating system functionalities, such as authentication, study plan generation, real-time collaboration, and AI-powered tools, leveraging programming languages like Java or Python for scalability and efficiency. The **Persistence Layer** provides secure data access and management through components like query handlers and Object-Relational Mapping (ORM) frameworks (e.g., Hibernate or MyBatis), ensuring robust interfacing between the Application and Data Layers. The **Data Layer** underpins the system by managing structured and unstructured data in relational databases (e.g., MySQL) and file repositories, integrating external services like Google Calendar through APIs to enhance system capabilities. These layers collaborate seamlessly, with the Presentation Layer sending user requests to the Application Layer, which processes them and interacts with the Persistence Layer to securely retrieve or store data in the Data Layer, ensuring smooth functionality across the system. For instance, when a student requests a personalized study plan, the Login Interface authenticates the user, the Application Layer's Study Plan Generator retrieves data from the Data Layer via the Persistence Layer, and the result is displayed on the Dashboard, demonstrating the system's modular and efficient workflow.

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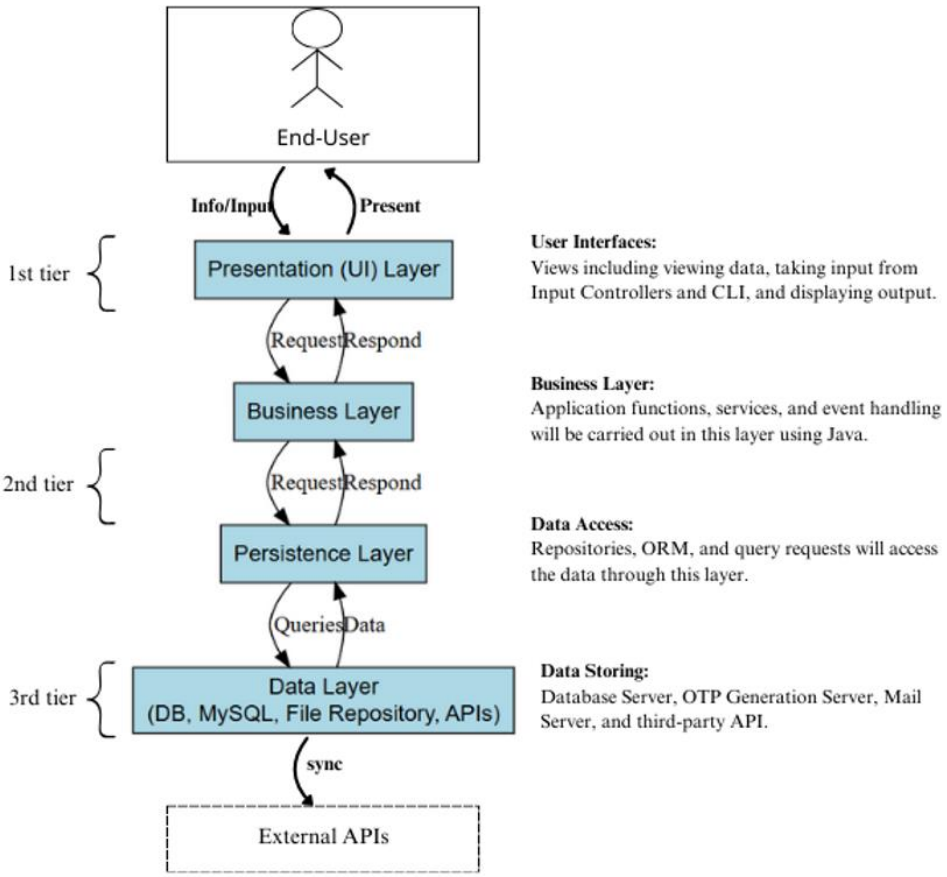


Figure 64 Three-tier EDUassist system architecture.

5.3 Subsystem Architecture

5.3.1 Main Interface Subsystem

The Main Interface Subsystem is the primary entry point for all users of the EDUassist system, including students and administrators. It is responsible for managing user interactions and providing access to the core functionalities of the system. This subsystem ensures a user-friendly interface that facilitates smooth navigation and interaction with the underlying components.

The subsystem consists of several interconnected components. The Homepage serves as the landing page for all users, offering quick access to the login and sign-up functionalities. The Login/Sign-Up Module manages user authentication, allowing registered users to securely log in and new users to create accounts. For users who have forgotten their credentials, the Password Retrieval Module provides an easy mechanism for resetting passwords via email or OTP. Once authenticated, users are directed to the Dashboard, which acts as the central hub for accessing personalized schedules, tasks, and study plans. This module connects seamlessly with the Notification Module, which delivers alerts and reminders for task deadlines, study plan updates, and other critical events, ensuring users remain informed.

The Main Interface Subsystem interacts with the Core Functionalities Subsystem to process user actions, such as retrieving a study plan or updating a task. It also communicates with the Data Subsystem to fetch user-specific information, such as schedules or notification data, which is displayed in the Dashboard. These interactions maintain a seamless flow of data and functionality across the system. **Figure 65** and

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Figure 66 display the structural relationships and workflows within the Main Interface Subsystem, illustrating its components and interactions.

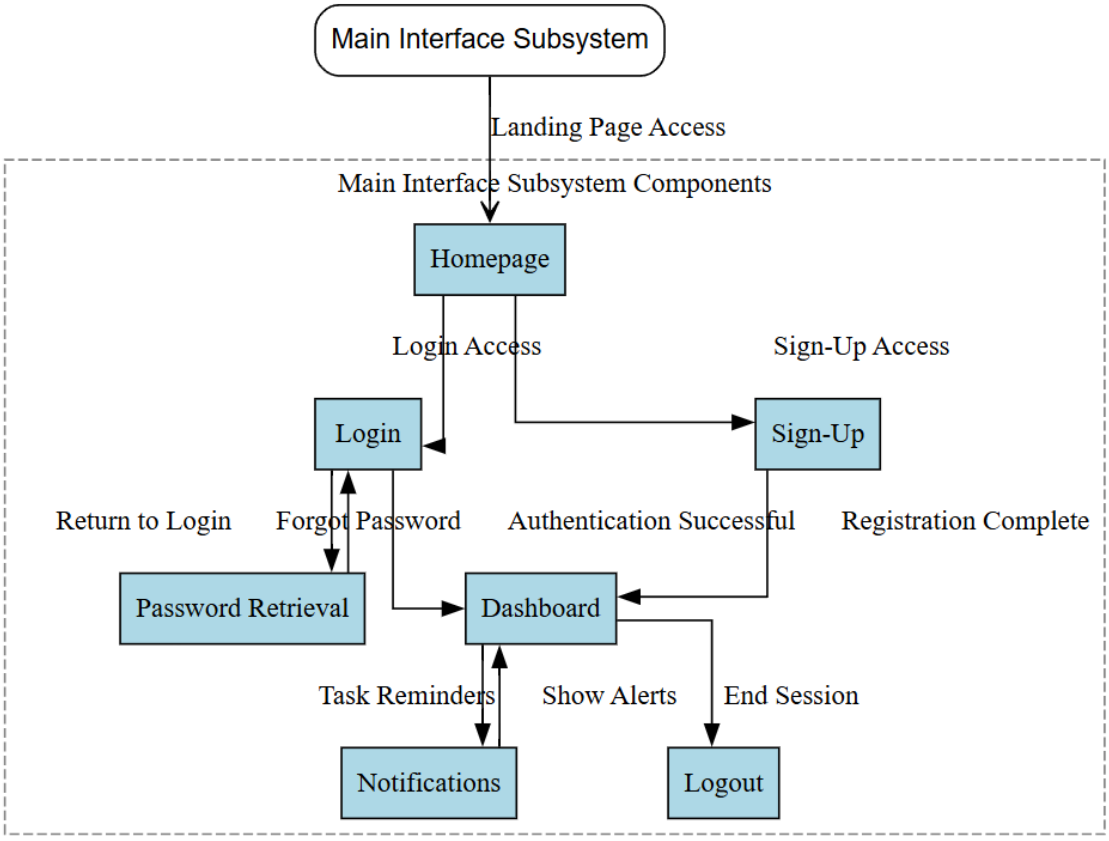


Figure 65 Main Interface Subsystem

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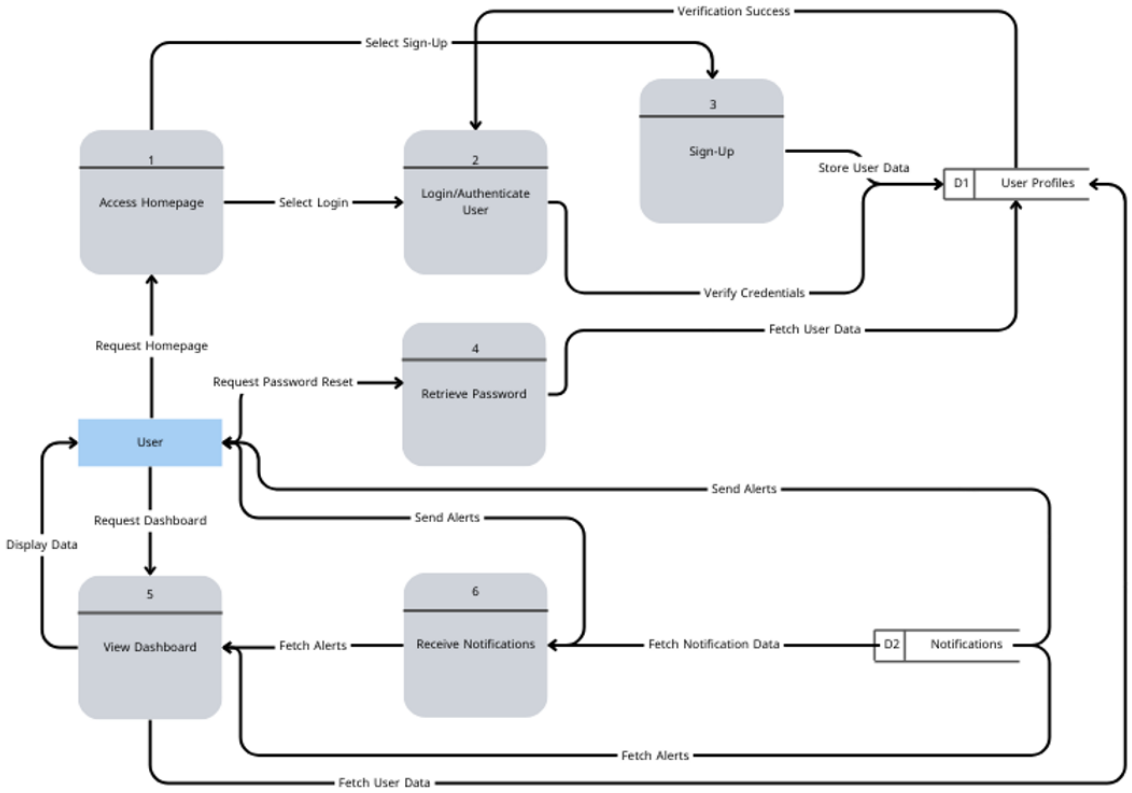


Figure 66 Main Interface Subsystem DFD

5.3.2 Admin Subsystem

The Admin Subsystem is responsible for providing administrative control over the EDUassist system. This subsystem is used by administrators to manage user accounts, configure system settings, and monitor system events. It plays a vital role in maintaining the integrity, functionality, and security of the platform.

The Admin Subsystem is composed of several key components that collectively enable administrators to oversee and configure the system effectively. The **User Management Module** allows administrators to create, update, and delete user accounts, as well as assign specific roles and permissions to users. The **System Settings Module** provides a centralized interface for configuring system-wide features, such as calendar integrations, notification preferences, and access policies. Additionally, the subsystem includes the **System Logs Module**, which records system events, such as user actions, errors, and system updates, to ensure accountability and facilitate troubleshooting.

This subsystem interacts extensively with the Data Subsystem to fetch and update user data, system configurations, and event logs. It also collaborates with the Core Functionalities Subsystem to enforce rules and validate user actions. These interactions ensure that the system operates smoothly while maintaining robust security and administrative oversight.

The design of the Admin Subsystem emphasizes modularity, security, and scalability. By isolating administrative functionalities into distinct components, the system is easy to manage, extend, and secure. **Figure 67** and **Figure 68** display the structural components and workflows within the Admin Subsystem, illustrating its interactions and responsibilities.

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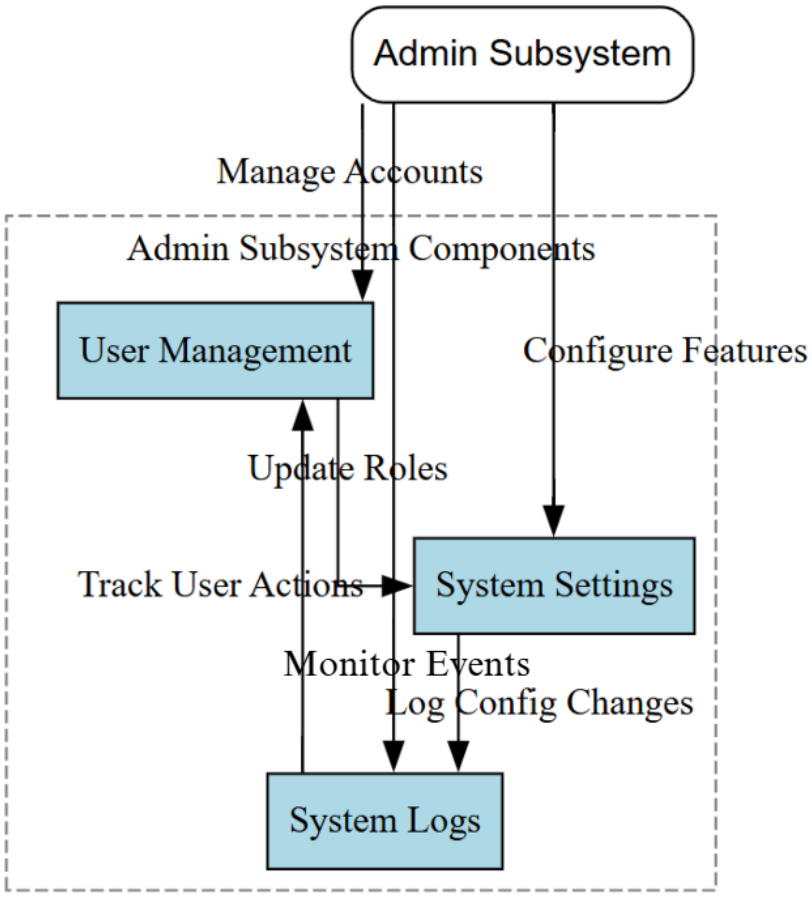


Figure 67 Admin Subsystem

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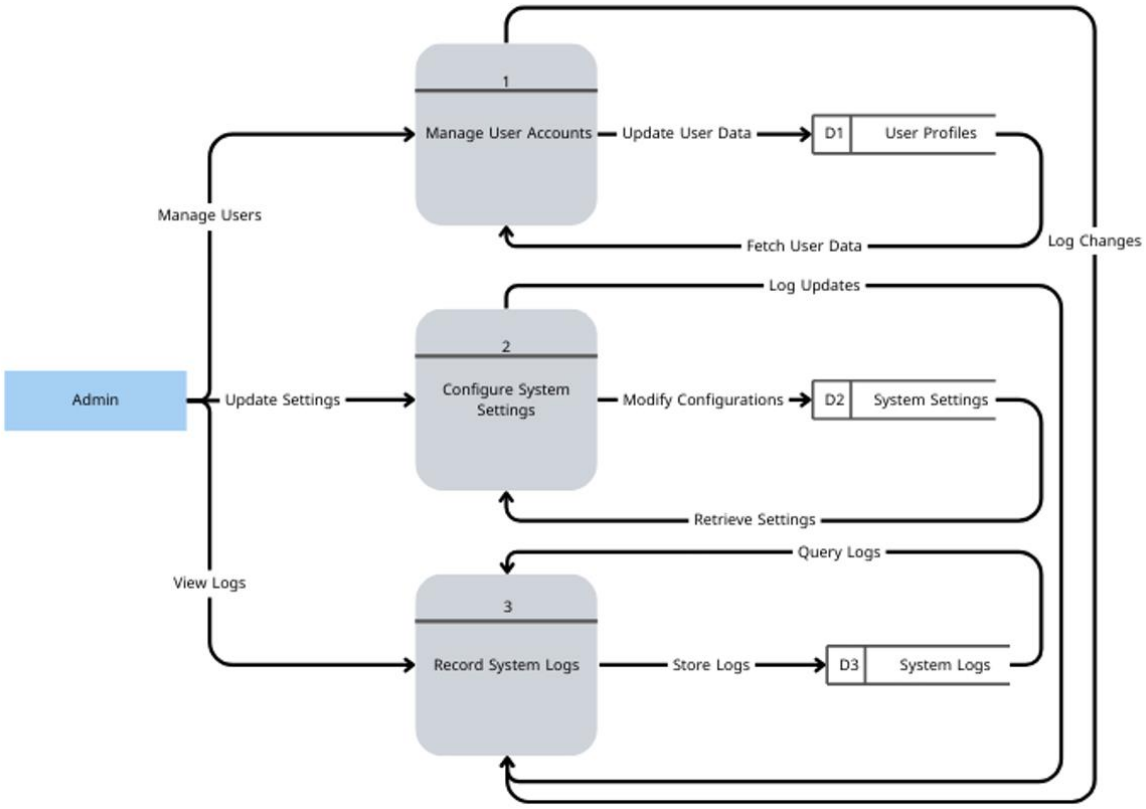


Figure 68 Admin Subsystem DFD

5.3.3 Data Subsystem

The Data Subsystem is responsible for managing all data storage and retrieval operations in the EDUassist system. It serves as the backbone of the platform, providing secure and efficient access to critical data such as user profiles, study plans, and notifications. The subsystem ensures data integrity, security, and optimized performance while enabling seamless interaction between the system's components.

This subsystem includes the **Security Manager Module**, which plays a key role in protecting sensitive data by enforcing encryption, secure access, and authentication mechanisms. The **Data Storage Module** manages the system's databases, ensuring efficient storage and retrieval of structured and unstructured data. Additionally, the **Data Retrieval Module** processes requests from other subsystems, such as fetching user profiles for the Main Interface Subsystem or retrieving logs for the Admin Subsystem.

The Data Subsystem interacts with all major components of the system, ensuring that data is stored securely and retrieved efficiently for processing. This design prioritizes modularity and scalability, enabling future enhancements and system expansion. **Figure 69** and **Figure 70** display the structure and workflows within the Data Subsystem, illustrating its interactions and responsibilities.

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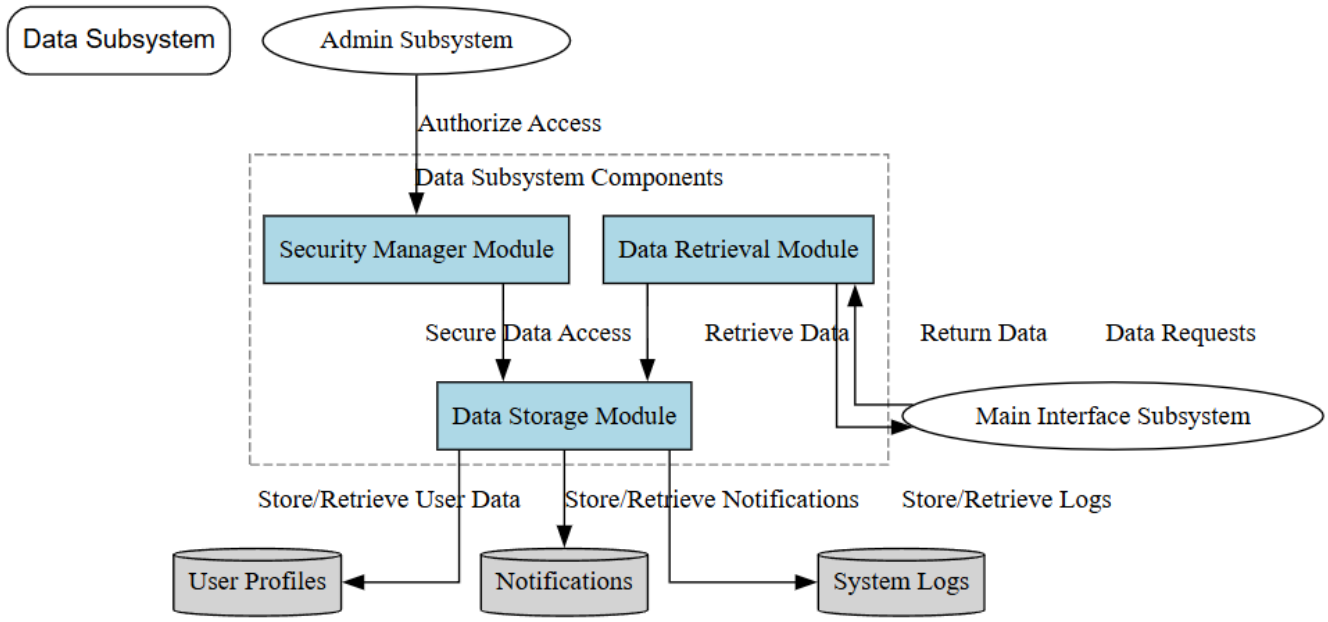


Figure 69 Data Subsystem

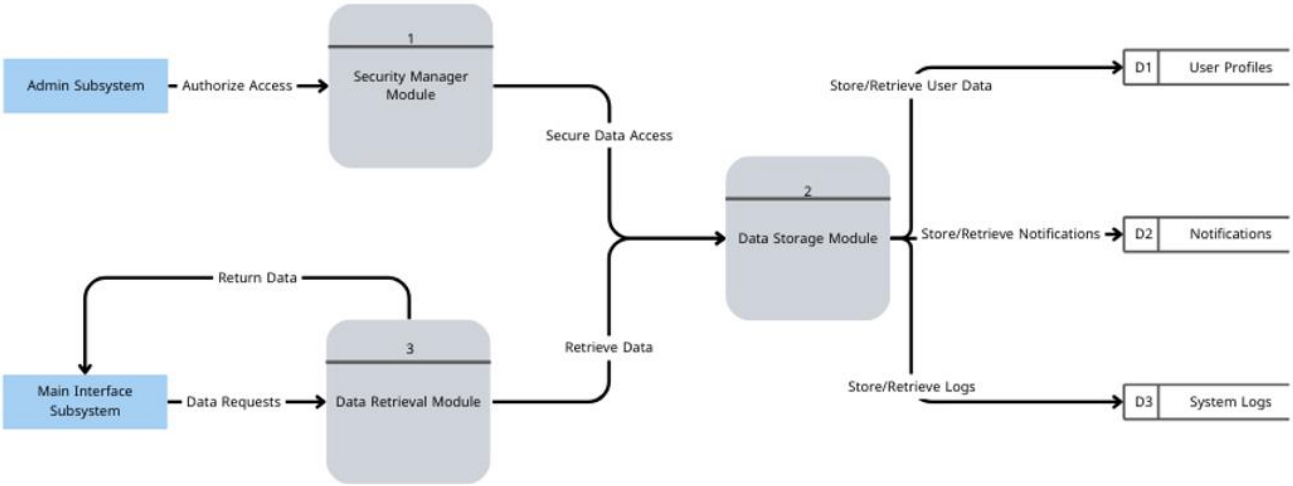


Figure 70 Data Subsystem DFD

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6. Data Design

This section presents the Data Description, Data Dictionary, Database Description.

6.1 Data Description

This subsection views the entities, their fields, datatype and constrains in EDUassist database as shown in **Table 6** below.

Entity	Attribute	Type	PK	FK	Constraints
User	UserID	INT(10)	✓		NOT NULL
	Username	VARCHAR(30)			NOT NULL
	Password	VARCHAR(30)			NOT NULL
	Email	VARCHAR(50)			NOT NULL
	Role	VARCHAR(20)			NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
Admin	AdminID	INT(10)	✓		NOT NULL
	Username	VARCHAR(30)			NOT NULL
	Password	VARCHAR(30)			NOT NULL
	Email	VARCHAR(50)			NOT NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
StudyPlan	StudyPlanID	INT(10)	✓		NOT NULL
	UserID	INT(10)		✓	NOT NULL
	PlanDetails	TEXT			NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
Task	TaskID	INT(10)	✓		NOT NULL
	UserID	INT(10)		✓	NOT NULL
	Title	VARCHAR(100)			NOT NULL
	Description	TEXT			NULL
	DueDate	DATETIME			NULL
	Status	VARCHAR(20)			NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
Notification	NotificationID	INT(10)	✓		NOT NULL
	UserID	INT(10)		✓	NOT NULL
	Message	TEXT			NOT NULL
	IsRead	BOOLEAN			NOT NULL
	CreatedAt	DATETIME			NOT NULL
Progress	ProgressID	INT(10)	✓		NOT NULL
	UserID	INT(10)		✓	NOT NULL
	StudyPlanID	INT(10)		✓	NOT NULL

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	Achievements	TEXT			NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
Resource	ResourceID	INT(10)	✓		NOT NULL
	Title	VARCHAR(100)			NOT NULL
	URL	VARCHAR(200)			NOT NULL
	Type	VARCHAR(50)			NOT NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
Whiteboard	WhiteboardID	INT(10)	✓		NOT NULL
	UserID	INT(10)		✓	NOT NULL
	Content	TEXT			NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
Video	VideoID	INT(10)	✓		NOT NULL
	UserID	INT(10)		✓	NOT NULL
	WhiteboardID	INT(10)		✓	NOT NULL
	VideoURL	VARCHAR(200)			NOT NULL
	CreatedAt	DATETIME			NOT NULL
	UpdatedAt	DATETIME			NOT NULL
Recommendation	RecommendationID	INT(10)	✓		NOT NULL
	UserID	INT(10)		✓	NOT NULL
	ResourceID	INT(10)		✓	NOT NULL
	Reason	TEXT			NULL
	CreatedAt	DATETIME			NOT NULL

Table 6 Data Description of EDUassent

6.2 Data Dictionary

This section presents the data dictionary that exist in EDUassist. Which are: entities and characterization for each field in **Table8** below.

Entity	Field	Description
User	UserID	Unique identifier for each user.
	Username	Name chosen by the user for login and display purposes.
	Password	User's password for authentication.
	Email	User's email address for communication and notifications.
	Role	Role of the user (e.g., student, admin).
	CreatedAt	Timestamp of when the user account was created.
	UpdatedAt	Timestamp of the last update to the user account.
Admin	AdminID	Unique identifier for each admin.

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	Username	Name chosen by the admin for login and display purposes.
	Password	Admin's password for authentication.
	Email	Admin's email address for communication and notifications.
	CreatedAt	Timestamp of when the admin account was created.
	UpdatedAt	Timestamp of the last update to the admin account.
StudyPlan	StudyPlanID	Unique identifier for each study plan.
	UserID	Identifier for the user associated with the study plan.
	PlanDetails	Detailed information about the study plan.
	CreatedAt	Timestamp of when the study plan was created.
	UpdatedAt	Timestamp of the last update to the study plan
Task	TaskID	Unique identifier for each task.
	UserID	Identifier for the user associated with the task.
	Title	Title or name of the task.
	Description	Detailed information about the task.
	DueDate	Deadline for completing the task.
	Status	Current status of the task (e.g., pending, completed).
	CreatedAt	Timestamp of when the task was created.
	UpdatedAt	Timestamp of the last update to the task.
Notification	NotificationID	Unique identifier for each notification.
	UserID	Identifier for the user receiving the notification.
	Message	Content of the notification message.
	IsRead	Indicates whether the notification has been read.
	CreatedAt	Timestamp of when the notification was created.
Progress	ProgressID	Unique identifier for each progress record.
	UserID	Identifier for the user associated with the progress record.
	StudyPlanID	Identifier for the study plan associated with the progress.

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	Achievements	List of achievements made by the user.
	CreatedAt	Timestamp of when the progress record was created.
	UpdatedAt	Timestamp of the last update to the progress record.
Resource	ResourceID	Unique identifier for each resource.
	Title	Title or name of the resource.
	URL	Link to the resource.
	Type	Type of resource (e.g., video, article).
	CreatedAt	Timestamp of when the resource was created.
	UpdatedAt	Timestamp of the last update to the resource.
Whiteboard	WhiteboardID	Unique identifier for each whiteboard.
	UserID	Identifier for the user associated with the whiteboard.
	Content	Content or notes written on the whiteboard.
	CreatedAt	Timestamp of when the whiteboard was created.
	UpdatedAt	Timestamp of the last update to the whiteboard.
Video	VideoID	Unique identifier for each video.
	UserID	Identifier for the user associated with the video.
	WhiteboardID	Identifier for the whiteboard associated with the video.
	VideoURL	Link to the video content.
	CreatedAt	Timestamp of when the video was created.
	UpdatedAt	Timestamp of the last update to the video.
Recommendation	RecommendationID	Unique identifier for each recommendation.
	UserID	Identifier for the user making the recommendation.
	ResourceID	Identifier for the resource being recommended.
	Reason	Reason for the recommendation.
	CreatedAt	Timestamp of when the recommendation was created.

Table 8 Data Dictionary of EDUassist

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6.3Database Description

This subsection presents the Entity Relationship Diagram (ERD), Entity Relationship (ER) mapping, and the SQL code for the EDUassist system using MySQL. The ERD is illustrated in **Figure 71** below. Additionally, the ER mapping is shown in **Figure 72**. At the end of this subsection is the SQL code .

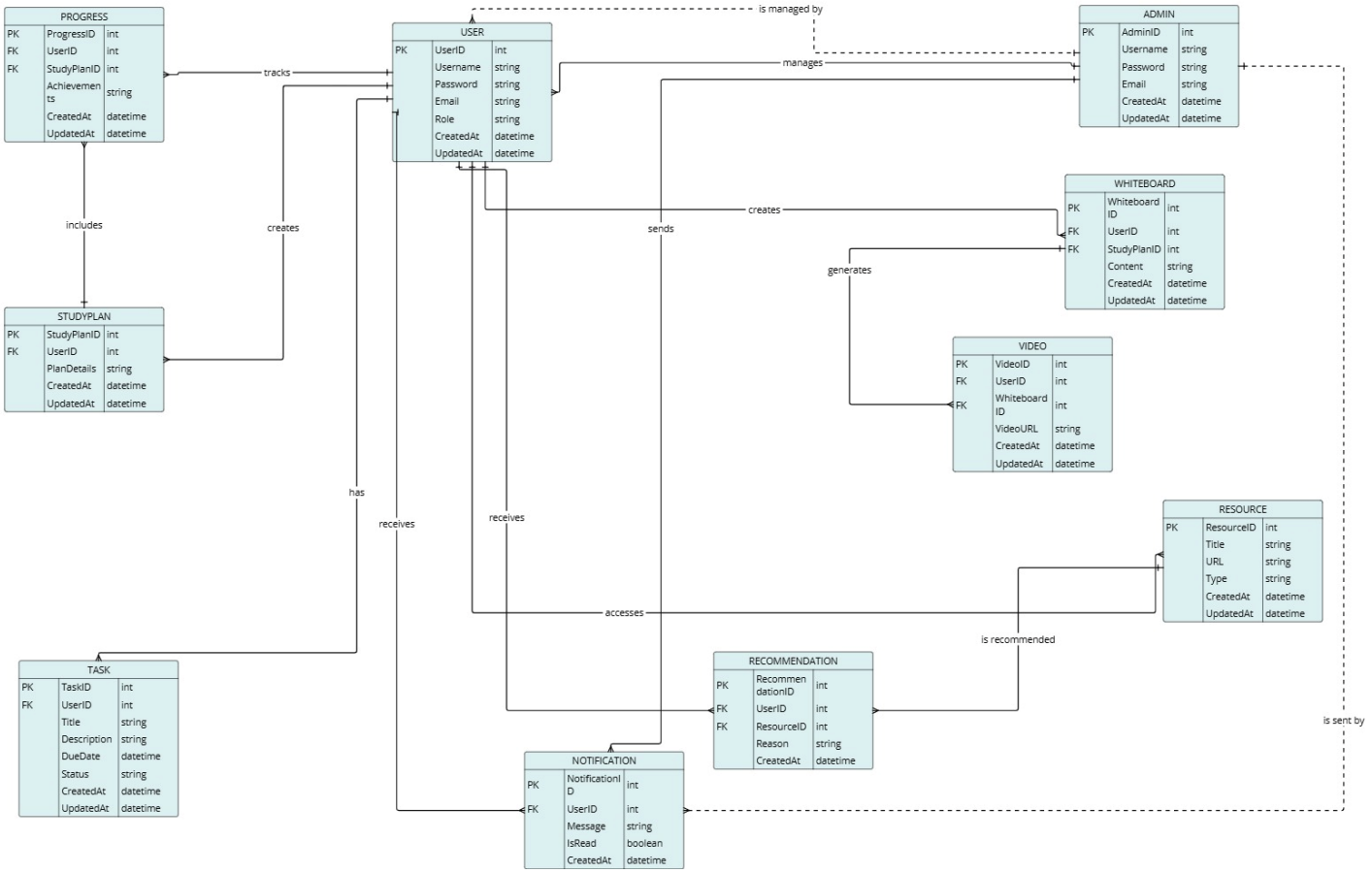


Figure 71 ERD for EDUassist

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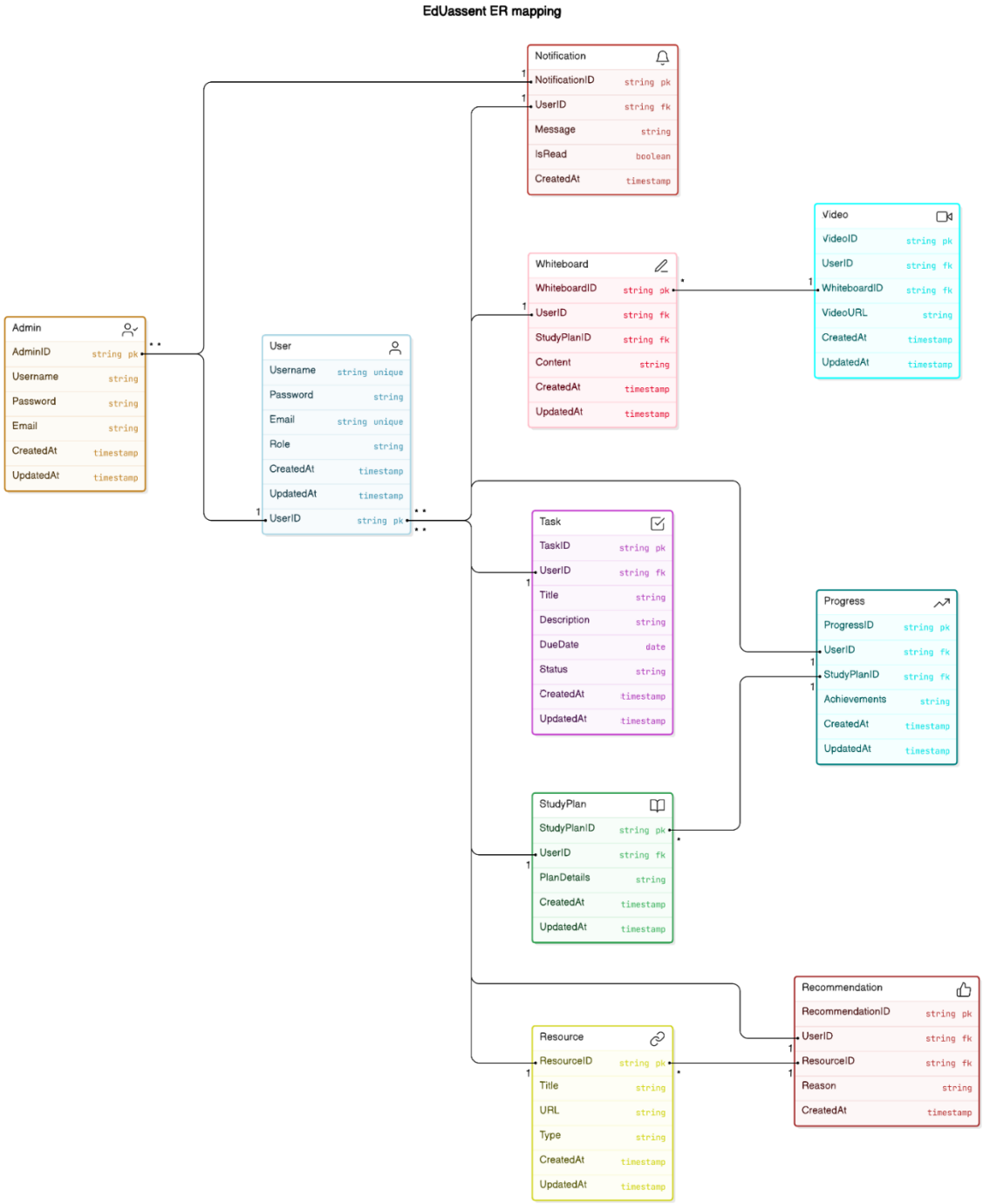


Figure 72 ER diagram Mapping

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6.3.1 Database Relation Code

```
-- Create User table
CREATE TABLE User (
  UserID INT PRIMARY KEY,
  Username VARCHAR(30) NOT NULL,
  Password VARCHAR(30) NOT NULL,
  Email VARCHAR(50) NOT NULL,
  Role VARCHAR(20) NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL
);
-- Create Admin table
CREATE TABLE Admin (
  AdminID INT PRIMARY KEY,
  Username VARCHAR(30) NOT NULL,
  Password VARCHAR(30) NOT NULL,
  Email VARCHAR(50) NOT NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL
);
-- Create StudyPlan table
CREATE TABLE StudyPlan (
  StudyPlanID INT PRIMARY KEY,
  UserID INT NOT NULL,
  PlanDetails TEXT NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL,
  FOREIGN KEY (UserID) REFERENCES User(UserID)
);
-- Create Task table
CREATE TABLE Task (
  TaskID INT PRIMARY KEY,
  UserID INT NOT NULL,
  Title VARCHAR(100) NOT NULL,
  Description TEXT NULL,
  DueDate DATETIME NULL,
  Status VARCHAR(20) NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL,
  FOREIGN KEY (UserID) REFERENCES User(UserID)
);
-- Create Notification table
CREATE TABLE Notification (
  NotificationID INT PRIMARY KEY,
  UserID INT NOT NULL,
  Message TEXT NULL,
  IsRead BOOLEAN NOT NULL,
  CreatedAt DATETIME NOT NULL,
```


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```
  FOREIGN KEY (UserID) REFERENCES User(UserID)
);
-- Create Progress table
CREATE TABLE Progress (
  ProgressID INT PRIMARY KEY,
  UserID INT NOT NULL,
  StudyPlanID INT NOT NULL,
  Achievements TEXT NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL,
  FOREIGN KEY (UserID) REFERENCES User(UserID),
  FOREIGN KEY (StudyPlanID) REFERENCES StudyPlan(StudyPlanID)
);
-- Create Resource table
CREATE TABLE Resource (
  ResourceID INT PRIMARY KEY,
  Title VARCHAR(100) NOT NULL,
  URL VARCHAR(200) NOT NULL,
  Type VARCHAR(50) NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL);
-- Create Whiteboard table
CREATE TABLE Whiteboard (
  WhiteboardID INT PRIMARY KEY,
  UserID INT NOT NULL,
  Content TEXT NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL,
  FOREIGN KEY (UserID) REFERENCES User(UserID));
-- Create Video table
CREATE TABLE Video (
  VideoID INT PRIMARY KEY,
  UserID INT NOT NULL,
  WhiteboardID INT NOT NULL,
  VideoURL VARCHAR(200) NOT NULL,
  CreatedAt DATETIME NOT NULL,
  UpdatedAt DATETIME NOT NULL,
  FOREIGN KEY (UserID) REFERENCES User(UserID),
  FOREIGN KEY (WhiteboardID) REFERENCES Whiteboard(WhiteboardID));
-- Create Recommendation table
CREATE TABLE Recommendation (
  RecommendationID INT PRIMARY KEY,
  UserID INT NOT NULL,
  ResourceID INT NOT NULL,
  Reason TEXT NULL,
  CreatedAt DATETIME NOT NULL,
  FOREIGN KEY (UserID) REFERENCES User(UserID),
  FOREIGN KEY (ResourceID) REFERENCES Resource(ResourceID));
```


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7 Component Design

EDUassist's component design section describes each component's purpose, behavior, and interaction within the system. The breakdown provides an overview of each component's responsibilities, processing logic, and constraints, so users can better understand how they contribute to system functionality.

7.1 Common Functions

7.1.1 Sign In

```
signIn() {
   Enter username and password
   If username and password match records in the database
      Then redirect user to the home page
   Else
      Show error message “Incorrect username or password, please try again.”
}
```

7.1.2 Forgot Password

```
forgotPassword() {
   Enter email or phone number to receive a recovery code
   If the provided information exists in the database
      Then send a recovery code to the user
   Else
      Show error message “Invalid email or phone number.”
}
```

7.1.3 User Management

```
manageUser() {
   Admin selects a user account to view or modify
   If the action is update
      Then update user details in the database
   If the action is delete
      Then remove user account from the database
   Show success message “User details updated” or “User removed successfully.”
}
```


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7.1.4 System Monitoring

```
systemMonitoring() {
   Retrieve system metrics such as CPU usage, memory, and response times
   If any metric exceeds threshold
      Then generate an alert for system administrators
   Display metrics dashboard to administrators
}
```

7.1.5 Integration Management

```
integrationManagement() {
   Admin selects integration to enable, disable, or configure
   If integration is enabled/disabled
      Then update integration settings
   Show success message “Integration settings updated successfully.”
}
```

7.1.6 Personalized Study Plans

```
generateStudyPlan() {
   Enter subjects, goals, and study preferences
   Generate a study plan based on input
   Show the study plan to the user for confirmation
   If confirmed
      Then save study plan to user profile
   Else
      Allow user to modify preferences
}
```

7.1.7 Collaborative Whiteboard

```
collaborativeWhiteboard() {
   User initiates or joins a whiteboard session
   Allow users to draw, write, or upload images in real time
   If changes are made
      Then synchronize updates across all participants
   Display collaborative whiteboard to users
}
```


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7.1.8 Task Tracker

```
trackTask() {
   User adds or updates tasks with due dates and priority levels
   Show task list to user with status indicators
   If task is marked complete
      Then update status in the task tracker
}
```

7.1.9 Video Generation

```
generateVideo() {
   User selects topics and study materials
   Process selected material to generate a video summary
   If video is generated successfully
      Then display video to user for download or playback
   Else
      Show error message “Failed to generate video.”
}
```

7.1.10 AI Visual Learning Tools

```
aiVisualLearning() {
   User selects concept or topic
   Generate visual representation using AI
   Display visual representation to the user
}
```

7.1.11 Resource Recommendations

```
recommendResources() {
   Analyze user’s past study patterns and goals
   Suggest relevant resources (e.g., articles, videos)
   Display recommendations to user
}
```

7.1.12 Resource Sharing

```
shareResource() {
   Select resource to share and enter recipient(s)
   If recipient(s) found
      Then send resource to recipient(s)
   Show confirmation message “Resource shared successfully.”
}
```


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7.1.13 Notifications

```
sendNotification() {
   Generate notification based on user's tasks or reminders
   If notification is enabled
      Then send notification to user
}
```

7.1.14 Visual Representation of Text

```
visualizeText() {
   Enter text to convert into a visual format
   Process text to create an infographic or diagram
   Display the visual representation to user
}
```

7.1.15 Role Management

```
manageRoles() {
   Admin assigns roles and permissions to users
   If role assignment is updated
      Then save changes in the database
   Show confirmation "Role and permissions updated."
}
```


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8 Detailed System Design

This section provides a comprehensive overview of the components described in the System Architecture section of the EDUassist. It delves into both higher-level components and essential lower-level subcomponents that contribute to the system's functionality. Each subsection will discuss the attributes of specific software components, with a focus on classification, functionality, interfaces, and interactions. Most components described in the System Architecture section will require a more detailed discussion, and other lower-level components and subcomponents may need to be described as well. Each subsection of this section will refer to or contain a detailed description of a system software component.

8.1 Classification , Definition and Responsibilities

The classification, definition, and responsibilities of each component will be presented in this section. Shown in **Table 9** below.

Components	Classification	Definition and Responsibilities
Common Functions		
Sign In	Function	Handles user authentication for users.
Login Interface		
Redirect to Homepage User/admin	Function	Allows access to the system after successful authentication.
Forget My Password Interface		
Password Recovery	Function	Facilitates secure password reset/ recovery using email for both admin and users.
Settings Interface		
Personalize Preferences	Function	Allows to modify notification settings, themes, and other preferences.
admin Functions		
Admin Homepage Interface		
Display System Metrics	Function	Shows system activity logs, user actions, and task status to the admin . links to key admin functionalities like user management and reports.
User Management Tap		
Search/Add/Edit/Delete / update Users	Function	Enables the admin to search for, add, modify, or remove user accounts.
Reports Generation Tap		
Create Reports	Function	Generates custom reports based on user and system activity.
Configure System Settings	Function	Allows the admin to customize system-wide settings and integrations. Provides the ability to tailor system functionality to meet institutional needs and preferences.
User Functions		

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Dashboard Interface		
Quick Access Links	Function	Offers quick links to commonly used tools and features.
Task and Assignment Tracker Interface		
Monitor Progress	Function	Tracks completion status of tasks and assignments.
Add/Edit/Delete Tasks	Function	Allows users to manage tasks and assignments effectively.
Create Notification	Function	Allows the system to generate notifications for users regarding important updates. Keeps users informed about deadlines, task updates, and other relevant information.
Retrieve Notifications	Function	Enables users to view their list of notifications. Ensures users stay up-to-date with important information and reminders related to their studies.
Calendar Interface		
Synchronize Events	Function	Links tasks and deadlines with external calendar tools (i.e., Google Calendar).
Profile Interface		
Manage Personal Details	Function	Lets users update personal information like name, email, and phone.
Whiteboard Interface		
Collaborate and Annotate	Function	Facilitates real-time collaboration and note-taking with other users.
Receive AI Suggestions	Function	Provides AI-driven recommendations for enhancing productivity.
Generate Video	Function	Allows users to create short explanatory videos from whiteboard content. Supports users in summarizing and clarifying complex topics for better understanding.
Resource Library Interface		
Access Learning Resources		Provides curated learning materials like quizez, videos that been generate in whitboard, and articles.
Create Study Plans	Function	Lets users design custom study plans based on goals and syllabus.
Access Resources	Function	Allows users to access a library of educational resources relevant to their studies. Facilitates learning by providing users with valuable educational materials.
Track Progress	Function	Monitors user progress on their study plans and tasks. Provides users with insights into their academic performance and completion rates.
import Study Plans	Function	Enables users to import their study plans in various formats (e.g., PDF, word).

Table 9 Component's classification, definition, and responsibilities

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8.2 Constraints and composition

This section includes the components constraints and compositions (Pre-condition and Post condition) of each component in the **Table 10** below.

Components	Constraints	Pre-condition	Post-condition
Common Functions			
Sign In	Requires valid credentials.	User provides correct username and password.	User is authenticated and redirected to the homepage.
Login Interface			
Redirect to Homepage User/admin	User must be authenticated.	Successful login attempt.	Homepage is displayed for the user or admin.
Forget My Password Interface			
Password Recovery	Requires valid email address linked to account.	User submits the correct email.	Password reset instructions are sent.
Settings Interface			
Personalize Preferences	Changes apply only to current user session.	User selects preferences to modify	Preferences are updated and saved.
admin Functions			
Admin Homepage Interface			
Display System Metrics	Restricted to admin users only.	Admin logs into the system.	Dashboard with metrics is displayed.
User Management Tap			
Search/Add/Edit/Delete / update Users	Requires admin permissions.	Admin accesses the user management section.	User records are updated or managed.
Reports Generation Tap			
Create Reports	Requires sufficient user activity data.	Admin selects report parameters.	Custom report is generated.
Configure System Settings	Restricted to admins with special roles.	Admin accesses the configuration panel.	System settings are updated.
User Functions			
Dashboard Interface			
Quick Access Links	Limited to tools and features assigned to user.	User logs into the system.	Dashboard is displayed with relevant links.
Task and Assignment Tracker Interface			

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Monitor Progress	Tasks must exist in the system.	User logs in and views the tracker	Progress on tasks is updated and displayed.
Add/Edit/Delete Tasks	Actions apply only to assigned tasks.	User selects a task action.	Tasks are modified as specified by the user.
Create Notification	Notifications must be relevant to tasks.	User defines a new notification.	Notification is generated and linked to the task.
Retrieve Notifications	Notifications must exist for the user.	User opens the notification list.	Notifications are displayed.
Calendar Interface			
Synchronize Events	Limited to supported external tools (Google Calendar).	User authorizes synchronization.	Events are synchronized successfully.
Profile Interface			
Manage Personal Details	Only editable by the account owner.	User accesses the profile settings.	Personal details are updated.
Whiteboard Interface			
Collaborate and Annotate	Requires stable internet connection.	User opens the whiteboard tool.	Whiteboard session is active for collaboration.
Receive AI Suggestions	AI module must be operational.	User interacts with the whiteboard.	AI suggestions are displayed.
Generate Video	Requires sufficient storage space.	User selects video generation.	Video is generated and saved.
Resource Library Interface			
Access Learning Resources	Requires user-specific progress data.	User interacts with study resources.	Relevant learning resources are displayed.
Create Study Plans	Templates are predefined by the system.	User inputs study goals and syllabus.	Study plan is created and saved.
Access Resources	Resources must be relevant to the user's tasks.	User searches or browses for resources.	Matching resources are displayed.
Track Progress	User tasks must be linked to a study plan.	User interacts with the tracker.	Progress data is updated and displayed.
import Study Plans	Accepts specific file formats only.	User uploads a compatible file.	Study plan is imported and integrated.

Table 10 Constraints and composition

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8.3 Uses/Interactions

This section outlines the primary functions of the EDUassist, illustrated through sequence diagrams in **Figures 73 to 92**. These diagrams depict the common functionalities, as well as the specific functionalities for both Admin and User roles.

- Sign In :

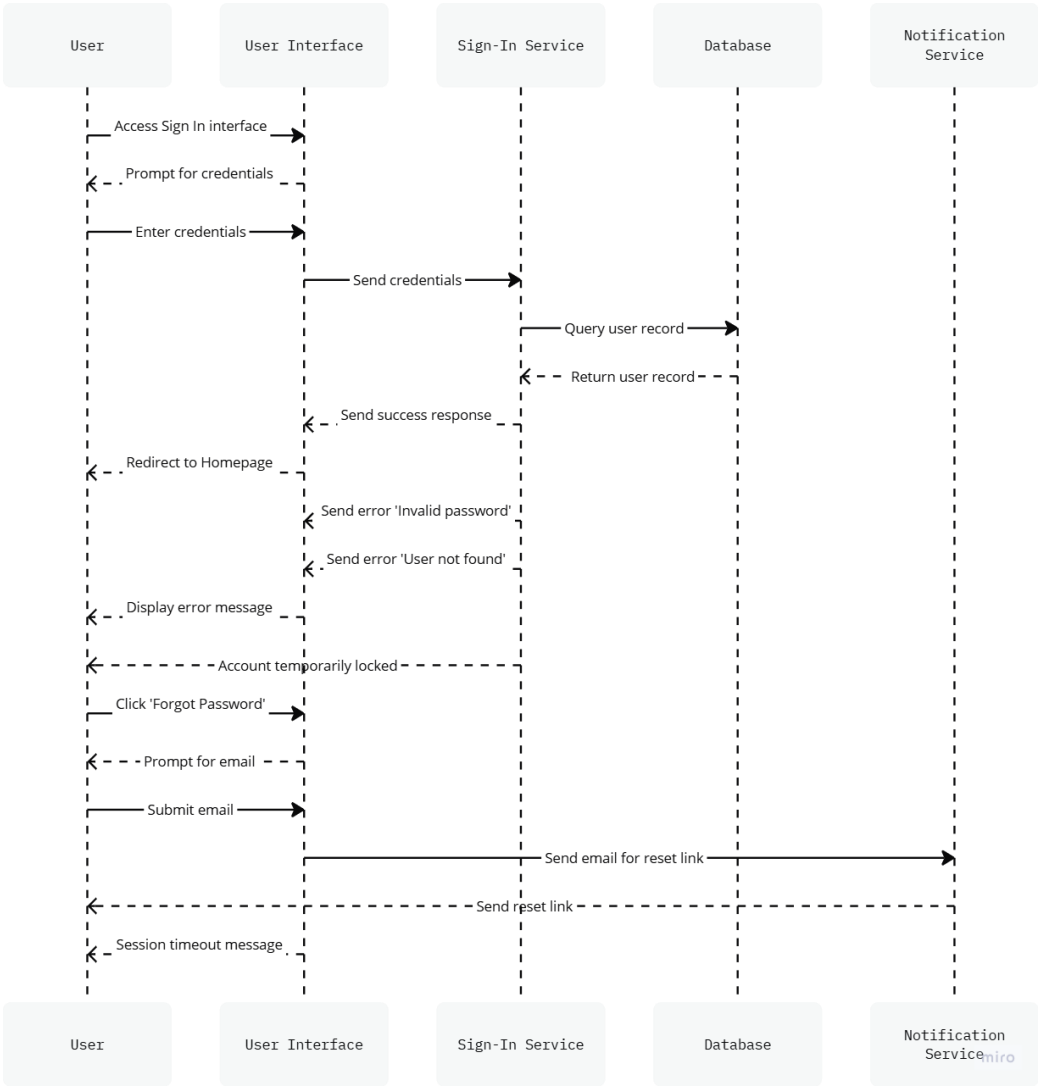


Figure 73 Sign in sequence diagram

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- Log in

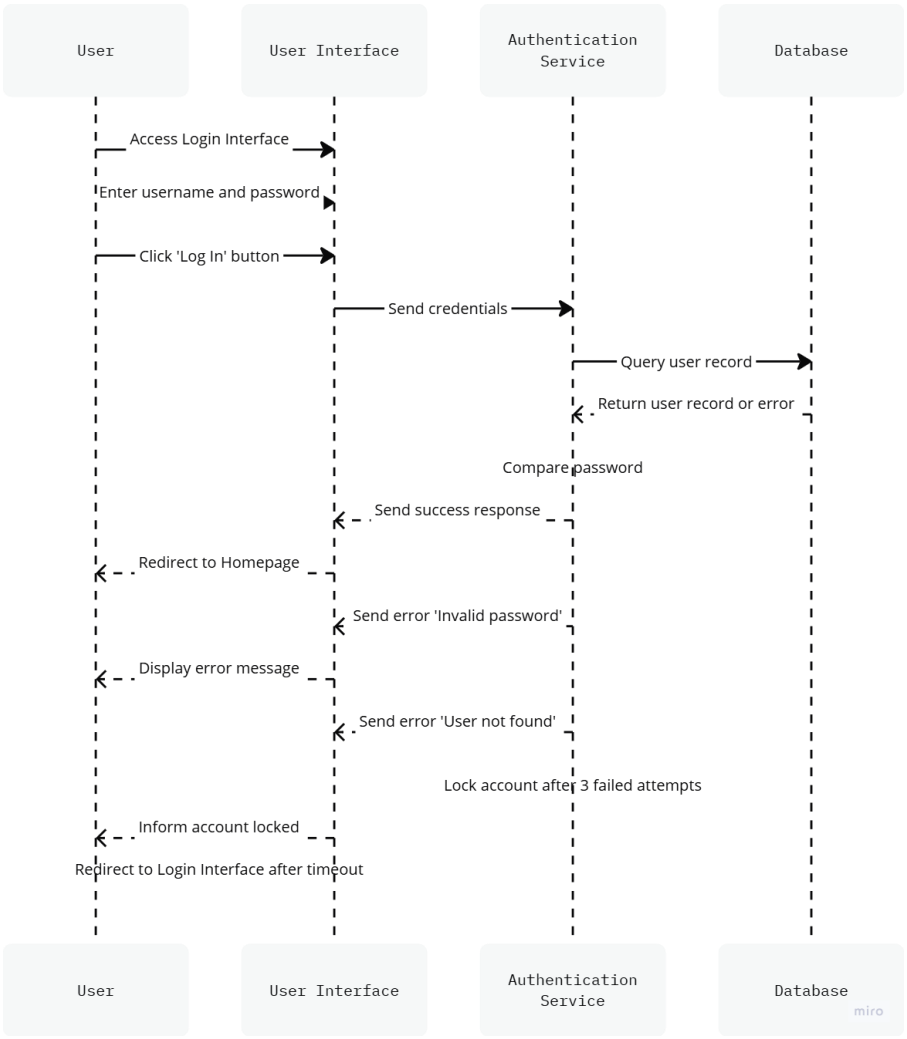


Figure 74 Login sequence diagram

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- Password Recovery :

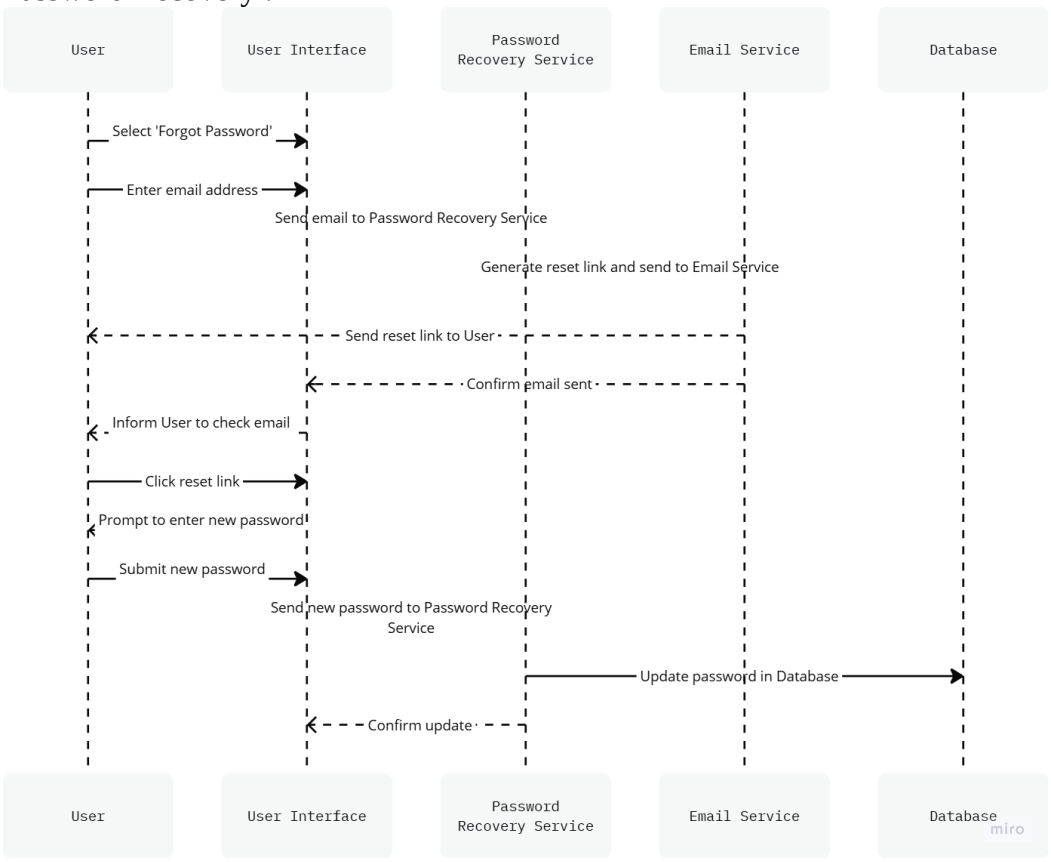


Figure 75 Password Recovery sequence diagram

- Personalize Preferences :

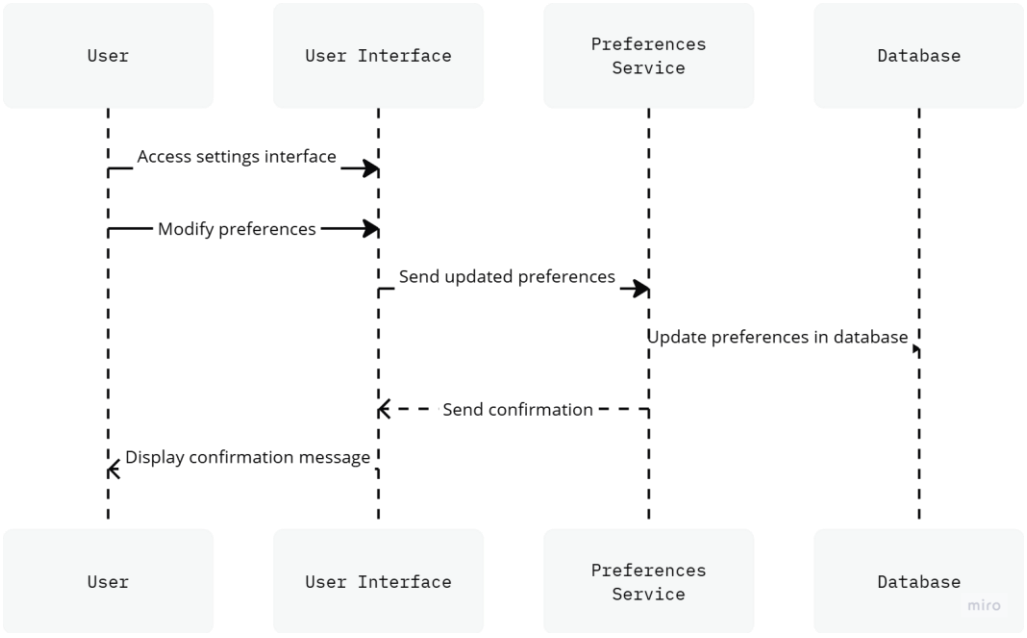


Figure 76 Personalize Preferences sequence diagram

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- Display System Metrics:

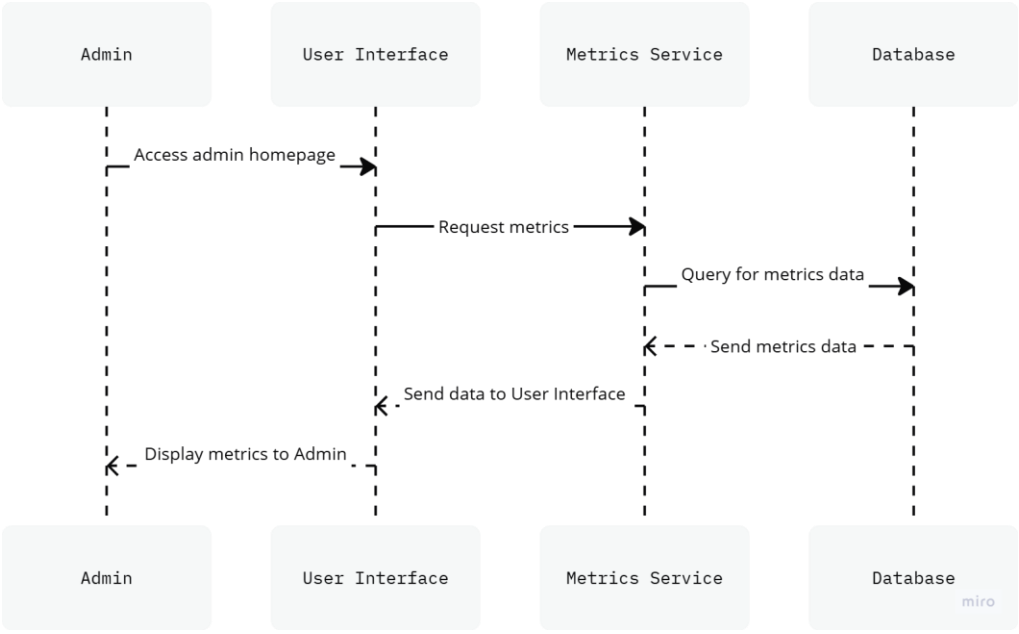


Figure 77 Display System Metrics sequence diagram

- Search/Add/Edit/Delete Users :

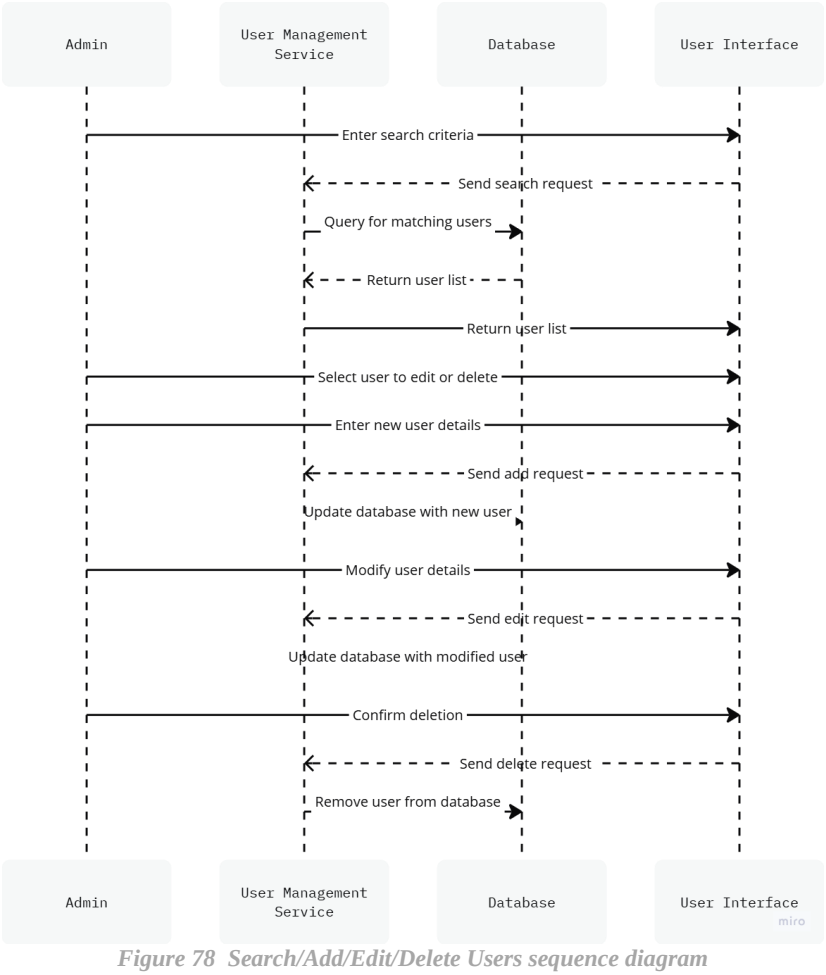


Figure 78 Search/Add/Edit/Delete Users sequence diagram

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- Create Reports :

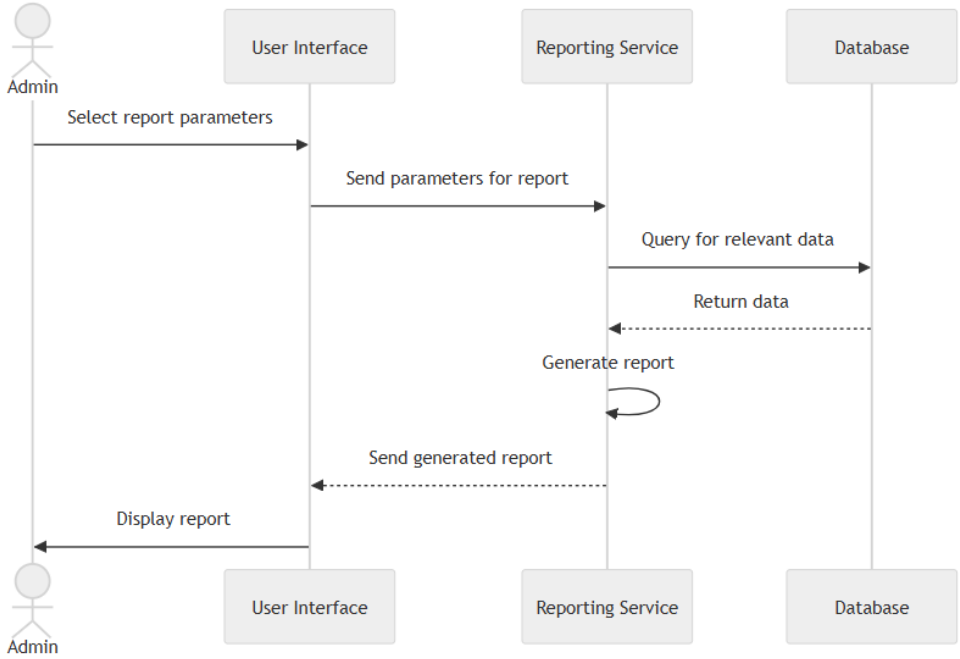


Figure 79 Create Reports sequence diagram

- Configure System Settings:

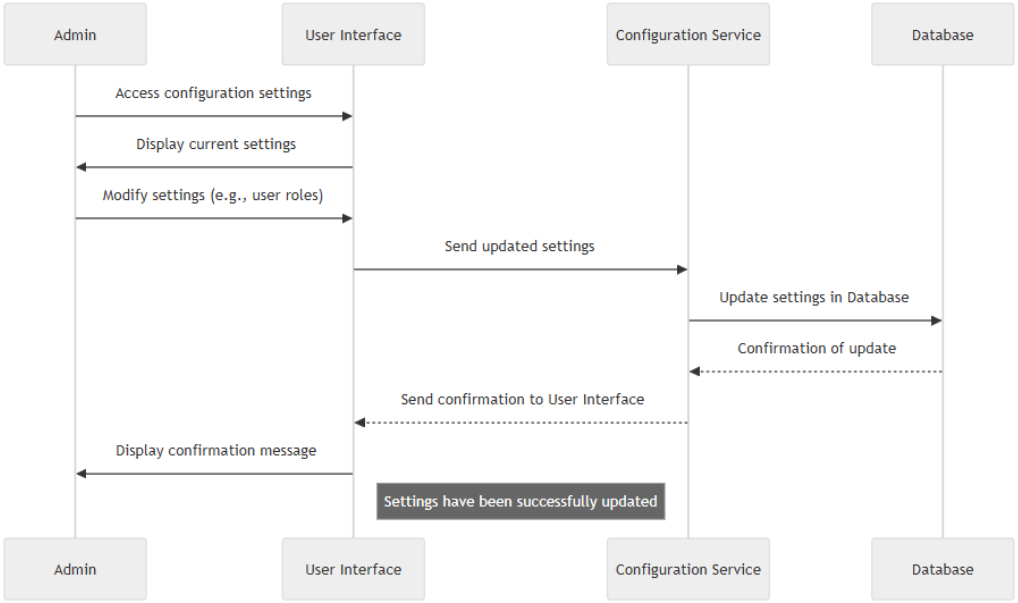


Figure 80 Configure System Settings sequence diagram

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- Dashboard Interface:

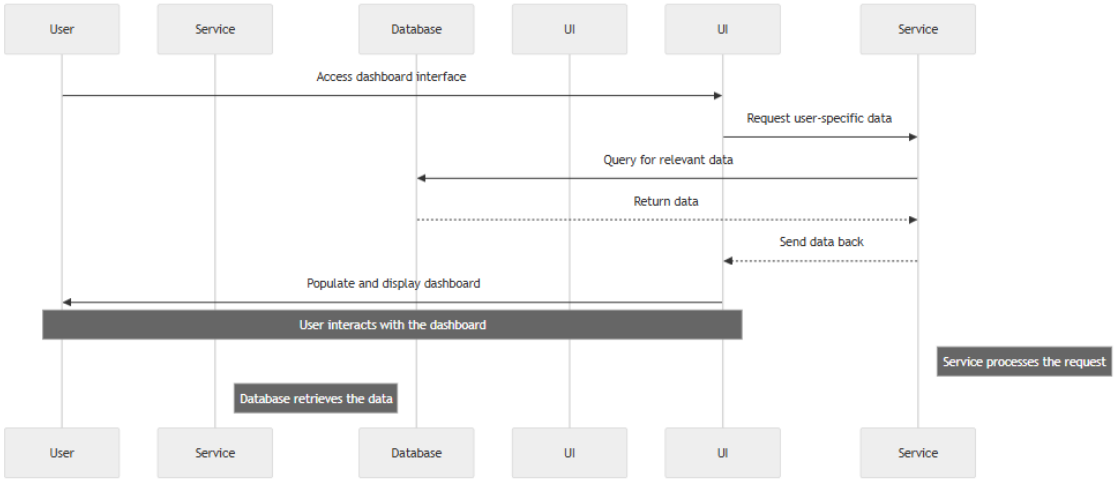


Figure 81 Dashboard Interface sequence diagram

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- Task and Assignment Tracker :

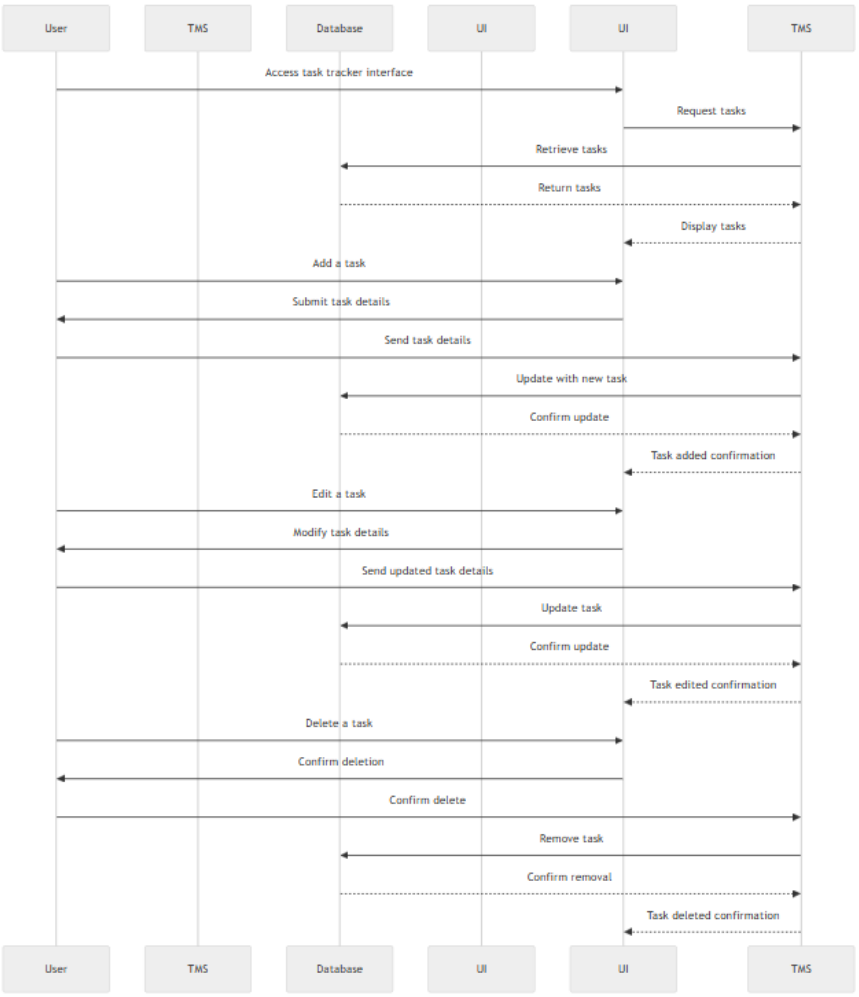


Figure 82 Task and Assignment Tracker sequence diagram

- Create Notification :

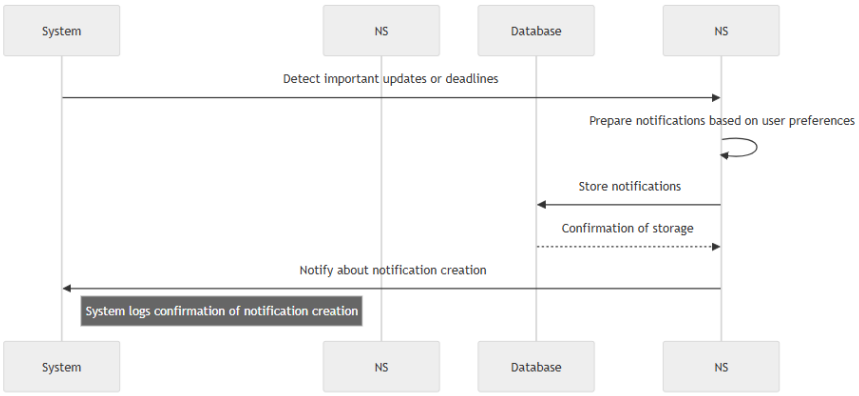


Figure 83 Create Notification sequence diagram

- Retrieve Notifications:

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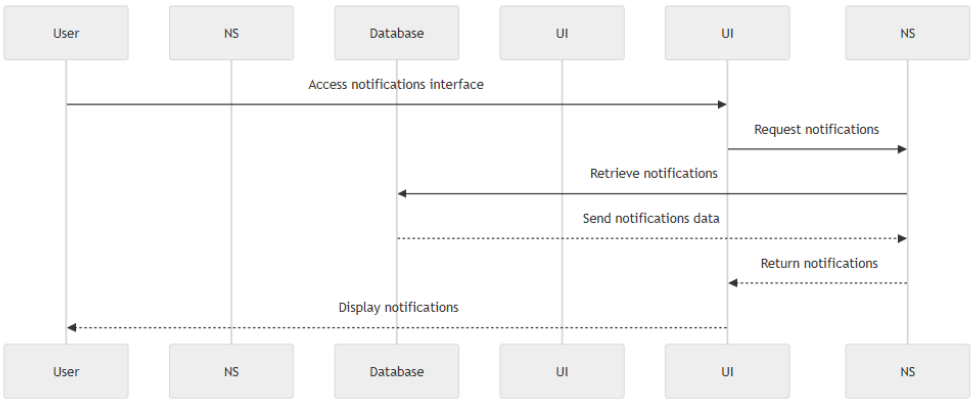


Figure 84 Retrieve Notifications sequence diagram

- Synchronize Events:

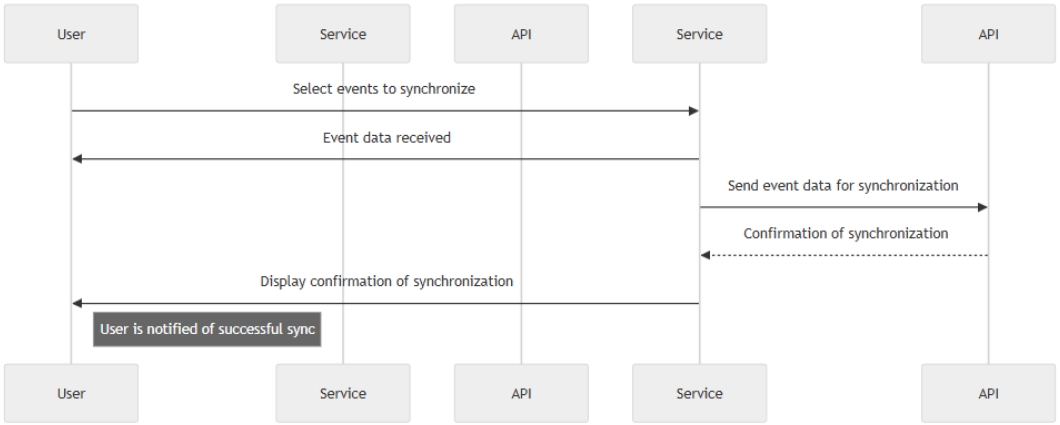


Figure 85 Synchronize Events sequence diagram

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- Manage Personal Details :

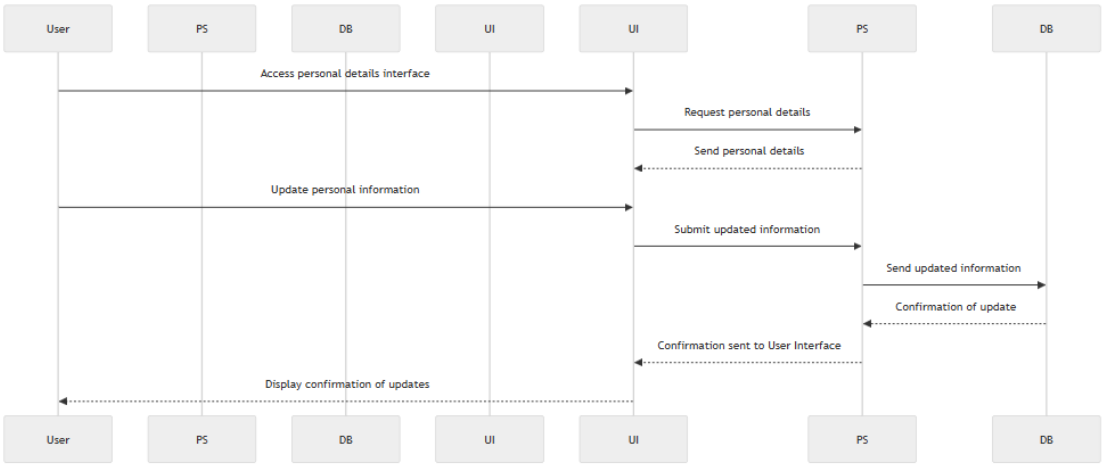


Figure 86 Manage Personal Details sequence diagram

- Collaborate and Annotate :

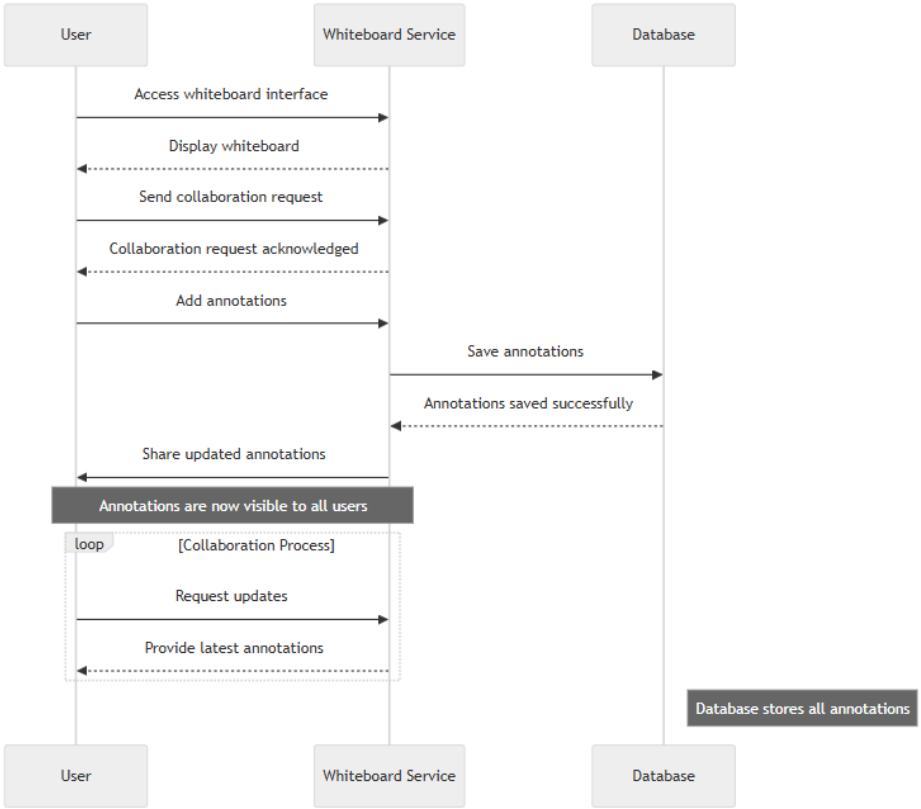


Figure 87 Collaborate and Annotate sequence diagram

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- Receive AI Suggestions:

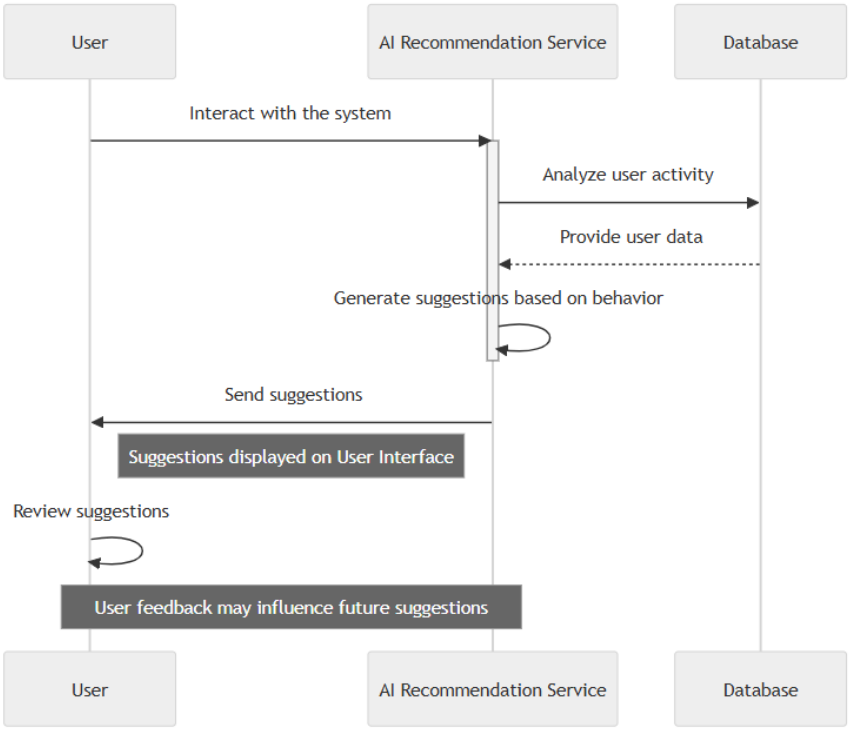


Figure 88 Receive AI Suggestions sequence diagram

- Generate Video :

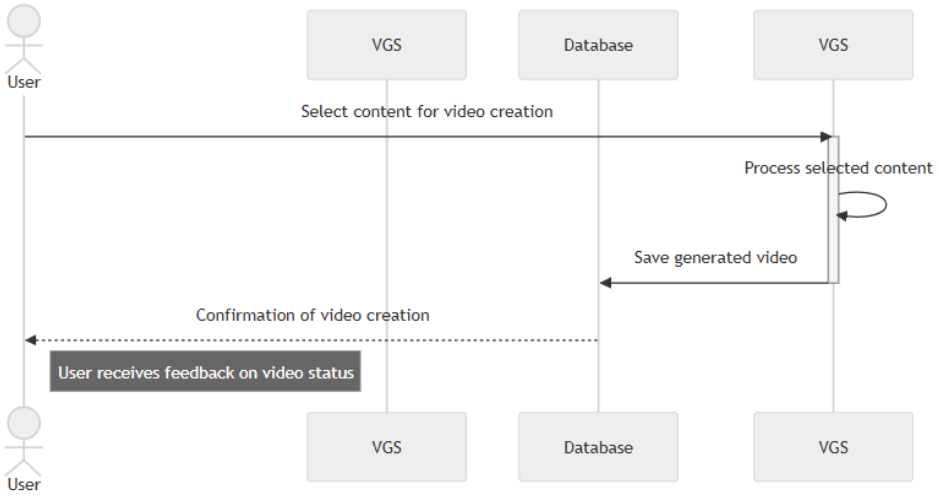


Figure 89 Generate Video sequence diagram

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- Access Learning Resources :

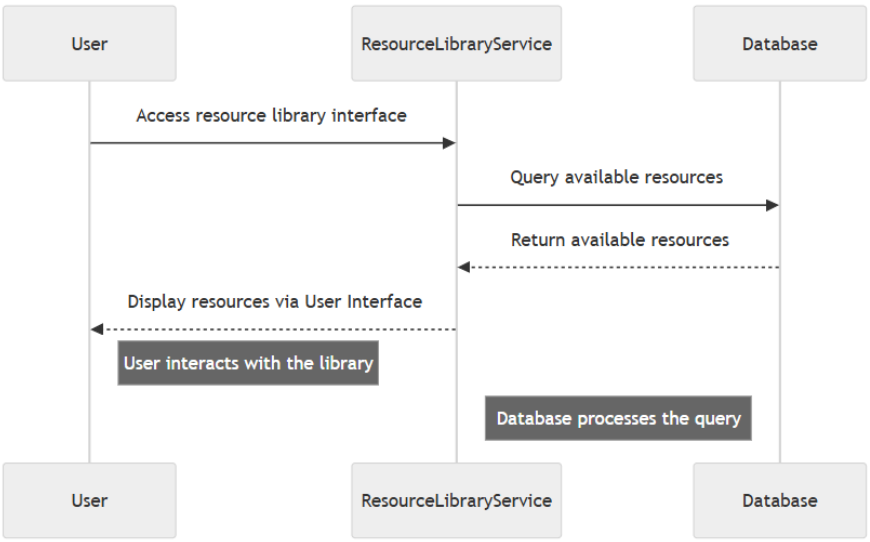


Figure 90 Access Learning Resources sequence diagram

- Create Study Plans :

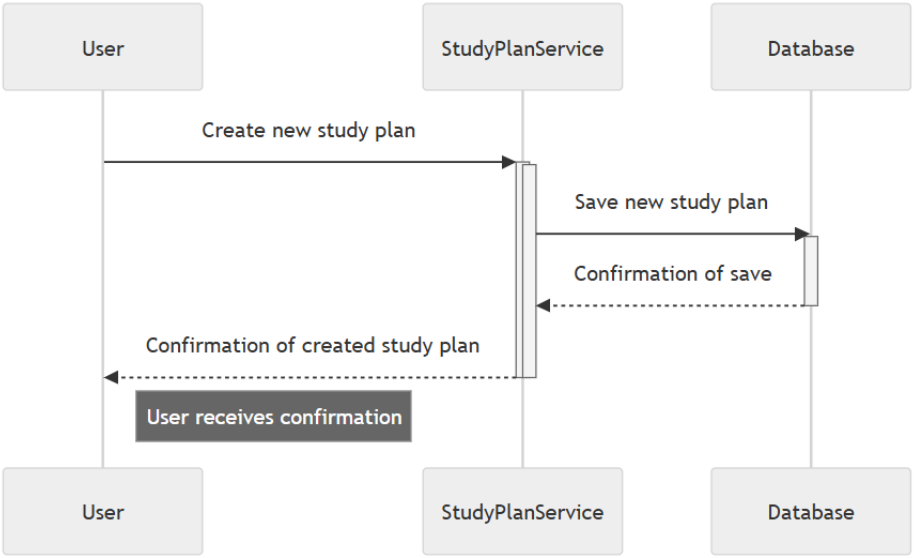


Figure 91 Create Study Plans sequence diagram

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- Import Study Plans :

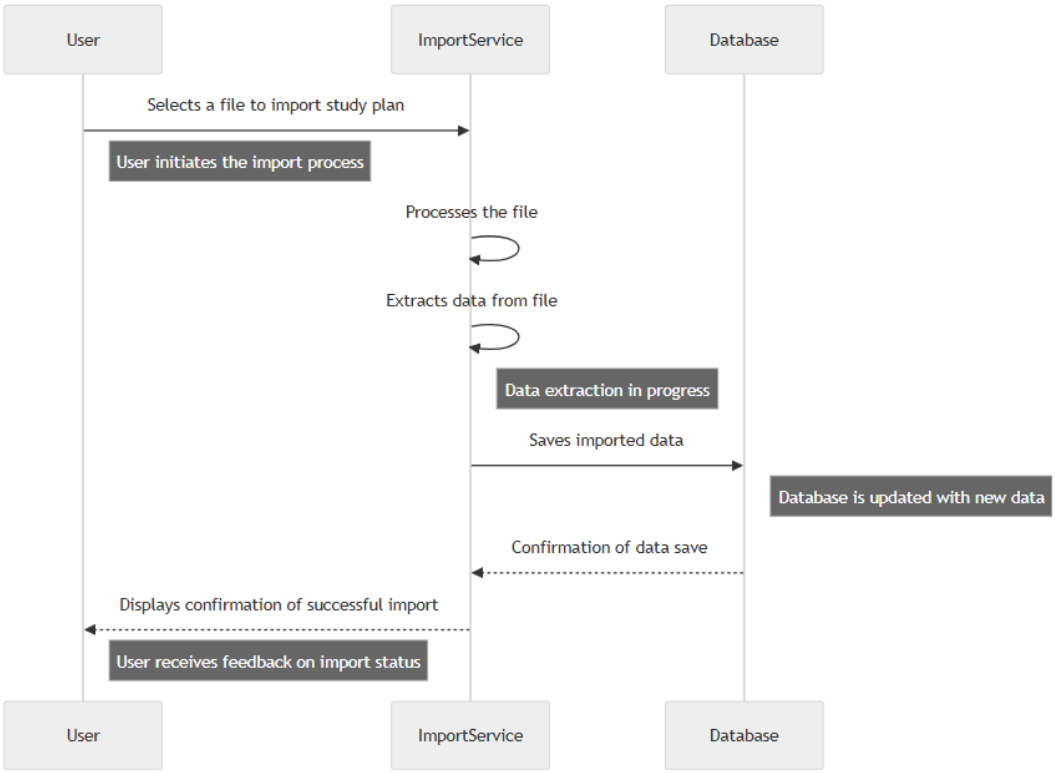


Figure 92 Import Study Plans sequence diagram

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8.4 Resources

This section will delineate the external entity resources pertinent to the EDUassist.

Table 11 below illustrates the specifications of the resources that are managed, influenced, or necessary for this entity.

Type	Resource
Databases	MySQL database is used for secure and efficient data storage and retrieval.
External APIs	Google Calendar API for scheduling, AI APIs for intelligent recommendations. ○ **Google Calendar API**: For managing user events and reminders. ○ **Google Drive API**: For storing and retrieving user documents and study materials.
Security Mechanisms	HTTPS, OTP generation, and secure password storage using encryption protocols.
Operating System	iOS and Android .
Memory Storage	40MB to 4 GB
Internet connection	Network
server	Google Cloud Platform (GCP) and Amazon Web Services (AWS)
Cloud Storage	Google Cloud Storage

Table 11 Resources specification

Table 12 shows the rase conditions situations that may occur and how could be resolved.

Race Condition	Solution
Multiple users attempt to edit the same study plan simultaneously.	Implement a locking mechanism to prevent concurrent edits to the same resource.
A user marks a task as completed while the Task Tracker is updating notifications.	Use transaction handling to ensure atomicity and consistency during updates.
Users simultaneously creating duplicate tasks in the Task Tracker.	Introduce unique task identifiers to prevent duplication.
Conflicting updates to personalized study plans by multiple devices.	Synchronize changes using a real-time conflict resolution protocol.

Table 12 Rase conditions

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Table 13 shows deadlock situations that may occur and how could be resolved.

Deadlock Situation	Description	Solution
Resource recommendation request conflicts with study plan update.	The Resource Library and Study Plan Management modules lock resources while waiting for each other.	Implement a timeout-based deadlock detection mechanism to abort and retry conflicting operations.
Simultaneous access to a single user account by admin and user.	Both parties attempt to update account details concurrently.	Prioritize admin operations and notify users of the pending update status.
AI video feature overload during video generation.	Simultaneous requests to the video generation interface causing resource lock.	Implement a request queue and distribute load using load balancers.

Table 13 Deadlock

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8.5 Processing

This subsection presents the components of EDUassist, along with their descriptions, inputs, and outputs

- **Common Functions:**

8.5.1 Sign In:

Table 14 below shows the description, input, output, and constraint of the Sign in component.

Description	Allows users to authenticate and access their account.
Input	Enter valid username and password.
Output	Redirects to the homepage or displays an error message if authentication fails.
Constraint	If the user provides invalid credentials, then access is denied, and an error message is shown.

Table 14 Sign in Functionality Process

8.5.2 Log in :

Table 15 below shows the description, input, output, and constraint of the Log in component.

Description	Redirects authenticated users to the relevant homepage (admin or user).
Input	Successful login attempt.
Output	Displays the homepage specific to the user role.
Constraint	If the user is not authenticated, then the redirection process fails, and they remain on the login page.

Table 15 : Login Functionality Process

8.5.3 Password Recovery :

Table 16 below shows the description, input, output, and constraint of the Password Recovery component.

Description	Enables users to recover access to their account via a password reset process.
Input	Provide an email/phone number linked to the account.
Output	Password reset link is sent to the provided email address.
Constraint	If the provided email is not registered, then the system cannot send a password reset link.

Table 16 Password Recovery

8.5.4 Personalize Preferences :

Table 17 below shows the description, input, output, and constraint of the Personalize Preferences component.

Description	Allows users to customize settings based on their preferences.
Input	User selects options in the settings interface.
Output	Preferences are updated and saved.
Constraint	If the user's session ends without saving, then the changes are discarded.

Table 17 Personalize Preferences

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- **Admin Functions :**

8.5.5 Display System Metrics:

Table 18 below shows the description, input, output, and constraint of the Display System Metrics component.

Description	Provides an overview of system activity and performance metrics to the admin.
Input	Admin logs into the dashboard.
Output	Metrics and reports are displayed on the admin homepage.
Constraint	If the user is not an admin, then access to system metrics is denied.

Table 18 Display System Metrics

8.5.6 Search/Add/Edit/Delete/Update Users :

Table 19 below shows the description, input, output, and constraint of the Search/Add/Edit/Delete/Update Users component.

Description	Allows admin to manage user accounts by performing CRUD operations.
Input	Admin provides search criteria or account details for modification.
Output	User accounts are created, updated, or deleted as specified.
Constraint	If the admin lacks necessary permissions, then the user management functionality is restricted.

Table 19 Search/Add/Edit/Delete/Update Users

8.5.7 Create Reports

Table 20 below shows the description, input, output, and constraint of the Personalize Preferences component.

Description	Allows admin to generate custom reports based on selected parameters.
Input	Admin selects report parameters.
Output	Custom report is generated and displayed.
Constraint	If there is insufficient user activity data, then the report cannot be generated.

Table 20 Create Reports

8.5.8 Configure System Settings :

Table 21 below shows the description, input, output, and constraint of the Configure System Settings component.

Description	Enables admins to configure system-wide settings.
Input	Admin accesses the configuration panel and modifies settings.
Output	System settings are updated.
Constraint	If the admin does not have a special role, then access to configuration is denied.

Table 21 Configure System Settings

- **User Functions**

8.5.9 Quick Access Links :

Table 22 below shows the description, input, output, and constraint of the Quick Access Links component.

Description	Provides users with quick access to tools and features.
Input	User logs in and navigates to the dashboard.
Output	Dashboard with relevant links is displayed.
Constraint	If tools or features are not assigned to the user, then they cannot access those links.

Table 22 Quick Access Links

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8.5.10 Monitor Progress:

Table 23 below shows the description, input, output, and constraint of the Monitor Progress component.

Description	Displays progress on tasks and assignments.
Input	User logs in and views the tracker.
Output	Progress updates and status are displayed.
Constraint	If no tasks exist in the system, then no progress can be displayed.

Table 23 Monitor Progress

8.5.11 Add/Edit/Delete Tasks :

Table 24 below shows the description, input, output, and constraint of the Add/Edit/Delete Tasks component.

Description	Enables users to create, modify, or delete tasks.
Input	User selects a task action.
Output	Task is created, updated, or deleted.
Constraint	If the task does not belong to the user, then they cannot modify it.

Table 24 Add/Edit/Delete Tasks

8.5.12 Create Notification:

Table 25 below shows the description, input, output, and constraint of the Create Notification component.

Description	Allows users to create task-related notifications.
Input	User defines a new notification.
Output	Notification is generated and linked to the task.
Constraint	If the notification is not related to a valid task, then it cannot be created.

Table 25 Create Notification

8.5.13 Retrieve Notifications :

Table 26 below shows the description, input, output, and constraint of the Retrieve Notifications component.

Description	Displays notifications relevant to the user.
Input	User opens the notification list.
Output	Notifications are displayed.
Constraint	If there are no notifications for the user, then the list remains empty.

Table 26 Retrieve Notifications

8.5.14 Synchronize Events :

Table 27 below shows the description, input, output, and constraint of the Synchronize Events component.

Description	Synchronizes the user's calendar with supported external tools (Google Calendar).
Input	User authorizes synchronization by sign in google email account
Output	Events are synchronized successfully.
Constraint	If the user denies authorization or not signin with their google account , then synchronization fails.

Table 27 Synchronize Events

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8.5.15 Manage Personal Details :

Table 28 below shows the description, input, output, and constraint of the Manage Personal Details component.

Description	Allows users to manage and update their personal information, such as name, email, and phone number.
Input	User provides new personal details through the profile settings interface.
Output	Updated personal details are saved and displayed in the user's profile.
Constraint	If the user provides invalid data (e.g., an incorrect email format or empty required fields), then the system prompts for correction and rejects the submission until valid data is provided.

Table 28 Manage Personal Details

8.5.16 Collaborate and Annotate :

Table 29 below shows the description, input, output, and constraint of the Collaborate and Annotate component.

Description	Provides a collaborative space for users to annotate and share content.
Input	User opens the whiteboard tool.
Output	Active collaboration session on the whiteboard.
Constraint	If the internet connection is unstable, then the collaboration session cannot proceed.

Table 29 Collaborate and Annotate

8.5.17 Receive AI Suggestions :

Table 30 below shows the description, input, output, and constraint of the Receive AI Suggestions component.

Description	Displays AI-generated suggestions on the whiteboard interface.
Input	User interacts with the whiteboard.
Output	AI suggestions are displayed.
Constraint	If the AI module is not operational, then no suggestions are displayed.

Table 30 Receive AI Suggestions

8.5.18 Generate Video:

Table 31 below shows the description, input, output, and constraint of the Generate Video component.

Description	Allows users to generate videos based on whiteboard content.
Input	User selects video generation.
Output	Video is generated and saved.
Constraint	If there is insufficient storage space, then video generation fails.

Table 31 Generate Video

8.5.19Access Learning Resources :

Table 32 below shows the description, input, output, and constraint of the Access Learning Resources component.

Description	Displays learning resources tailored to the user's progress..
Input	User interacts with the study resources interface.
Output	Relevant learning resources are displayed.
Constraint	If user progress data is unavailable, then relevant resources cannot be displayed.

Table 32 Access Learning Resources

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8.5.20 Create Study Plans :

Table 33 below shows the description, input, output, and constraint of the Create Study Plans component.

Description	Allows users to create personalized study plans.
Input	User provides study goals and syllabus information.
Output	Study plan is created and saved.
Constraint	If the study plan template is not predefined, then it cannot be created.

Table 33 Create Study Plans

8.5.21 Access Resources :

Table 34 below shows the description, input, output, and constraint of the Access Resources component.

Description	Provides access to study resources relevant to the user's tasks.
Input	User searches or browses for resources.
Output	Matching resources are displayed.
Constraint	If no relevant resources exist, then the search result is empty.

Table 34 Access Resources

8.5.22 Track Progress :

Table 35 below shows the description, input, output, and constraint of the Track Progress component.

Description	Tracks the user's progress on assigned tasks and study plans.
Input	User interacts with the progress tracker.
Output	Progress data is updated and displayed
Constraint	If tasks are not linked to a study plan, then progress tracking is unavailable.

Table 35 Track Progress

8.5.23 Import Study Plans :

Table 36 below shows the description, input, output, and constraint of the Import Study Plans component.

Description	Allows users to upload and integrate external study plans
Input	User uploads a compatible file.
Output	Study plan is imported and integrated into the system.
Constraint	If the file format is not supported, then the study plan cannot be imported.

Table 36 Import Study Plans

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8.6 Interface/Exports

This section outlines the primary activities associated with the EDUassist system, as detailed in **Tables 37 to 50**. The functionality of the EDUassist will be clearly defined to ensure that it operates as intended. It is essential to fully understand the classifications, definitions, responsibilities, constraints, and compositions of each system component. The constraints, pre-conditions, and post-conditions are discussed in section 8.2, while section 8.5 covers the input, output, and descriptive specifications of the EDUassist system.

- **Sign In:**

Attribute	Details
Description	Enables users to securely sign in to the system using their credentials (username and password).
Data	Username and password provided by the user.
Stimulate	User inputs valid credentials into the sign-in form and clicks the "Sign In" button.
Response	The system authenticates the user and redirects them to their personalized dashboard upon successful Sign in .
Comment	If invalid credentials are entered, the system will display an error message and allow retry attempts.

Table 37 Sign In

- **Login:**

Attribute	Details
Description	Authenticates the user and redirects them to the appropriate homepage (admin or user).
Data	Credentials verified from the user database.
Stimulate	Successful sign-in triggers this interface to display the homepage.
Response	Redirects users to their designated homepage based on their role.
Comment	If authentication fails, the user is sent back to the sign-in interface with an error notification.

Table 38 Login

- **Forget My Password:**

Attribute	Details
Description	Provides a mechanism for users to reset their password by verifying their email.
Data	User's registered email address.
Stimulate	User submits a password recovery request through the form.
Response	Sends a password reset link or code to the provided email.
Comment	If the email does not exist in the database, the system will notify the user.

Table 39 Forget My Password

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- **Settings:**

Attribute	Details
Description	Allows users to update and personalize their application preferences, such as theme, language, and notification settings.
Data	User-selected preferences.
Stimulate	User modifies settings through the interface.
Response	Saves changes and applies them immediately or on the next session.
Comment	If no changes are made, the default settings remain active.

Table 40 Settings

- **Admin Homepage:**

Attribute	Details
Description	Provides administrators access to system metrics and management tools.
Data	Real-time data on system performance and user activity.
Stimulate	Admin logs into the admin dashboard.
Response	Displays widgets and reports summarizing system statistics.
Comment	Metrics are updated dynamically or upon admin request.

Table 41 Admin Homepage

- **User Management:**

Attribute	Details
Description	Enables administrators to search, add, edit, delete, or update user accounts.
Data	User records stored in the database.
Stimulate	Admin interacts with user management features.
Response	Updates the user database accordingly.
Comment	Unauthorized changes are restricted by system permissions.

Table 42 User Management

- **Reports Generation:**

Attribute	Details
Description	Allows administrators to create and configure reports based on user activity and system data.
Data	Selected parameters for report generation.
Stimulate	Admin defines the report criteria and generates the report.
Response	The system outputs a custom report in the desired format.
Comment	Reports can only be generated if sufficient data exists.

Table 43 Reports Generation

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- **Dashboard:**

Attribute	Details
Description	Displays quick access links to tools, notifications, and features relevant to the user.
Data	User role and assigned tasks.
Stimulate	User logs into the system and views the dashboard.
Response	Displays a tailored dashboard with relevant options.
Comment	Tools and links are displayed based on user permissions.

Table 44 Dashboard

- **Task and Assignment Tracker:**

Attribute	Details
Description	Allows users to monitor and manage their tasks and assignments.
Data	Task details, progress metrics, and deadlines.
Stimulate	User interacts with tasks via the tracker interface.
Response	Updates the task status or displays relevant information.
Comment	Tasks cannot be updated if they are marked as "Completed" or "Archived."

Table 45 Task and Assignment Tracker

- **Calendar:**

Attribute	Details
Description	Synchronizes events with external calendars like Google Calendar to manage schedules.
Data	Calendar event data.
Stimulate	User authorizes synchronization with an external calendar.
Response	Updates the user's calendar with synchronized events.
Comment	Synchronization is limited to supported calendar platforms.

Table 46 Calendar

- **Profile:**

Attribute	Details
Description	Allows users to manage personal details such as name, email, and contact information.
Data	User profile details.
Stimulate	User accesses the profile interface and edits their information.
Response	Saves updated profile details.
Comment	If incorrect details are entered, the system prompts the user to revise them.

Table 47 Profile

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- **Whiteboard:**

Attribute	Details
Description	Facilitates collaboration by allowing users to share ideas and annotate in real time.
Data	Shared whiteboard content.
Stimulate	User opens the whiteboard and interacts with tools or participants.
Response	Updates the shared whiteboard in real time.
Comment	Requires an active internet connection for collaboration.

Table 48 Whiteboard

- **Resource Library:**

Attribute	Details
Description	Provides access to study materials, tools, and resources for learning.
Data	Resource metadata and user progress.
Stimulate	User browses, searches, or downloads resources.
Response	Displays resources relevant to the user's needs.
Comment	Resources are filtered based on user preferences and role.

Table 49 Resource Library

- **Import Study Plans:**

Attribute	Details
Description	Allows users to upload and integrate external study plans.
Data	Uploaded file in a supported format.
Stimulate	User uploads a study plan file.
Response	Integrates the study plan into the system.
Comment	If the file format is unsupported, the import fails, and the user is notified.

Table 50 Import Study Plan

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8.7 Detailed Subsystem Design

This section outlines the main components of the EDUassist and illustrates how data flows between them. The flow chart below from **Figure 93 to 105** depicts the interactions within the system for both students and administrators.

- **Login Interface**

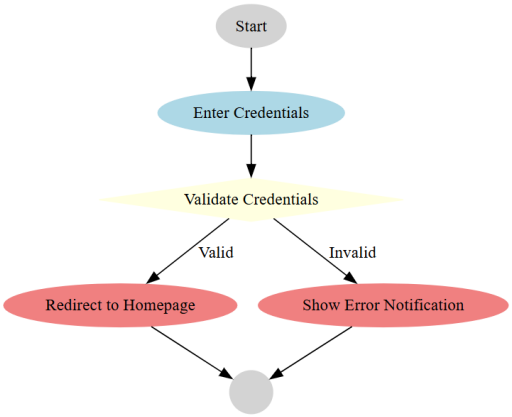


Figure 93

- **Forget My Password**

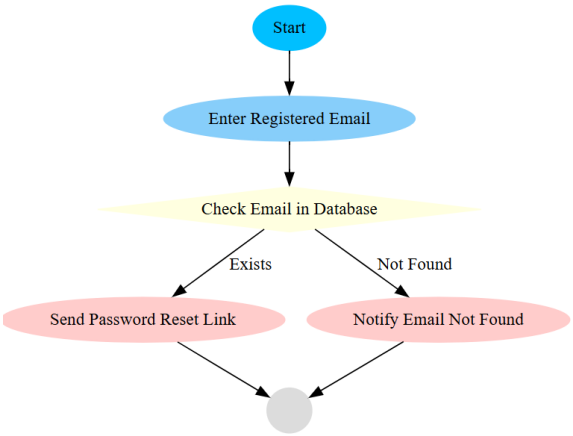


Figure 94

- **Settings Interface**

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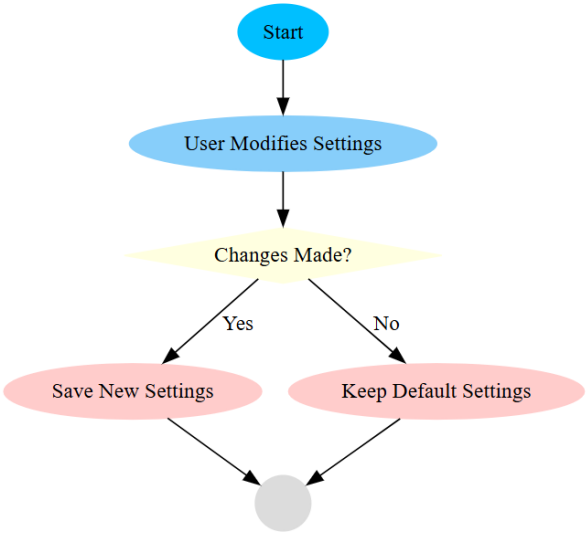


Figure 95

- **Admin Homepage**

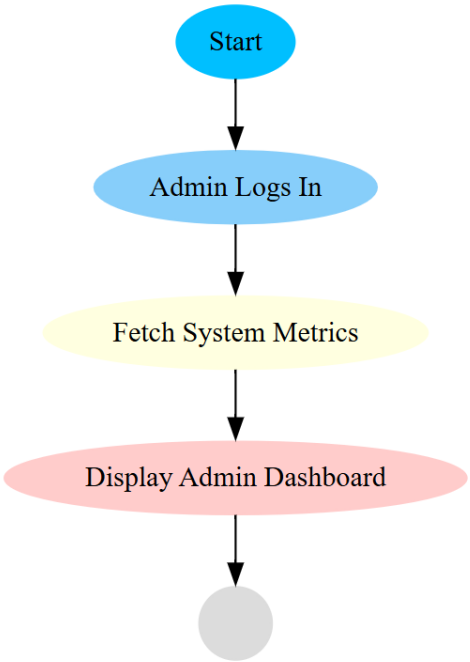


Figure 96

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- **User Management Tab**

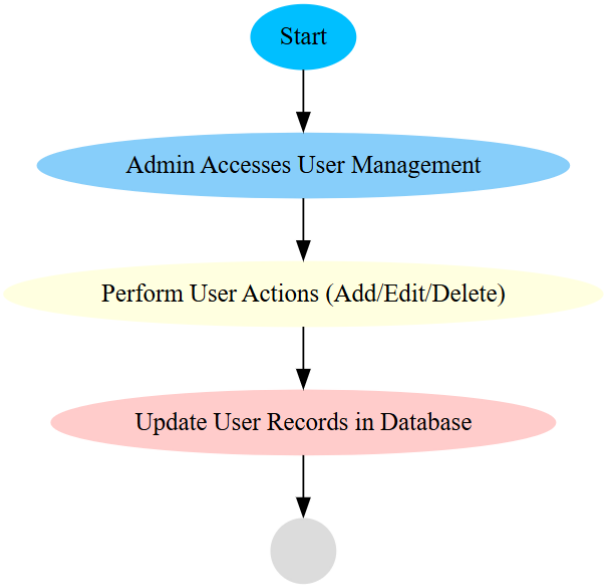


Figure 97

- **Reports Generation**

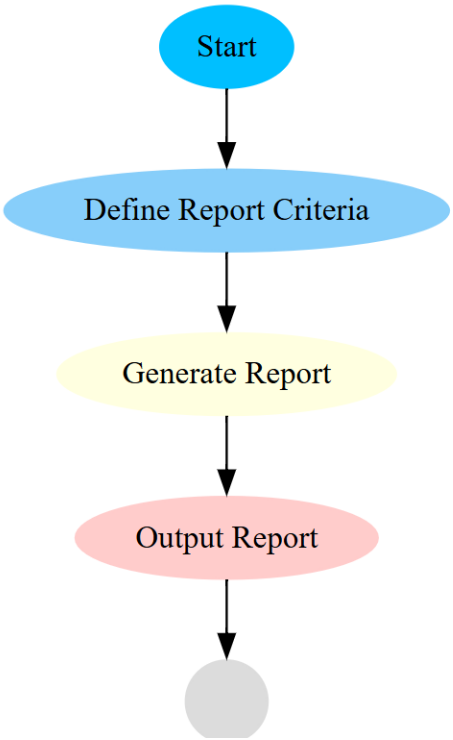


Figure 98

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- **Dashboard**

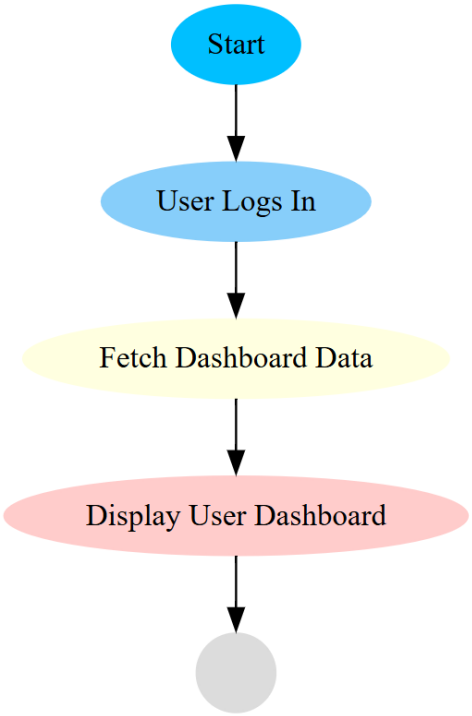


Figure 99

- **Task and Assignment Tracker**

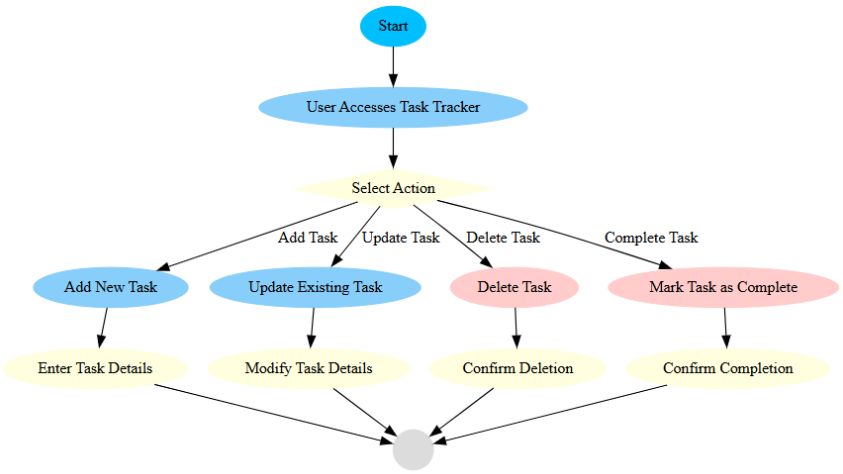


Figure 100

- **Calendar**

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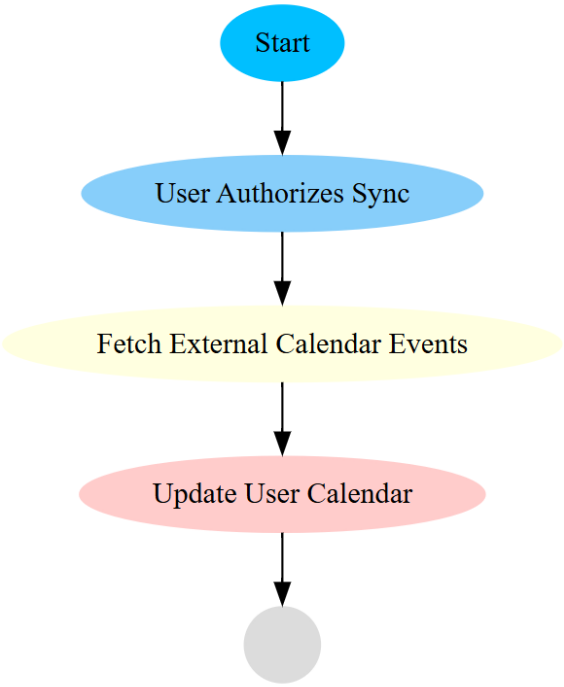


Figure 101

- **Profile**

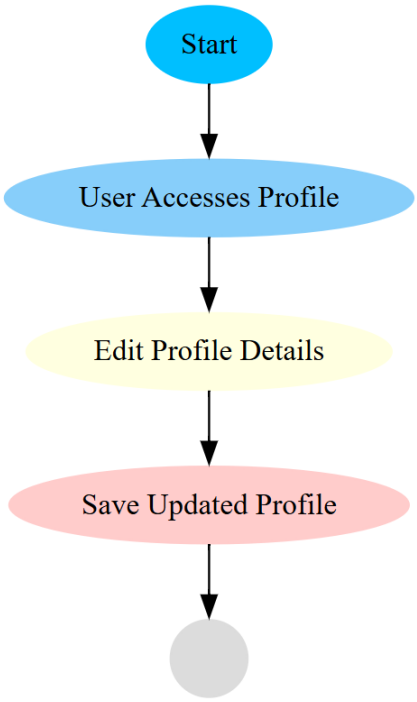


Figure 102

- **Whiteboard**

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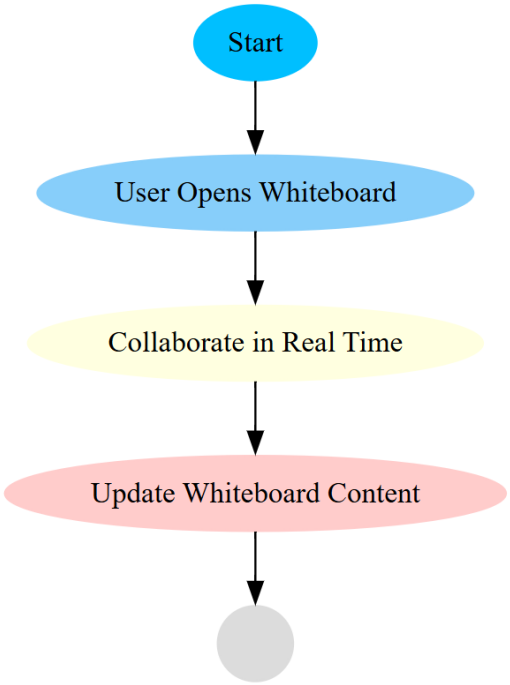


Figure 103

- **Resource Library**

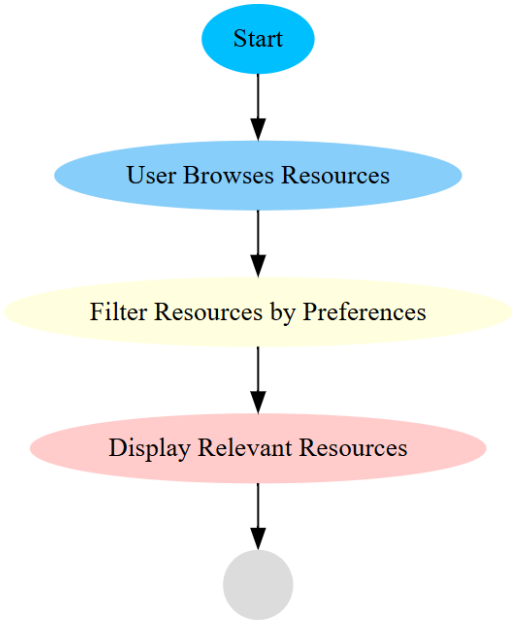


Figure 104

- **Import Study Plan**

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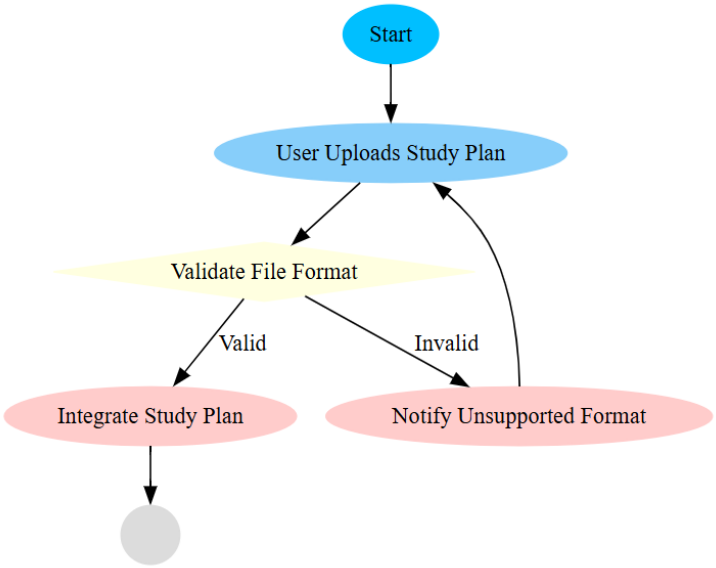


Figure 105

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9 Other Design Features

All the design features of the EDUassist project have been illustrated in a detailed manner in the above Sections and Subsections.

10 Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) is a tool that connects the functional requirements from the System Requirements Specification (SRS) to the design components in the System Design Description (SDS). It ensures that every requirement is addressed in the system's design, helping us track progress and verify that nothing is overlooked. This makes it easier to validate the design and ensure the system meets its goals.

Association ID in SRS	Technical Assumption(s) and/or Customer Need(s)	Functional Requirement	User	Association ID in SDS	System Components
3.2.1.1	Admin and students can log in securely using valid credentials.	Sign In	Admin/Student	7.1.1	Authentication Module
3.2.1.2	Admin and students can reset their password via email.	Forgot Password	Admin/Student	7.1.2	Password Recovery Module
3.2.2.1	Admins can manage user accounts: create, modify, and delete users.	User Management	Admin	7.2.1	User Management Module
3.2.2.2	Admins can monitor system usage, generate analytics, and view reports.	System Monitoring	Admin	7.2.2	System Monitoring Module
3.2.2.3	Admins can configure external integrations like Google Calendar	Integration Management	Admin	7.2.3	External Integration Management API

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	or LMS systems.				
3.2.3.1	Students can access a personalized study plan tailored to their academic goals.	Personalized Study Plans	Student	7.3.1	Study Plan Generator Module
3.2.3.2	Students can collaborate using AI-assisted real-time whiteboards.	Collaborative Whiteboard	Student	7.3.2	AI-Powered Whiteboard Module
3.2.3.3	Students can track and manage academic tasks and deadlines.	Task Tracker	Student	7.3.3	Task Tracking Module
3.2.3.4	Students can generate short explanatory videos from whiteboard content.	Video Generation	Student	7.3.4	Video Processing Module
3.2.3.5	Students can access AI-powered visual learning content based on their syllabus and progress.	AI Visual Learning Tools	Student	7.3.5	Visual Content Engine
3.2.4.1	Students can receive AI-driven recommendations for study resources based on their preferences.	Resource Recommendations	Student	7.4.1	AI Recommendation Engine
3.2.4.2	Students can share resources and collaborate on projects using	Resource Sharing	Student	7.4.2	Resource Sharing Module

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	the platform's tools.				
3.2.5.1	Admin and students can access notifications for tasks, deadlines, and updates.	Notifications	Admin/Student	7.5.1	Notification Management Module
3.2.6.1	Students can convert textual content into visual formats such as charts or graphs.	Visual Representation of Text	Student	7.6.1	Text Visualization Module
3.2.7.1	Admins can adjust roles and permissions for users.	Role Management	Admin	7.7.1	Role and Permission Management Module

Table 51 Requirements Traceability Matrix

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11. System Prototype

11.1.1Design and Development Process

The design and development of the EDUassist prototype utilized a high-fidelity prototyping methodology. This approach was chosen to create an interactive and visually representative model of the platform, allowing stakeholders a to engage with the design in a manner that closely resembles the final product. The prototype was developed using Figma, a collaborative interface design tool that facilitates real-time feedback and iteration.

11.1.2 Functionalities Demonstrated

The prototype showcases several **core functionalities** essential to the eduassist system, including:

- **User Registration and Authentication**: Demonstrating how users can create accounts and log in securely.
- **Task Management**: Illustrating the process of creating, editing, and tracking tasks.
- **Resource Library**: Displaying access to educational materials and study plans.
- **Profile Management**: Showing how users can update personal details and customize preferences.
- **Calendar**: Providing users with the ability to manage schedules and deadlines.
- **Share Whiteboard**: Supporting collaborative work by enabling users to share whiteboard sessions.
- **Admin Dashboard**: Displaying system metrics, user management tools, and reports for administrators.
- **Recommendation .**
- **Upload files**

Additionally, there are **incomplete or future features** planned for implementation, such as:

- **Advanced Analytics Dashboard**: To provide insights on user engagement and performance.

11.1.3 Testing and Feedback

Testing Methodology

As a preliminary prototype, testing focused on gathering basic insights from initial user impressions:

- **Component Testing:** Each screen was reviewed to ensure visual coherence and consistency.
- **User Feedback Session**: A small group of educators and students were shown the prototype and provided feedback on its design and clarity.

User Feedback

Feedback from the session highlighted the following:

- **Positive Aspects**: Users liked the clean layout and simple design.
- **Areas for Improvement**: Users suggested adding clearer labels for buttons and refining the transitions between different sections.

Incorporation of Feedback

Based on the feedback:

- Button labels were clarified to ensure better understanding.

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- Transition designs were refined to enhance visual flow between pages.

These improvements set the stage for further iterations, ensuring the prototype aligns with user expectations and system goals.

11.1.4 Challenges and Limitations

In the development of the eduassist prototype using Figma, several real challenges and limitations were encountered that affected the design process:

1. **Limited Interaction Capabilities**: While Figma allows for the creation of interactive prototypes, complex interactions such as multi-step processes or conditional logic cannot be fully simulated. This limitation may lead to a gap between user expectations and the actual flow of the final application.

2. **Dynamic Content Representation**: Figma does not support the integration of dynamic content, which means that features reliant on user-generated data or real-time updates cannot be effectively demonstrated. For example, showcasing live notifications is not possible, potentially obscuring key functionalities of the eduassist system.

3. **Learning Curve for New Users**:For team members who are not familiar with Figma, there may be a steep learning curve. Understanding the tool's features and functionalities can take time, which may delay the prototyping process, especially if team members are new to design tools.

These challenges highlight the importance of understanding the limitations of prototyping tools like Figma. While it serves as a valuable resource for visualizing design concepts and facilitating collaboration, it is crucial to remain aware of these constraints to ensure effective communication of the eduassist project's vision and functionality to stakeholders.

11.2 Prototype flow diagram

The prototype serves a crucial role in validating the eduassist project idea. It allows stakeholders to visualize the intended user experience and gather early feedback, fostering a collaborative development environment.

The following **Figures from 106 to 112** are screenshots illustrate the prototype's design and functionality, showcasing the user interface and flow of interactions within the system.

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Figure 106 EDUassist Flow 1

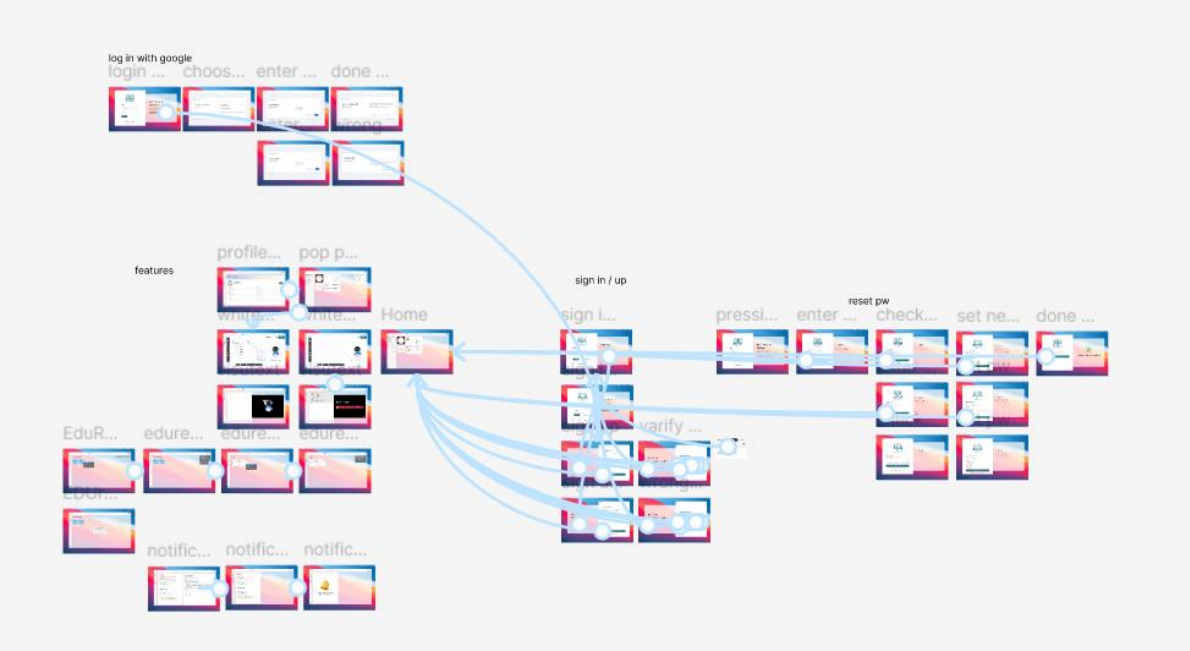


Figure 107 EDUassist Flow 2

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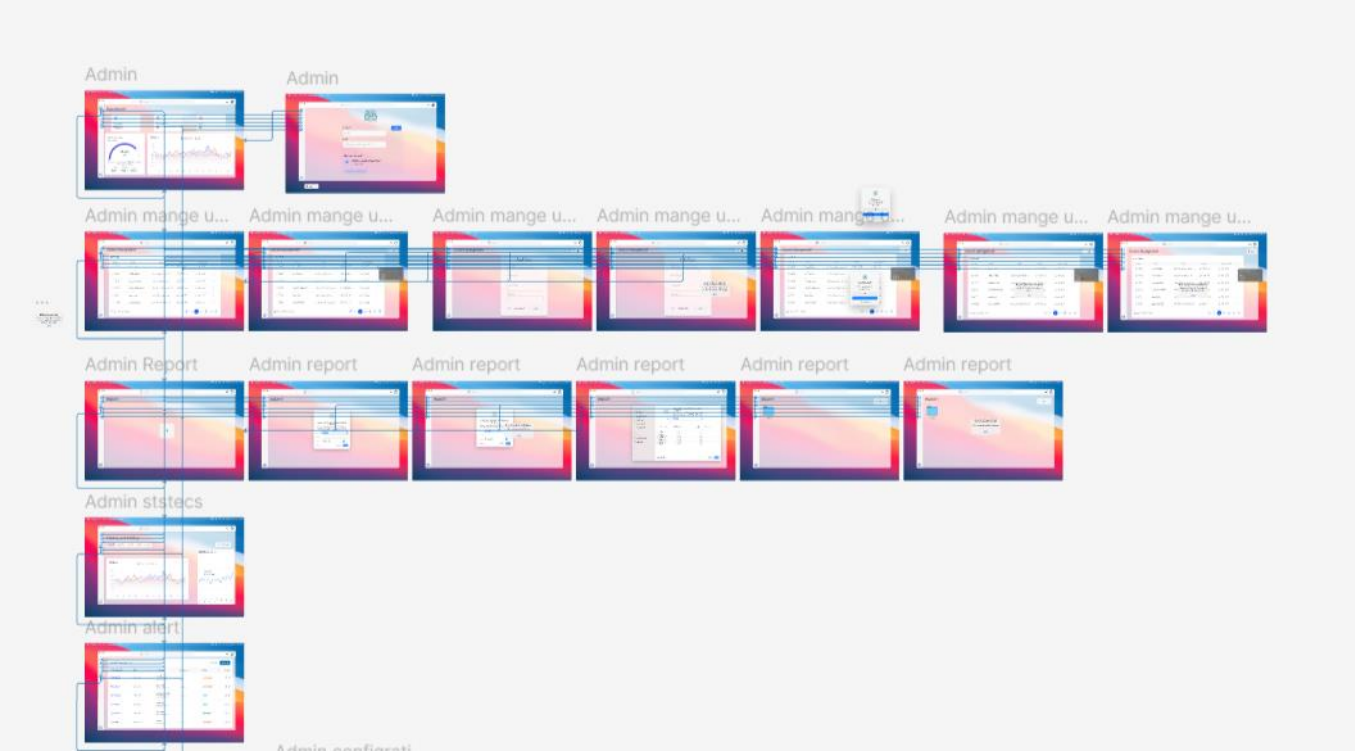


Figure 108 EDUassist Flow 3

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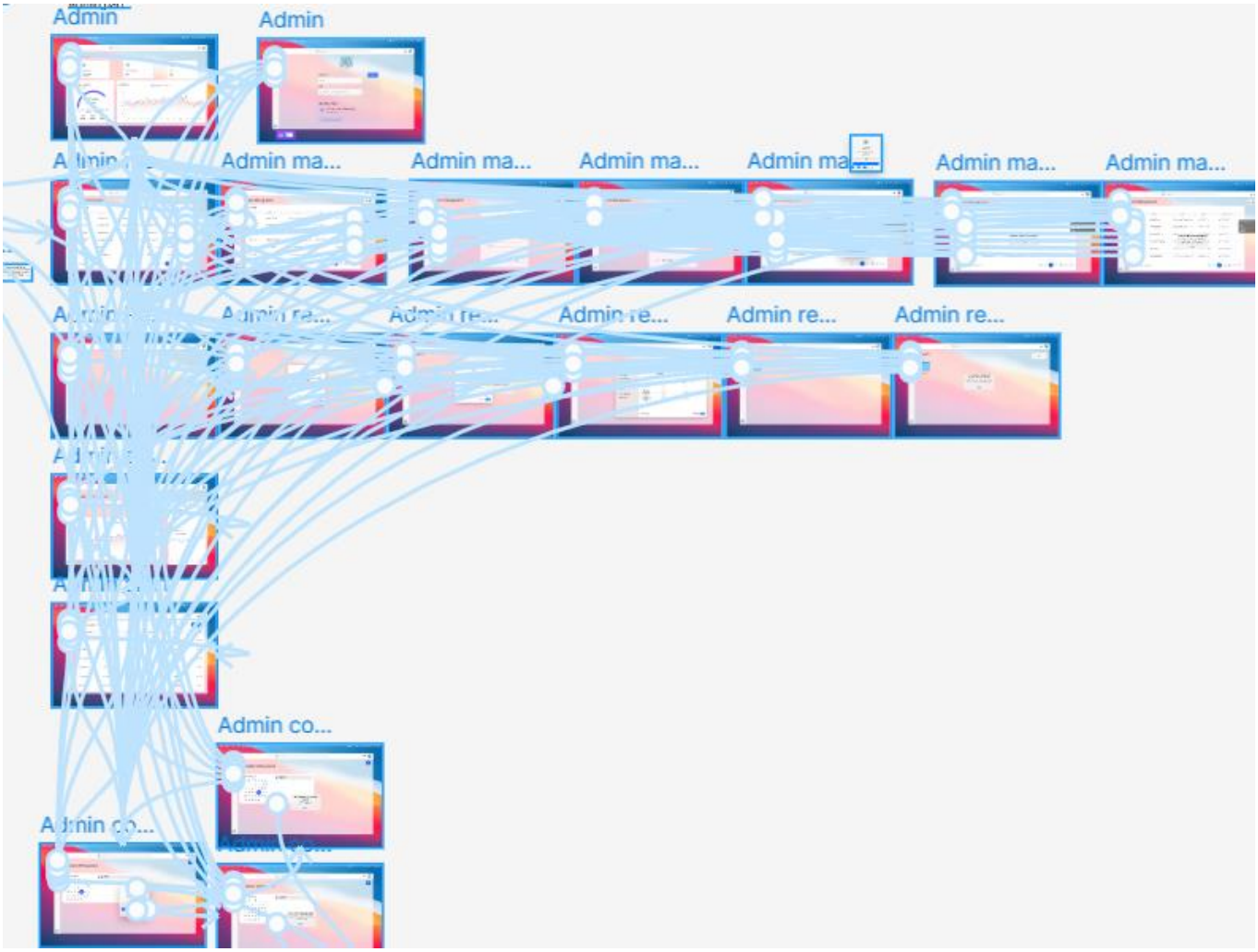


Figure 109 EDUassist Flow 3-1

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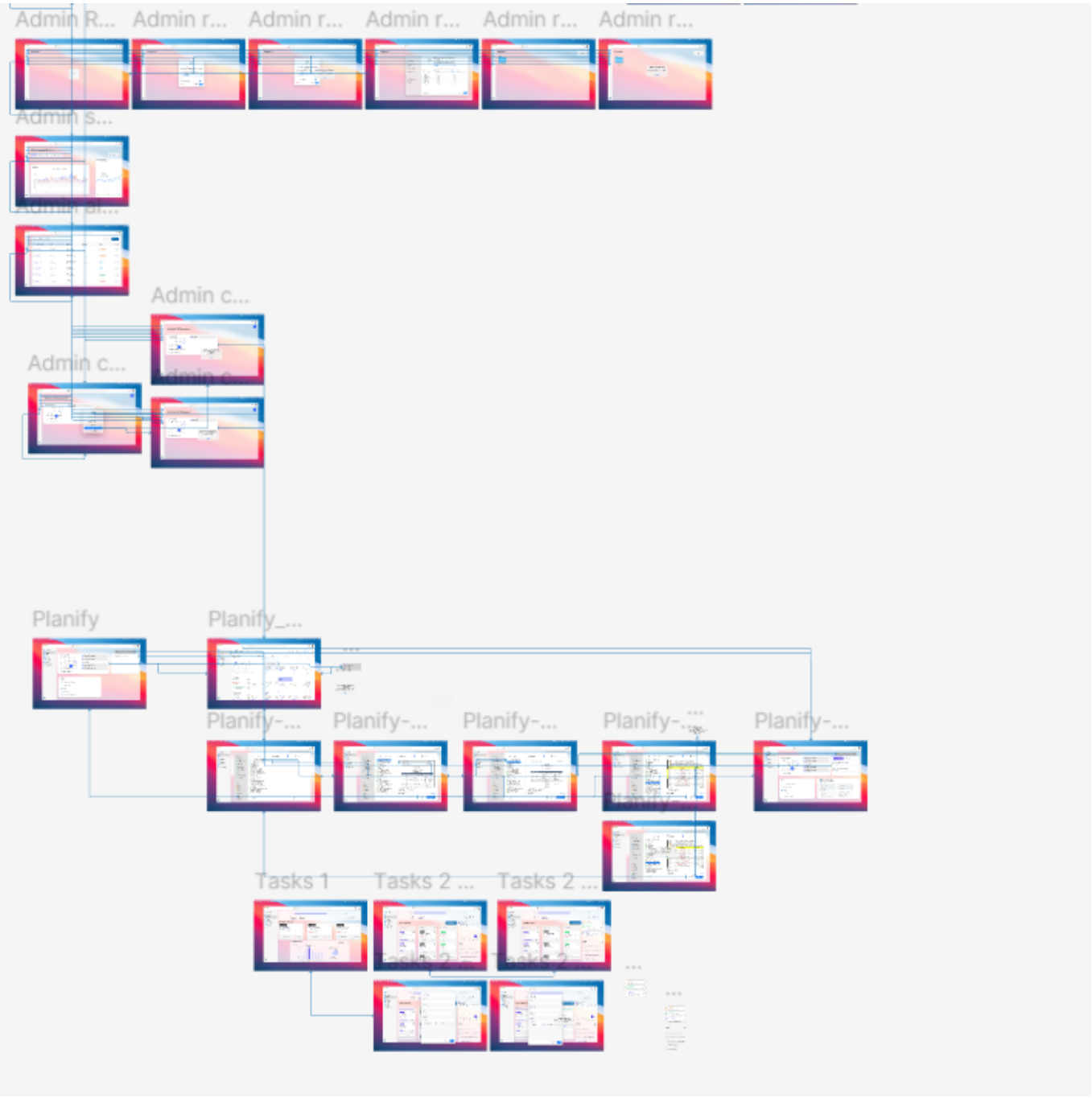


Figure 110 EDUassist Flow 4

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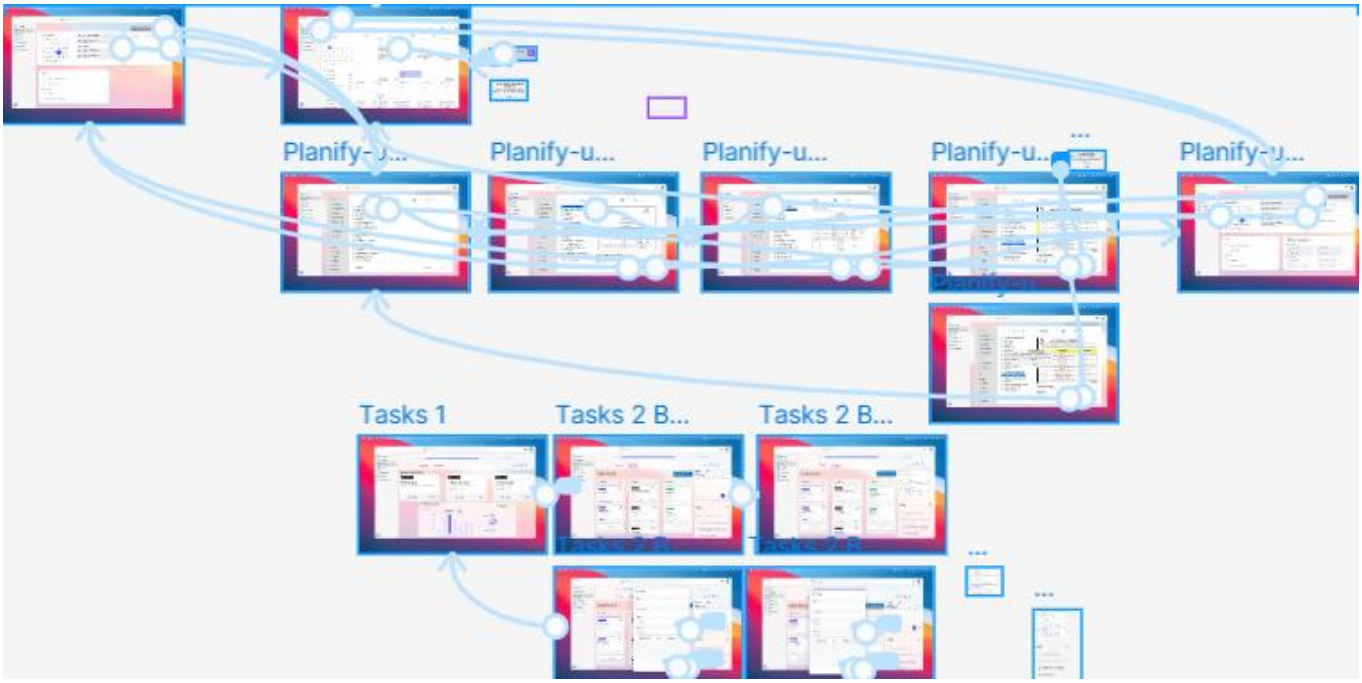


Figure 111 EDUassist Flow 4-1

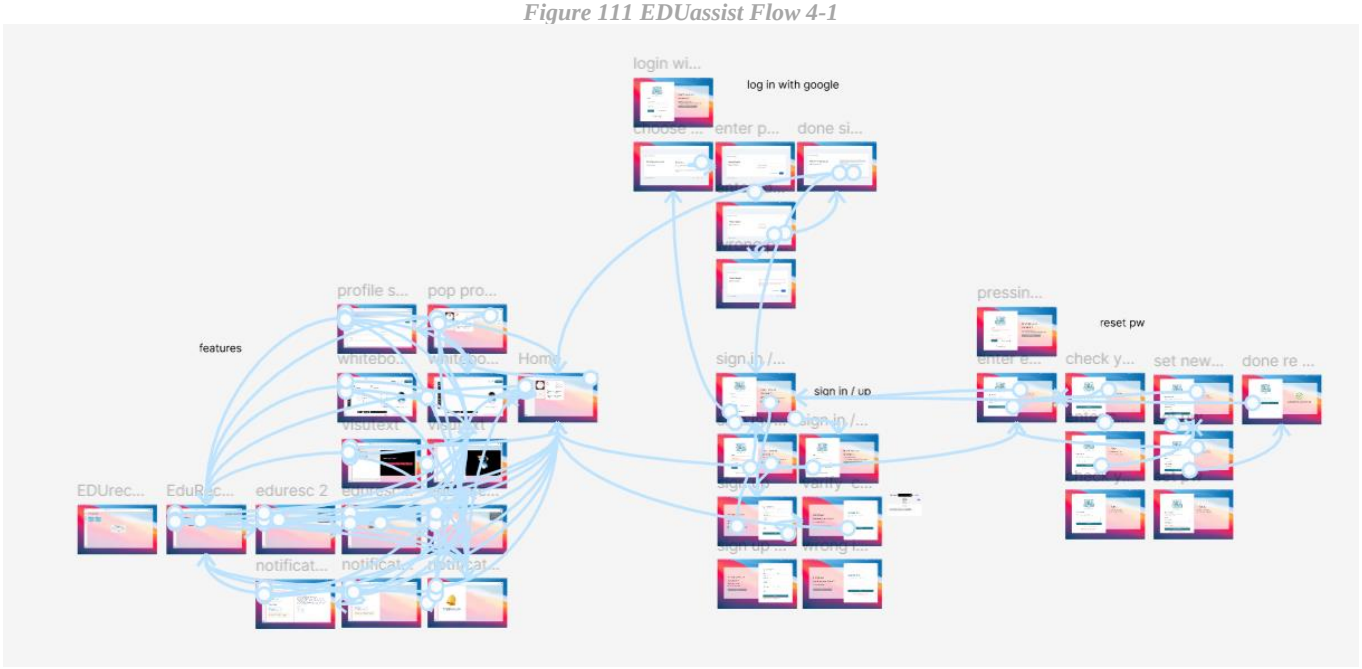


Figure 112 EDUassist Flow 5

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